

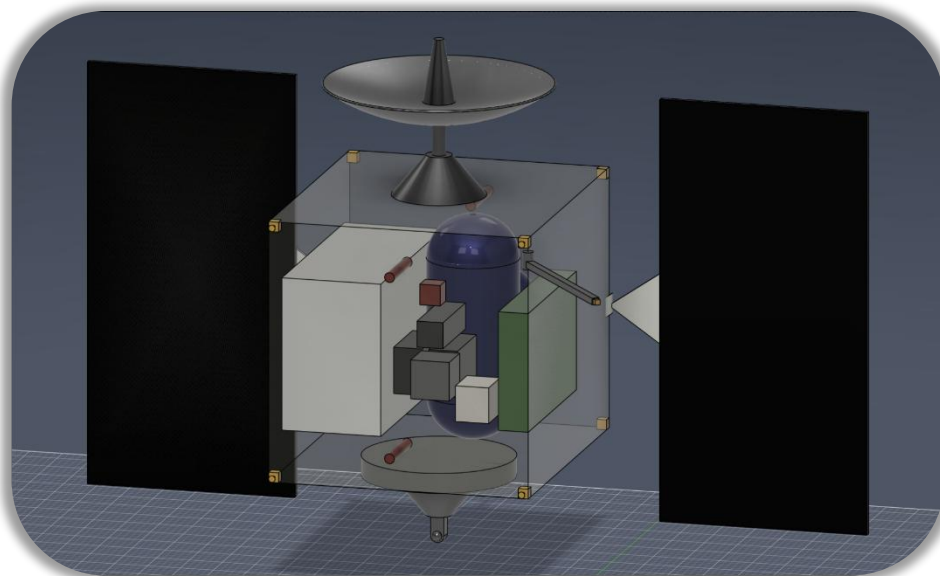
HEIMDALL

Heliophysics Earth Interface and Mars Data Acquisition Long-range Link

2026 AIAA Request for Proposal Response: Heliophysics Science
Observer & Communication Platform

ILLINOIS TECH

Armour College of Engineering



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A. Executive Summary

The Heliophysics-Earth Interface and Mars Data Acquisition Long-range Link (HEIMDALL) mission is proposed by the Illinois Institute of Technology's Scarlet Cows in response to AIAA's Spacecraft Design Request for Proposal (RFP). This RFP challenges undergraduate students to design a mission capable of progressing understanding of our Sun, support conjunction absent communications for future missions between the Moon and Mars, and meet standard project constraints, such as a \$400 Million-dollar maximum budget and a launch date prior to 2036 for operation through the late 2040s. The RFP lists two supporting documents, which give their own scientific goals and objectives.

The National Academies of Science, Engineering, and Medicine's *The Next Decade of Discovery in Solar and Space Physics* (DS) provides a context of what we understand currently about our sun, its effects on the surrounding heliosphere, and lists Priority Science Goals (PSGs) that should be answered in the next decade of space exploration. The document uses these PSGs to define what measurements need to be taken and their respective scientific instrument (Table B-5 in the survey). These Needed Capabilities provide the Scarlet Cows with the scientific requirements used in defining and determining our scientific payload.

NASA's Moon to Mars Strategy and Objectives (M2M) describes the current limits on communication to missions spanning the area between Earth and Mars and lists a wide range of requirements for future missions to reach Mars. While many of them are beyond the scope of a single mission, the Scarlet Cows used relevant objectives to define the communications requirements supporting future manned and unmanned missions to Mars.

The Scarlet Cows proposed four possible operation plans. Based on locations optimal for solar weather observation, Lagrange points around Mercury, Venus, and Earth were considered. Mercury and Venus Lagrange points were determined to be too costly in fuel and funds, leaving Earth Lagrange Points 4 (L4) and 5 (L5). Both are viable options for solar weather observation and enhanced communication, but because ESA VIGIL is headed to L5 in the mid 2030s, the Scarlet Cows decided to operate in L4. When discussing possible novelties to separate HEIMDALL from VIGIL, the Scarlet Cows considered the use of Trojan asteroids within Earth's L4. Using a crater on a Trojan asteroid as an antenna has been considered for many years as a possibility, but the limited knowledge of Trojan asteroids in L4, particularly spin rates, makes landing encounters too risky, leaving the Scarlet Cows with a final operation plan. HEIMDALL will operate from Earth LaGrange point 4.

To begin HEIMDALL's mission, a Falcon 9 will launch from Cape Canaveral by the end of 2035, carrying HEIMDALL into a 300 km parking orbit. Over a period of 5 days, calibration and instrument checks and verification will occur, and if all checks come back nominal, a go-no-go will be issued to begin the next phase of flight, a heliocentric transfer orbit. HEIMDALL will begin its transfer to the Sun-Earth Lagrange Point 4, a gravitationally stable region in space that offers a full view of solar phenomena between the Sun, Earth, and Mars. This transfer period will occur over approximately two years, as HEIMDALL moves 60 degrees ahead of Earth. The transfer orbit will be accomplished using the second stage of Falcon 9, which can act as a boost stage. After the two-year transfer, HEIMDALL will be approximately 60 degrees ahead of Earth and can perform its final phasing maneuver, a circularization to match Earth's orbit. At this point, the Falcon 9 second stage will separate from HEIMDALL and be discarded. After separation, HEIMDALL will orient itself, so the Sun observing instruments and solar arrays face the Sun and begin collecting data. During operation, space weather data, such as solar energetic particles (SEP), coronal mass ejections, and magnetic field changes can be collected and sent to Earth. The data can be processed and used to direct decisions relevant to missions affected by solar weather activity. Operations will last until 2049, after which HEIMDALL can continue to operate with additional funding, offer reduced data transmissions, or be decommissioned. Decommission would simply include an end of data transmission and monitoring as added debris in L4.

With a concept of Operations established, and comparing with active satellites relevant to our mission, a Science Traceability Matrix (STM) was created to define HEIMDALL's scientific objectives. The scientific and communications goal of the HEIMDALL is:

Detect, characterize, and provide timely warning of heliophysical transients, including coronal mass ejections, solar flares, and solar energetic particle events, as well as ambient solar-wind structures, including plasma disturbances and magnetic field variations along the Sun–Mars transit corridor to quantify radiation, plasma, and magnetic-field hazards to Mars-bound crews and vehicles.

The STM allows the Scarlet Cows to determine what observables, and in turn performance requirements, are necessary to achieve this goal.

A scientific payload trade study was conducted for eight instrument types across eighteen total instruments. The eighteen instruments were chosen from the STM based on relevance to our mission and available data found for the

trade criteria. For the heliophysics mission, the specific instruments needed to perform our proposed scientific objectives are analyzed in a trade study to determine which ones would best support the mission. HEIMDALL's core scientific payload will consist of a coronagraph, energetic particle telescope, and heliospheric imager, with the option to add a radio burst monitor, solar wind analyzer, X-ray flux monitor, magnetograph, and a fluxgate magnetometer to extend the number of scientific observables. These instruments, verified with a trade study, will each provide a piece of data necessary to detect space weather capable of impacting future Mars missions.

To determine a communications payload capable of establishing and maintaining consistent communications between Earth and Mars, a trade study was conducted. Two paths can be taken when it comes to satellite communication: technology that has been used consistently over the past decade of space missions, or experimental technology that may be commonplace by the end of the mission. HEIMDALL will accommodate both routes using a combination of X-band and laser communications. X-band, which has an extensive heritage background, will be used for necessary command and minimum data transmissions. Laser transmission, which is projected to be the next stage of interstellar communications, is capable of large and efficient data transmissions, at the cost of greatly increased pointing accuracy requirements. Accommodating a combination of both communications systems allows HEIMDALL to be both reliable and future compatible, expanding the range of utility past the required time frame, and is worth the extra accuracy required by the laser transmitter.

The "brain" of HEIMDALL will be required to not only process and send/receive data, but also to monitor every subsystem to ensure nominal operation. To do so, HEIMDALL will use the Voyager Technologies IPC-7000 with a Proton400k dual-core processor. This on-board computer (OBC) contains a powerful processor that maintains high levels of radiation resistance, making it suitable for the hazardous conditions found at Sun-Earth L4; however, this OBC does not contain a watchdog that automatically identifies unit failure, and continues to run the operations in a different built-in OBC. To account for this problem, HEIMDALL will have redundancy consisting of two Voyager Technologies IPC-7000 with Proton400k dual-core processors; only one OBC will be running at a time, while the other one will be the cold spare in the case of a dangerous CME or other anomaly that damages the working OBC. Since HEIMDALL will be located at L4 for approximately 15 years, the internal hardware oscillator found in the chosen OBC will maintain the master clock synchronized (corrected by ground station uplinks) and distribute a 1 pulse per second signal across payload operations. Data management is optimized to handle approximately 7 GB of

science telemetry generated daily. The OBC can process and compress raw solar images before routing the data to a dedicated 1 TB Solid-State-Buffer storage.

HEIMDALL's propulsion requirements are limited to regular station keeping once in position due to the use of the Falcon 9 upper stage as a boost stage, which greatly reduces the mass and ΔV requirement placed on the propulsion system. Analyses focusing on mass conservation, heritage equipment, and integration simplicity resulted in equipping HEIMDALL with monopropellant thrusters, which balanced the heritage requirement with the minimum propellant mass requirement. A benefit of choosing the Earth L4 position is that, once in position and the orbit is matched, fuel spent on station keeping is kept to a minimum, allowing for minimal mass or greatly extended operations. Because of the accuracy requirements imposed by laser communications, HEIMDALL will perform station-keeping maneuvers every one to two months to limit the accumulation of ΔV changes due to gravity and solar radiation pressure. Based on the conducted trades, a mission ΔV requirement of $54.352 \frac{m}{s}$ is necessary for the 15-year mission.

Taking advantage of constant solar exposure, HEIMDALL utilizes solar arrays from Rocket Lab in order to power the scientific payload, communication equipment, and supporting subsystems. In total, the satellite requires about 470 W to function, with the majority of the power going to the scientific payload and communication systems. Two of the specified solar arrays will be sufficient to provide this power requirement and will be deployed once in position. Despite the absence of eclipses, a minimum battery capacity is implemented to support operation and provide power in the case of an emergency. With a battery weighing 6.3 kg, HEIMDALL will be able to supply more than enough power required for operation.

The structural composition of the satellite is key to ensuring that the HEIMDALL satellite survives the approximately 15-year mission duration, where HEIMDALL will monitor the Sun as well as be a communication link between Earth, Mars, and future missions. To account for this issue, the materials for HEIMDALL were carefully selected, and it was concluded that a hybrid structure would be the best fit for HEIMDALL and the hazardous conditions found at Sun-Earth L4. The primary structure will be made out of Al-Li 2195 alloy, which exhibits good tensile strength, excellent cryogenic performance, and radiation resistance. The secondary structure will be made out of aluminum honeycomb cores with carbon fiber reinforced polymer to ensure that HEIMDALL survives the impact of small to medium-sized debris found at Sun-Earth L4, such as space dust. Finally, more critical components, such as joints and separation rings, will be made out of Ti-6Al-4V. This titanium alloy exhibits excellent acoustic performance that will translate

to more protection for the scientific and communication payload during liftoff and separation from the launch vehicle. The combination of these materials will provide a balance between mass savings and structural rigidity to ensure that HEIMDALL successfully completes all phases of its mission.

At L4, HEIMDALL is in a thermal environment that is dominated by the Sun's solar radiation. In this near-perfect vacuum, thermal convection and Earth radiation effects are negligible. Knowing this allows for heat collection, transportation, and dissipation to be carefully monitored and conducted to be able to balance HEIMDALL at a temperate equilibrium. This ensures all HEIMDALL's components stay within their respective temperature ranges, preventing any components from overheating or undercooling. The thermal control system is a mainly passive system with some active heaters for more precise temperature control for sensitive components.

With HEIMDALL's mission fully defined, analyses can be conducted on the risks of the mission and the cost of production and operation. In terms of risk, any sort of mechanical or electrical failure would leave part of the satellite inoperable indefinitely, so risk mitigation strategies focus on lowering the likelihood as much as possible. These strategies include increased radiation shielding, electromagnetic shielding, and redundancy in communication hardware, so Earth will always have some sort of method to communicate with and control HEIMDALL. Using NASA's Potential Cost Estimating Capability (PCEC), HEIMDALL will have a maximum total project cost of \$396.9 Million, meeting the budget constraints of the RFP. Since the largest limiting budget factor is operations costs, HEIMDALL could remain functional beyond the proposed mission scope with increased funding.

B. Introduction

A human expedition to Mars has been a global engineering and scientific objective for decades. However, multiple risks must be addressed before humans can safely make the journey to Mars. Among the most significant of these risks are the hazards posed by the Helio physical environment between Earth and Mars, including but not limited to radiation, energetic particles, and transient space weather events, many of which peak in the eleven-year solar cycle. In addition to the challenges posed by the deep space environment, communication interruptions and blackouts between Earth and Mars present a major challenge for crew safety and mission coordination.

As stated in the AIAA RFP (Table 1), the mission must provide both scientific instruments to monitor solar and space weather phenomena and a line-of-sight communications capability between Earth and Mars. This statement demonstrates the core objectives of the HEIMDALL mission. The main goal of HEIMDALL is to support a future crewed mission to Mars by providing Helio physical hazard information and improved communication availability across deep space. The services HEIMDALL will provide support future missions to Mars in the following ways: 1) monitoring and detecting solar and space weather phenomena that may threaten spacecraft and crew safety, 2) characterizing radiation, plasma, and magnetic field hazards along the Sun–Mars transit corridor, and 3) reducing communication blackouts between Earth and Mars to support long duration operations and crew safety. By meeting these science and communication objectives outlined in the RFP, HEIMDALL will improve crew safety during transit to and from the Red Planet.

Table 1: AIAA Requirements and Constraints.

AIAA Requirements
The satellite shall provide scientific instruments to support our understanding of the local cosmos, including the ability to monitor and detect solar and space weather phenomena, and enable the capability to provide early warning to crewed missions.
Ability to provide line of sight between the Earth and Mars to enable near constant communications capability during the REP specified mission period.
The mission design shall provide satellite and system analysis to show the capability for line-of-sight communication between Earth and Mars for the full Earth-Mars synodic cycle between 2036 and 2049.
AIAA Constraints
The cost of the mission and capability development in support of its activities shall not exceed \$400 Million US Dollar (in FY25), including development, hardware, launch, and operation cost of the mission through the primary mission phase.
The mission should launch no later than December 31, 2035, with the system designed to operate in the late 2040's

B.1 Mission Statement

HEIMDALL will detect, characterize, and provide timely warning of helio-physical transients, including coronal mass ejections, solar flares, and solar energetic particle events, as well as ambient solar-wind structures, including plasma disturbances and magnetic field variations along the Sun–Mars transit corridor to quantify radiation, plasma, and magnetic-field hazards to Mars-bound crews and vehicles.

C. Requirements and Constraints

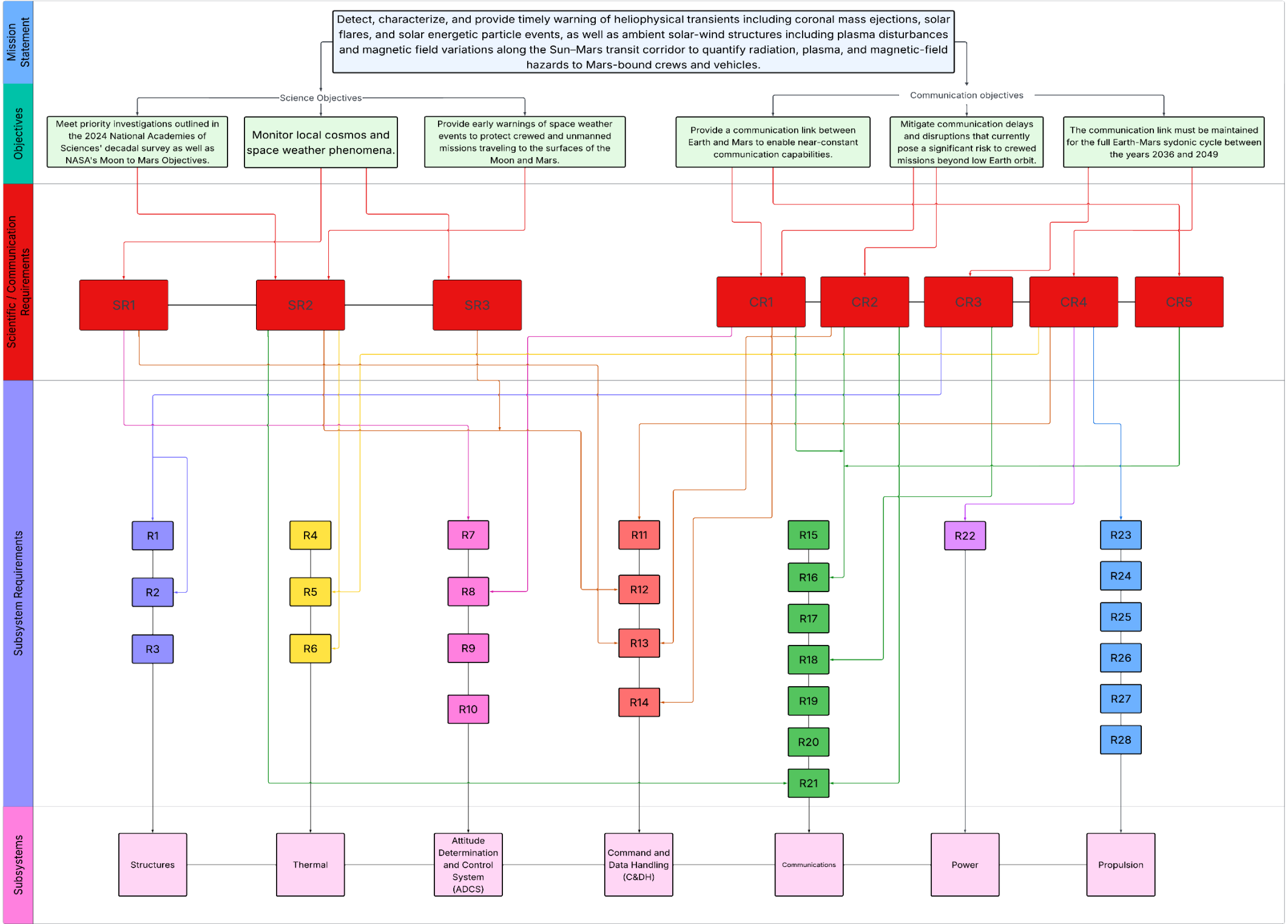


Figure 1: Requirements flow down diagram for the HEIMDALL mission. The different HEIMDALL's subsystems work together to attempt to accomplish all science and communication objectives.

The structure of Figure 1 is the following: the mission statement is what the HEIMDALL team wants to accomplish with its mission; this mission statement has to meet the scientific and communication objectives indicated by AIAA, along with the M2M and Decadal Survey objectives, which are found in the second level. Next, in order to achieve these objectives, the HEIMDALL satellite must complete the scientific and communication functional requirements. Lastly, HEIMDALL's subsystems must work collectively to ensure that the satellite has enough resources/technology/capacity to meet the science and communication requirements. The legend for the subsystem requirements can be found in Appendix 1.

The functional and performance requirements for space weather and communications objectives were developed around the Decadal Survey and Moon to Mars objectives; the R&Cs were organized in Table 2 and Table 3, respectively:

Table 2: R&C's for the space weather (Science) objectives.

Space Weather Functional Requirements	
SR1	The payload instrumentation shall provide measurements of coronal mass ejections (CMEs).
SR2	The payload instruments shall measure the projection of CMEs and provide an estimation of its direction with a predetermined point accuracy to generate a space weather forecast model.
SR3	The payload instruments shall measure the interplanetary magnetic field vector in the standard Radial-Tangential-Normal coordinate frame.
Space Weather Performance Requirements	
The payload instruments will measure CME's acceleration over a range of 500 km/s ² to 3,000 km/s ² .	
The payload instruments will measure the angular width and position angle of CMEs to an accuracy of $\pm 1^\circ$.	
The payload instruments will provide an estimated arrival time at targets (Earth, Mars, Moon, and satellite traveling within these bodies) with an accuracy window of ± 6 hours.	
The payload instruments will measure solar electron fluxes within a range of 100 keV to 15 MeV.	
The payload instruments will measure proton and ion fluxes across an energy range of 10 MeV to 500 MeV.	
The payload instruments will characterize lower energy seed populations at energy levels between 10 keV to 500 keV.	
The payload instruments will measure magnetic field magnitudes over a range of ± 25 nT.	

Table 3: Requirements and Constraints for the communication objectives.

Communications Functional Requirements	
CR1	The satellite shall provide a clear line of communication between Earth and Mars preventing lapses in communication.
CR2	The payload instruments shall provide position, navigation, and timing services for future crewed missions to Mars.
CR3	The satellite shall maintain communication functionality during nominal satellite operations and expected environmental conditions.
CR4	The communications hardware shall meet thermal limits and radiation tolerance requirements for the mission-specific environment.
CR5	The satellite shall operate in an approved deep space communications band (X-band) and comply with licensing requirements.
Communications Performance Requirements	
The satellite must be able to communicate with up to a range of 400 million kilometers.	
The satellite must maintain antenna pointing accuracy sufficient to support high-gain communications with the surfaces of Earth and Mars, respectively.	
The satellite must be compatible with the intended ground network in modulation, coding, and protocols.	
The communications system will operate within the satellite power budget, including peak transmit modes.	
The communications system must meet satellite mass and volume allocations.	

D. Concept of Operations

HEIMDALL will be located at the Sun-Earth Lagrange point L4 to facilitate a communication line between Mars and Earth while still being able to provide an accurate space weather forecast to protect future space missions with and without crews. Figure 2.1 and 2.2 shows a mission diagram for each of the phases for the HEIMDALL satellite.

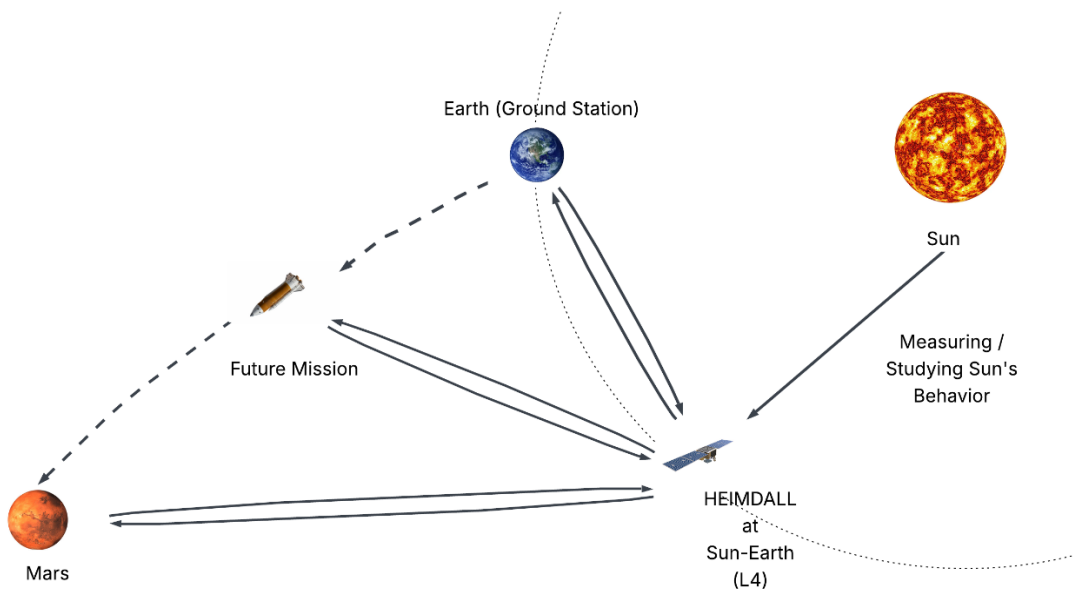


Figure 2.1: HEIMDALL communication links once Sun-Earth L4 is reached.

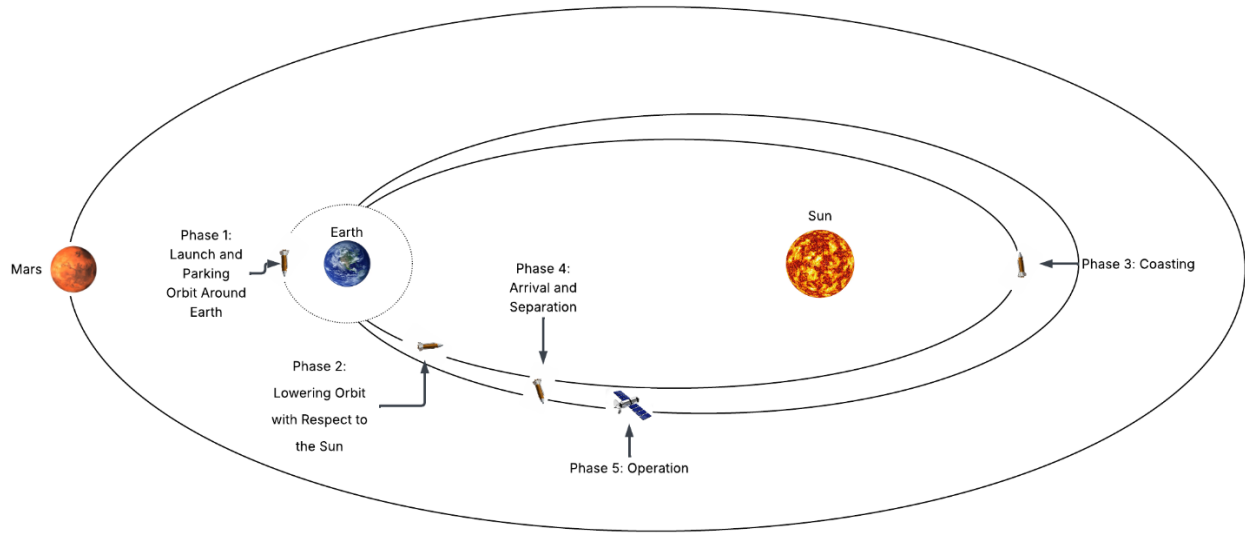


Figure 2.2: Operation Map for HEIMDALL phases.

D.1 Phase 1: Launch and Parking Orbit Around Earth

Cape Canaveral will be the launch site for this mission, as SpaceX's Falcon 9, our candidate launch vehicle capable of meeting our launch requirements, is located and launches from this site. HEIMDALL must be designed in a way that meets all criteria specified by the SpaceX Launching Guide. The HEIMDALL satellite will be delivered to a low Earth parking orbit of 300 km with an inclination of 23.43° . This is the inclination of the ecliptic plane of the Earth. Orbiting the Earth at the ecliptic plane ensures the satellite's heliocentric orbit in later phases at the correct inclination with respect to the solar system without additional fuel usage by the second stage of the launch vehicle. The most fuel-efficient launch window occurs at a RAAN of 0° ; at this RAAN, the difference between the ecliptic and orbital planes is the smallest. To achieve the RAAN of 0° , we simulate our launch using Ansys System Took Kit with the launch date set to be October 1st, 2034. The determined launch window is anticipated to be at 10:26:56.000 UTCG. After entering a low Earth orbit at 300km, HEIMDALL begins its second phase of the mission. To ensure the deployment at Sun-Earth L4, HEIMDALL will go through a series of calibrations to ensure that the equipment is in its full capabilities. This phase will last for 5 days, after which the third phase will begin.

D.2 Phase 2: Lowering Orbit with Respect to the Sun

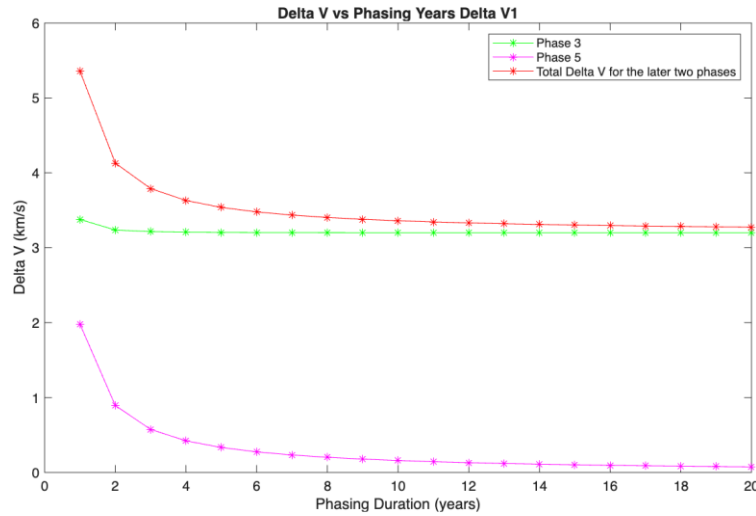


Figure 3: Phasing Maneuver duration vs Delta V change.

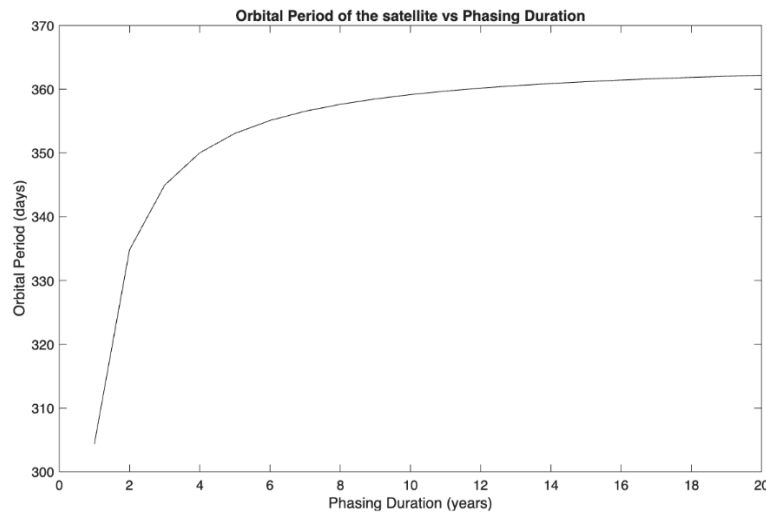


Figure 4: Orbital period of the satellite as a function of the orbits needed to arrive at Sun-Earth L4.

Figure 3 shows that as we increase the time it takes for HEIMDALL to reach L4, the amount of total fuel consumption for the two maneuvers decreases. Consequently, this saves mass and cost for the mission. Figure 4 shows the corresponding orbital period of the satellite as a function of the orbits needed for it to be at L4. Based upon the previous observation, HEIMDALL has opted to do a two-satellite-year phasing maneuver as it sees a significant drop in total fuel consumption but still be able to save time for mission operation. After five days of calibrations and system check at LEO, HEIMDALL performs its first maneuver since launch. This maneuver takes place on October 1st, 2034. The maneuver is carried out by the second stage of the launch vehicle. Once the HEIMDALL satellite finishes all

calibration on October 1st, 2034, at 10:49:26.058 UTCG its phasing maneuver begins. HEIMDALL is set to do a second burn to put it into an orbit of 337.15 days. At the time of the burn, HEIMDALL will be at 9.31° behind Earth's velocity vector. This burn is calculated to be $3236.6 \frac{m}{s}$.

D.3 Phase 3: Coasting

The launch vehicle's second stage protects the satellite and completes two full orbits with a total duration of 668 days before reaching Sun-Earth's L4. Also, during the phasing maneuver, HEIMDALL will run periodic tests of the payload equipment to check that everything is still functioning correctly.

D.4 Phase 4: Arrival and Separation

After HEIMDALL completes two full Heliocentric orbits, it will be in the L4 region located 60 degrees ahead of Earth. Once HEIMDALL has arrived at this desired point, the second stage will perform its last ΔV impulse to relocate itself into the same orbit as Earth's. The amount of ΔV needed is the difference between the Earth's heliocentric velocity and the satellite velocity at that exact location. The calculation shows us how much fuel HEIMDALL needs to burn to get a ΔV of $814.9 \frac{m}{s}$ tangential to its orbit required to raise its orbit back to Earth's heliocentric orbit. After this burn, HEIMDALL will place itself into the L4 region and orbit the Sun at approximately the same speed as Earth. Note that since Earth's orbit and the satellite's orbit are not circular but ellipses, the heliocentric velocities of Earth and HEIMDALL vary depending on the distance between them and the Sun, and the distance between the satellite and Earth will vary as well. Fortunately, since L4 is a stable region, minimal station-keeping maneuvers are needed. After arriving at L4, we are planning to operate until 2049. Software was used to simulate and visualize this maneuver. Figure 5 shows the location of HEIMDALL after two full orbits around the Sun. HEIMDALL, after orbiting the Sun for 668 days, is located at the L4 region. With this maneuver the HEIMDALL satellite satisfies one of the mission objectives established in Figure 1 which states that the mission has to be launched in late 2034 and be operable between late 2035 and late 2049.

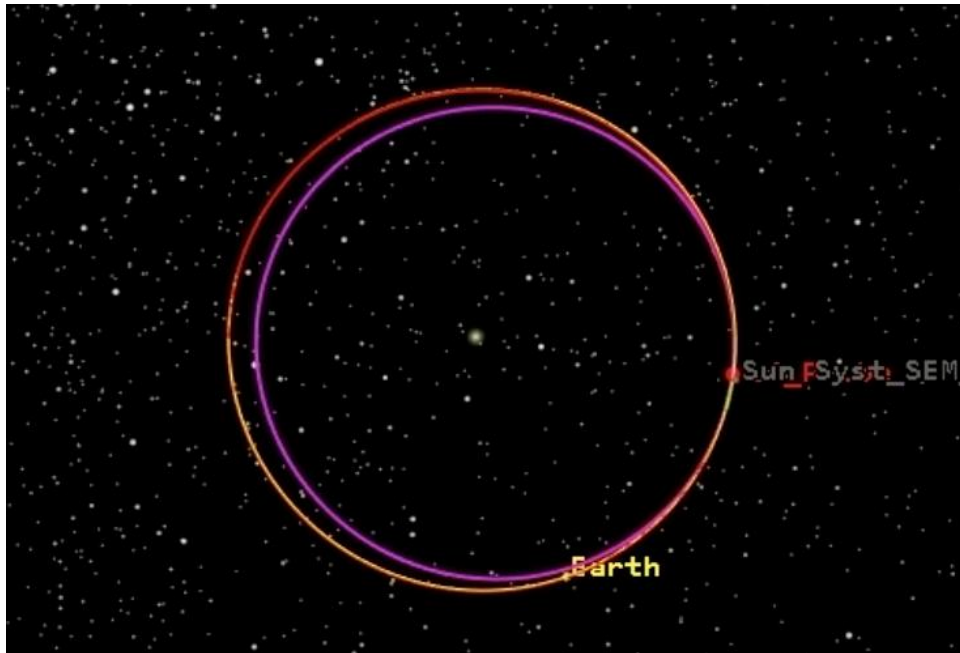


Figure 5: STK simulation of the phasing maneuver planned for HEIMDALL.

D.5 Phase 5: Operation

After the separation from the second stage of the launch vehicle Falcon 9, HEIMDALL enters its operational phase of the mission. During this phase, HEIMDALL deploys its solar panels. Its attitude determination and control system corrects its orientation. During this phase, the solar panels must face the Sun as HEIMDALL's main source of power comes from the solar array. Heliophysics equipment is placed on the same side of the satellite as the solar panels so that it can take pictures to analyze and characterize solar activities. The antenna mounted on HEIMDALL supports communication between Earth, Mars, and future missions. HEIMDALL stays in the L4 region for 12 years (until 2049).

D.6 Phase 6: End of Life

After 2049, HEIMDALL will remain located at Sun-Earth L4 and two potential scenarios have to be considered: the first one is if the HEIMDALL equipment is still in good conditions with no signs of failure, then HEIMDALL will continue monitoring the Sun and serve as a communication link between Earth, Mars, and crewed (or uncrewed) missions. However, if the scientific / communication payload presents severe damage, then HEIMDALL will be

completely abandoned at L4 or to be directed into a super synchronous graveyard orbit or completely, in the case where HEIMDALL is sent to a graveyard orbit, extra fuel must be considered to execute this maneuver.

E. Historical Analysis

The HEIMDALL team took inspiration from many past missions with similar goals and objectives, to successfully monitor space weather and solar activity, missions such as Parker Solar Probe [32] that contain instruments capable of measuring magnetic fields, plasma, and energetic particles with high fidelity. Also, the Parker Solar Probe used advanced thermal protection systems capable of surviving dangerous conditions similar to what HEIMDALL will face at L4. Another mission that was relevant to our design was the STEREO (Solar Terrestrial Relations Observatory) [33]. This mission had success measuring CMEs and solar flares from an angle offset from the Earth-Sun line. With this in mind, the idea of placing HEIMDALL at Sun-Earth L4 became the ideal to prove early-warning views of the solar surface before it rotates to face Earth. When it comes to communications, missions like MRO (Mars Reconnaissance Orbiter) [34], MAVEN, (Mars Atmosphere and Volatile Evolution) [35], and ESCAPADE (Escape and Plasma Acceleration and Dynamics Explorers) [36] served as a starting point to develop a long-range communication system for the HEIMDALL satellite.

E.1 Alternative Concepts

One of the first concepts that HEIMDALL considered was sending a constellation of satellites to Venus' Lagrange points L4 and L5. This direction was ultimately abandoned because of two reasons: the cost of sending two satellites to these points would surely surpass the budget specified in the AIAA request for proposal, and that while we would get measurements of the Sun from a much closer distance, when designing a clear line of communication between Earth, Venus, our Moon, and Mars finding ; sending two satellites to Venus' Lagrange points L4 and L5 would potentially add complexity and jeopardize one of the main goals of this project.

Next, the HEIMDALL team considered using one of the Earth's Trojan asteroids in L4 (2010TK7 and 2020XL5) as a landing site and having a direct platform to observe and measure the Sun's activity. The team considered that, with the help of robots, we could potentially use a crater found in any of these Trojan asteroids as a big antenna, opening the line of communication between Earth, Moon, and Mars. However, these trojan asteroids are close to the Sun, and the intense light makes it difficult to have a clear, detailed view of the surface of these asteroids [27]. It would not

only take years to find a good image of these Trojan asteroids, but also to study the behavior and spin rate of these objects.

Most of the missions involving Sun monitoring, such as NOAA [31] establish their satellites on Earth's L1, which results in a less expensive and easier maneuver. The only known mission that is planning a different orbital maneuver is the VIGIL satellite, currently being developed by the European Space Agency and scheduled to launch in 2031. VIGIL satellite will be positioned at Earth's L5, and it will be able to monitor and measure CMEs 4 to 5 days before they hit Earth [28]. The HEIMDALL team took this mission concept as an inspiration and decided to go to Earth's L4 to offer even more protection to Earth, and to spacecraft being launched to the Moon or Mars. Satellites located at L1 see dangerous CMEs directly in front of Earth, whereas HEIMDALL will be 60 degrees ahead of Earth's orbit, allowing us to have a side view of the Sun-Earth line and precisely track the direction, speed, and trajectory of dangerous CMEs. The positioning of HEIMDALL is key to ensuring the safety of any kind of mission because that way we can provide highly accurate arrival predictions of potential hazards [29]. Figure 6 summarizes the criteria that the HEIMDALL team followed in order to select Sun-Earth L4 as the best candidate for this mission.

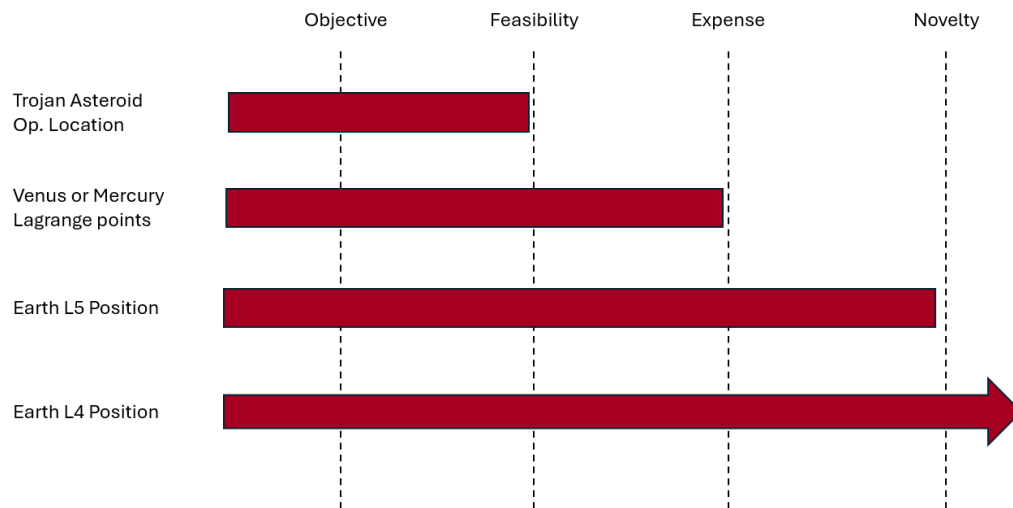


Figure 6: Alternative mission concepts. The HEIMDALL team evaluated multiple mission architectures and identified the Sun-Earth L4 orbit as the optimal candidate. This orbital placement guarantees the fulfillment of all mission objectives while maintaining strict budget compliance. The mission profile is highly feasible (leveraging maneuver profiles similar to the VIGIL satellite) while simultaneously establishing a pioneering, unprecedented mission concept at L4.

F. Science Traceability Matrix

The Science Traceability Matrix (STM) covers the high-level science objective with observables, measurement requirements, and displays the instruments selected from the Scientific Payload Trade Study, which is covered in section G. For the science mission, the primary objective is to detect, characterize, and provide timely warning of heliophysical transients, including coronal mass ejections (CMEs), solar flares, and solar energetic particle (SEP) events, as well as ambient solar-wind structures, including plasma disturbances and magnetic field variations along the Sun-Mars transit corridor. Through research, these areas of space weather were deemed the most critical to Mars-bound mission success due to their potential impact on spacecraft electronics and the human body. HEIMDALL's science mission can directly impact the success of future Mars-bound missions through a more timely and advanced understanding of the heliosphere, aligning with science requirements SR1, SR2, and SR3 found in Table 2.

The STM breaks the mission objective into various measurable observables, including CME detection, CME propagation, solar X-ray and flare activity, energetic particle flux, radio burst signature, plasma conditions, magnetic-field behavior, and solar source-region context. Each of the observables is tied to a specific measurable parameter, like CME speed, angular width, propagation direction, SEP onset time, particle energy range, solar-wind velocity, plasma density, magnetic-field strength, and radio burst frequency drift. This ensures that each instrument supports a specific aspect of the mission.

This complete list of equipment covers scientific measurables beyond what is needed to complete the mission objective. Thus, a minimum instrument configuration is defined as the least amount of equipment needed to ensure mission success. These instruments include a white-light coronagraph, a helio-spheric imager, and an energetic particle telescope/spectrometer. The white-light coronagraph provides the initial CME propagation direction, the helio-spheric imager extends this capability by tracking CME propagation through interplanetary space, improving estimates of the trajectory and arrival times, and the energetic particle telescope/spectrometer provides direct monitoring of SEP radiation conditions, which is one of the most immediate hazards to Mars-bound crews.

Additional instruments in the STM would improve event classification, CME tacking, magnetic-field characterization, and source-region identification. However, these instruments enhance extended scientific ability rather than meeting the minimum mission requirements. Overall, the STM provides the outline of mission objectives for scientific payload selection. It defines the measurement architecture, identifies the minimum viable payload, and shows a path for expanding scientific capabilities.

Table 4: Science Traceability Matrix (STM) for the HEIMDALL satellite.

Science/Comms Objectives	Measurement Requirements			Instrument				
	Observables	Parameters	Performance Requirements	Instrument Type	Proposed Instruments	Instrument Performance	Parameters For Mission	Size / Weight / Power
Detect, characterize, and provide timely warning of heliophysical transients including coronal mass ejections, solar flares, and solar energetic particle events, as well as ambient solar-wind structures including plasma disturbances and magnetic field variations along the Sun–Mars transit corridor to quantify radiation, plasma, and magnetic-field hazards to Mars-bound crews and vehicles.	Coronal Mass Ejection (CME) Detection	White-light coronal brightness evolution and CME front morphology; onset time; plane-of-sky leading-edge speed; acceleration; angular width; propagation direction proxy; brightness/mass proxy; expansion rate.	Visible-band coronagraph with defined spectral range, FOV, cadence, spatial resolution, and SNR sufficient to detect CME onset and front evolution.	White-light Coronagraph	CCOR-1	CCOR-1 heritage: 480–730 nm; FOV 3.7–17 R _⊙ ; image size 2048 × 1920 pixels.	Continuous Sun-pointing observations, low stray-light environment, and downlink latency compatible with event warning products.	CCOR-1: 35" × 23" × 15"; 25 lb; 25 W; 25 kbps.
	CME Propagation and Projection	Heliospheric white-light elongation–time track of the CME front; radial distance vs. time; speed; acceleration/deceleration; direction relative to Mars transit line; ETA; arrival speed; interaction with HSS/CIR.	Heliospheric imager with wide-angle FOV, stable photometry, and cadence/latency sufficient to update arrival forecasts. Recommended operational cadence ≤60 min; latency ≤120 min.	Heliospheric Imager (HI)	SOLOHI	SOLOHI heritage: 3968 × 3968 effective image size; ~1.3 arcmin resolution; spectral bandpass broad visible/NIR ~500–850 nm; operational cadence ≤60 minutes; end-to-end latency ≤120 minutes.	Continuous time-series viewing geometry for the Sun–Mars line, uninterrupted solar-disk view, and end-to-end latency ≤120 min.	SOLOHI: 684.0 × 455.1 × 115.6 mm; 16.6 kg; 13.5 W; ~53.5 Gbits/orbit. X-ray monitor envelope; weight/power not given.
	Solar X-ray / Flare Observables	Soft X-ray flux of the full solar disk in 0.1–0.8 nm and 0.05–0.4 nm bands; flare start time; rise time; peak flux; flare classification; timing relative to particle release.	Full-disk X-ray monitor with defined passbands, dynamic range, and cadence sufficient for flare alerting.	X-ray Flux Monitor	VIGIL heritage X-ray monitor	X-ray flux spectrum (1.55–24.8 keV) across 6 channels with high resolution; photon flux range 2×10 ⁻⁸ to 2×10 ⁻³ W/m ² ; 10 degree field of view for total solar disk coverage.	Uninterrupted solar-disk viewing and low-latency alert generation.	10 × 10 × 32 cm; weight and power not given in source.
	Solar Energetic Particles (SEP)	Electron, proton, and ion differential flux versus energy and time; SEP onset time; peak flux; event duration; fluence; spectral hardness; radiation-storm severity.	Energetic particle detector with defined energy coverage, dynamic range, and cadence adequate for real-time radiation monitoring.	Energetic particle telescope / spectrometer	STIS heritage / similar energetic particle detector	Ions: 25–6000 keV; electrons: 25–250 keV; solid-state spectrometer; dual telescopes for high/low flux measurement.	Continuous monitoring during cruise and telemetry prioritization for radiation alerts.	SEP detector heritage example: mass 2.3 kg; power 3.8 W; <1 kbps.
	Shock / Radio Observables	Radio dynamic spectrum, including Type II and Type III bursts, intensity vs. frequency/time, and polarization; shock presence; shock start time; Type II drift rate; electron-escape timing.	Radio spectrometer with continuous decametric/hectometric coverage, sufficient spectral/time resolution, and deployable antennas.	Radio burst monitor / radio spectrometer	WAVES heritage (e.g., WIND @ L1)	4kHz–14 MHz; measures complete wave polarization and solar-wind electron plasma frequency; 80° × 60° FOV apertures.	RF-quiet spacecraft environment and accommodation for long antenna deployment.	Extremely long wire antennas (50 m, 7.5 m, 5.28 m ×2), 12 m radial boom.
	In-situ Solar Wind Plasma Observables	Bulk speed (V _{sw}); number density (N _p); ion temperature (T _i); electron temperature (T _e , if measured); alpha fraction/composition; dynamic pressure; shock arrival signatures; ICME sheath/ejecta discrimination; HSS/CIR arrival and duration.	Solar-wind plasma analyzer with energy-per-charge coverage, FoV, cadence, and dynamic range sufficient to resolve ambient flow and shock structure.	Solar-wind ion / plasma analyzer	PLA on VIGIL heritage Background	Velocity dynamic range 200–2500 km/s; density dynamic range 0.2–150 cm ⁻³ ; temperature dynamic range 40,000–1,000,000 K; field of view (FoV) ±22.5° (azimuth, in ecliptic plane) × 45° (elevation).	Ram-facing viewing geometry and near-continuous sampling through solar-wind disturbances.	Estimates based on Solar Orbiter SWA-EAS on which VIGIL will be based: 16.5 kg; 78.75 L volume; 14 W; 14,000 bps.
	Magnetic-Field Observables	Full vector magnetic field B; B ; directional change; fluctuation level; shock jump signature; IMF polarity; flux rope / magnetic-cloud rotation; turbulence level; plasma beta when combined with plasma data.	Vector magnetometer with defined range, cadence, noise floor, and boom-mounted for magnetic cleanliness.	Fluxgate magnetometer	Magnetometer (MAG) / Fluxgate Magnetometer (FGM) pair	±440 nT range; 8 Hz cadence; 928 bps data rate, boom separation 3.9 m and 5.6 m.	Dedicated boom, magnetic cleanliness program, and stable attitude knowledge.	MAG heritage: 0.65 kg; 1.71 W; boom placement at 3.9 m and 5.6 m.
	Solar Source-Region Context	Full-disk EUV imaging and photospheric vector magnetic field for eruption source-region identification, magnetic connectivity, and contextual forecasting support.	Optional enhancement measurement set; retain only if the mission wants source-region context beyond threshold hazard detection and characterization.	EUV imager / magnetograph	PMI on VIGIL	Vector magnetograms; spatial resolution ~2–2.5 arcsec; FOV full solar disk + margin; magnetic-field accuracy 10 G.	Maintain solar-disk view; include only if enhancement science is in scope.	PMI; optics unit envelope 500 × 400 × 700 mm, 22.6 kg; electronics unit envelope 225 × 223 × 175 mm, 5.6 kg; 40 W.

G. Scientific Payload Trade Study

For our heliophysics mission, the specific instruments needed to perform our proposed scientific objectives are analyzed in a trade study to determine which ones would best support the mission. An instrument trade study was conducted for eight instrument *types* and 18 total instruments within those 8 categories. The 18 instruments were chosen based on relevance to our mission and available data found for the trade criteria. The top-rated instruments resulting from the study can be observed in the STM above in more detail.

A Technology Readiness Assessment (TRA) was conducted to study the instrumentation proposed to determine what *current* technology best fits the defined mission. The Technology Readiness Level (TRL) scores for each instrument were estimated using *Table 2.3.2-1: TRL Definition and Decomposition by Factor*, since most TRL scores were not listed in their respective literature. Scores were specifically based on whether the instrument was heritage and if it was made for use in a closely related orbit to our L4 mission. All Sun-Earth Lagrange point orbits were rewarded with a matching orbit type score, which then boosted the TRL level that we estimated. The instruments used in LEO or GEO did not receive a ‘Yes’ in Table 7 for a matching orbit type in the trade study.

Five criteria were chosen to conduct the trade study, which we decided were most relevant in this initial process. The criteria include: TRL, Orbit Type, Mass, Size, and Power in Table 5. These parameters add up to 1, and the weights of each are explicitly considered. TRL, Orbit Type, and Size each weigh 0.25, as we believed these were the most important parameters that would contribute to our design-making process for instruments to use. Orbit Type is our means of defining whether an instrument was used in a relevant orbit, such as our L4 orbit, and Size is the expected volume of the instrument. Our mission is under a strict budget; we prefer to consider instruments that have already been developed, proven to function in relevant environments, and smaller in size to be able to save money. Power and Mass were weighed at 0.10 and 0.15, respectively. These two parameters are still important in our decision-making, but they are easier to work around when evaluating instruments. The allocation of the criteria weights is shown in Table 5.

Table 5: Science Payload Weight Criteria

Criterion	Type	Weight ($\Sigma=1.0$)	Direction	Scoring Notes
TRL	Discrete	0.25	Higher is better	NASA TRL (map using lookup table)
Mass (kg)	Quantitative	0.15	Lower is better	Lower mass preferred (spacecraft constraints)
Power (W)	Quantitative	0.1	Lower is better	Lower power preferred (power budget)
Size (m ³)	Quantitative	0.25	Lower is better	Lower volume preferred (packaging constraints)
Orbit Type Match (Yes/No)	Binary	0.25	Yes=1, No=0	Does candidate match mission orbit? Yes→1, No→0

The trade study was completed for 8 different *types* of scientific instrumentation shown in the Legend in Table 6: two coronagraphs, three magnetometers, three plasma analyzers, three x-ray flux measurement systems, two heliospheric imagers, one magnetograph, two energetic particle telescopes/spectrometers, and two radio burst monitors. Table 7 shows the instrument data that was retrieved from their respective mission descriptions, input into the STM, and then used for the trade study. The parameters in red font had to be approximated from relevant mission findings since their values could not be found in their documentation.

Table 6: Instrument Legend and Input Parameters

Legend
Coronagraphs
Magnetometers
Plasma Analyzers
X-Ray Flux
Helio spheric Imagers
Magnetographs
Energetic Particle Telescopes
Radio Burst Monitors

Table 7: Science Payload Instrument Parameters

Candidate	TRL (1-9)	Mass (kg)	Power (W)	Size (m ³)	Orbit Match (Yes/No)
Compact Coronagraph CCOR-1 on GOES-U	9	25	22.5	0.18954	No
Roman Coronagraph	6	303	340	3.366	Yes
Magnetometer FDM Pair on SWFO-1	8	0.65	1.71	5.6	Yes
MAGIC AMR Vector on RadCube	8	0.023	0.48	0.000005292	No
Cesium Scalar on Magsat	7	7.7	9	0.01042368	No
CoDICE on IMAP	8	16.1	25.1	0.0434	Yes
Falcon SEED on STPSat-6	8	4.3	3.4	0.002	No
PLA on VIGIL	9	25	5	1	Yes
XFM on VIGIL	9	1	1	0.0032	Yes
HEL1OS on Aditya-L1	9	8.1	14.55	0.01507768	Yes
EXIS on GOES-R	8	30	40	0.08436	No
Heliospheric Imager on Solar Orbiter	9	16.56	13.5	0.0777843	No
Heliospheric Imagers on STEREO	8	16.3	10	0.12012	No
Magnetograph PMI on VIGIL	8	28.2	40	0.14	Yes
EPT-HET on Solar Orbiter	8	1.57	5.2	0.00289344	No
STIS on SWFO-L1	9	2.3	3.8	2	Yes
RPW on Solar Orbiter	8	13.6	7	5	No
WAVES on WIND	8	10	1	337.1963	Yes

Scoring each parameter in the trade study was normalized using a 0-1 scale. This was computed with a linear scaling method. The total score was computed using a weighted scale where the instruments were ranked according to their weighted scores. The higher the score of each of the weighted final scorings of the instrument parameters, the more those instruments fit into our mission. These scores can be seen in Table 8.

If the instrument value was the maximum value, then that parameter score received a 0, meaning that the instrument does not fit our team's mission. For example, if an instrument weighed 300 kg, and there were no other weights larger, that instrument would receive 0 in the category score. The closer to 1 that the parameter scores received, the more likely those instruments reflected what our mission requires and how well they aligned with the specified criteria in the study. The total weighted score was computed by summing the product of the weight of the criteria and the normalized score. The total weighted scoring can be observed in Table 8. The TRL score was computed with a TRL

Lookup table, and the other scores were computed using Table 37 shown in Appendix 2. The total weighted score shows each instrument's averaged scores across the five criteria. The higher this score, the better those instruments are for the mission.

Table 8: Instrument Parameter Scorings

Candidate	TRL Score	Mass Score	Power Score	Size Score	Orbit Score	Total Weighted Score	Rank
Compact Coronagraph CCOR-1 on GOES-U	1	0.9176	0.9351	0.9994	0	0.7310	9
Roman Coronagraph	0.6	0.0000	0.0000	0.9900	1	0.6475	18
Magnetometer FDM Pair on SWFO-1	0.9	0.9979	0.9964	0.9834	1	0.9702	5
MAGIC AMR Vector on RadCube	0.9	1.000	1.000	1.000	0	0.7250	10
Cesium Scalar on Magsat	0.8	0.9747	0.9749	1.000	0	0.6937	17
CoDICE on IMAP	0.9	0.9469	0.9275	0.9999	1	0.9598	6
Falcon SEED on STPSat-6	0.9	0.9859	0.9914	1.0000	0	0.7220	12
PLA on Vigil	1	0.9176	0.9867	0.9970	1	0.9856	4
XFM on Vigil	1	0.9968	0.9985	1.0000	1	0.9994	1
HEL1OS on Aditya-L1	1	0.9733	0.9586	1.0000	1	0.9918	3
EXIS on GOES-R	0.9	0.9011	0.8836	0.9997	0	0.6985	16
Heliospheric Imager on Solar Orbiter	1	0.9454	0.9617	0.9998	0	0.7379	8
Heliospheric Imagers on STEREO	0.9	0.9463	0.9720	0.9996	0	0.7140	14
Magnetograph PMI on Vigil	0.9	0.9070	0.8836	0.9996	1	0.9493	7
EPT-HET on Solar Orbiter	0.9	0.9949	0.9861	1.0000	0	0.7228	11
STIS on SWFO-L1	1	0.9925	0.9902	0.9941	1	0.9964	2
RPW on Solar Orbiter	0.9	0.9552	0.9808	0.9852	0	0.7127	15
WAVES on WIND	0.9	0.9671	0.9985	0.0000	1	0.7199	13

To better visualize this rating system, the 18 instruments were then plotted on a color-coded bar graph shown in Figure

7. The highest rated instruments can be determined, thus helping to identify which would be best for our mission.

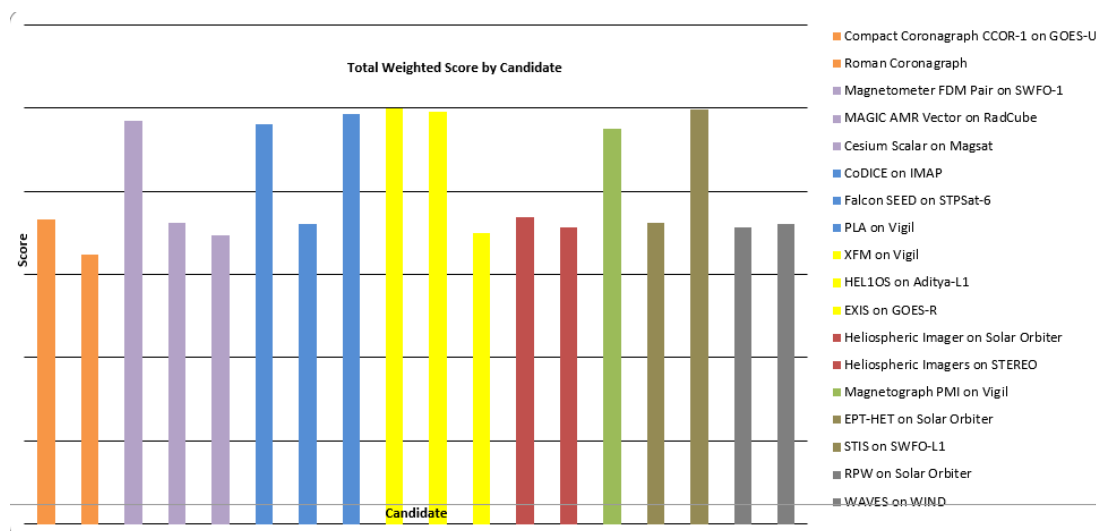


Figure 7: Bar Graph of the 18 Rated Instruments

HEIMDALL should be able to use the best-rated of each instrument category to monitor and detect a wide variety of heliophysics objectives shown in the STM table. As explained in the STM in the section above, the three instruments that are a minimum requirement to accomplish our science objectives are the coronagraph, heliospheric imager, and the energetic particle telescope/spectrometer. This trade study outlined which of those specific instruments would be selected to use in the payload.

This scientific payload trade study helps narrow down a selection of the proper instruments required to accomplish our proposed mission objectives. First, the AIAA requirements of designing a spacecraft equipped with the right instruments to support our understanding of the local cosmos while monitoring and detecting solar/space weather phenomena to be able to provide early warning for crewed missions. Any selection of the instruments in the trade study will accomplish the monitoring and detection of solar weather, which will also act as an early warning system. Additionally, the instruments selected in the study will accomplish the Space Weather Requirements shown in Table 2. Altogether, these instruments provide an insight into monitoring and detecting solar and space weather.

H. Communication Equipment Trade Study

For our communications mission, four modes of transmission were considered in a trade study to determine performance benchmarks relevant to our planned mission architecture. Utilizing our best judgment, two communication modes were selected to be specialized for scientific data/satellite health and bulk data transmission, respectively.

Conducting a communications trade study is slightly more straightforward in comparison to scientific instrumentation, as our communication subsystem essentially performs two tasks: receiving data onboard our satellite at Earth-Sun L4 from ground stations or other spacecraft and sending transmission data from our satellite to those sources (uplink/downlink).

Table 9: Various performance parameters of four communications modes, derived from previously flown missions' published data.

Candidate	TRL	High-End Mass	Peak Power	Size (m^3)	Maximum Effective Range	Bitrate at Max Range	Frequency Range/Wavelength	Atmospheric Interference (Y/N)
S-band RF	9	35 kg	100 W	5	0.00215 AU	1.6 kbps	2-4 GHz	N
X-band RF	9	45 kg	100 W	5	10+ AU	1-5 kbps	8-12 GHz	N
Ka-band RF	9	55 kg	100 W	5	2.5 AU	331 kbps	12-18 GHz	Y
Near-Infrared Laser	9	35 kg	75 W	2	2.58 AU	6.25 Mbps	1550 nm	Y

Notably, all four communications modes (S-band RF, X-band RF, Ka-band RF, and Laser) possess a Technology Readiness Level of 9. This benchmark was considered critical to the communications subsystem, as utilizing our mission as a testbed for unproven communications equipment would jeopardize all other mission objectives.

Mass, power, size, effective range, bitrate at range, and frequency/wavelength were determined via heritage NASA missions' reported performance. These factors were seen as critical due to their impact on other subsystems' requirements and overall SWaP. Power for all RF systems was budgeted at 100W for peak power consumption. Relative size for all RF systems included an approximate 4 cubic meters for a 2m HGA, the largest we could comfortably pack in a fixed configuration. Mass, maximum range, and bitrate at maximum range were taken from published performance specifications of previously flown missions [16, 17, 20, 22].

Table 10: Selection criteria and assigned weights for communication modes.

Criterion	Type	Weight (sum=1.0)	Direction	Scoring Notes
Bitrate	Quantitative	0.25	Higher is better	Predicted bitrate at maximum range for communications
Mass (kg)	Quantitative	0.2	Lower is better	Lower mass preferred (spacecraft constraints)
Power (W)	Quantitative	0.2	Lower is better	Lower power preferred (power budget)
Size (m ³)	Quantitative	0.1	Lower is better	Lower volume preferred (packaging constraints)
Range (AU)	Quantitative	0.25	Higher is better	Does maximum range exceed 1AU?

Weights were assigned by prioritizing in descending order: range, bitrate at range, mass, power, and size. Corresponding to their weights, bitrate and range were considered critical to communications performance and overall mission success.

Table 11: Total and sub scores of selected communication modes.

Candidate	Bitrate Score	Mass Score	Power Score	Size Score	Range Score	Total Weighted Score	Rank
S-band RF	0.000256	1.0000	0.0000	0.0000	0.0002	0.2001	3
X-band RF	0.00048	0.5000	0.0000	0.0000	1	0.3501	2
Ka-band RF	0.05296	0.0000	0.0000	0.0000	0.25	0.0757	4
Near-Infrared Laser	1	1.0000	1.0000	1.0000	0.258	0.8145	1

Scores were calculated in a similar manner to our scientific trade study, where a score of 0 corresponds to a failing or outlying parameter in a negative manner, and a score of 1 represents a positive outlying parameter. In essence, the higher the score, the better. Notably, Near-Infrared Laser outperformed every other communications system in estimated weight and power consumption.

As shown in Table 11, the Near-Infrared Laser outperformed all other systems, followed by X-band RF, S-band RF, and Ka-band RF. This aligns well with our mission objectives and accurately demonstrates the high bitrate of Laser and the long-range efficacy of X-band.

Once scientific instrument data rates were determined, minimum bitrates could be concluded pre-modulation. The minimum bitrate for our mission's scientific objectives (Table 11) excluded S-band immediately. This discovery

simplified our selection of frequency spectra, although we included it as a viable option for the purpose of conducting a more thorough trade study.

Atmospheric interference can be considered synonymous with rain fade in RF transmission, or light scattering in laser transmission. Due to our mission's need to communicate directly with Earth ground stations (lacking a planned relay satellite in Earth orbit), atmospheric interference is detrimental to overall mission performance, especially in scientific data transmission and satellite health and telemetry.

Due to the high likelihood of atmospheric interference in both Ka-band RF and Laser communication, and with the elimination of S-band RF from consideration, X-band RF was selected as our 'workhorse' communications mode, reserved for the ferrying of scientific, health, and telemetry data, as well as for mission tasking updates and communication window scheduling.

With X-band serving as a reliable means of low-bitrate communication, we were able to select a bulk transmission mode from Ka-band RF and Laser communications. In weighing the effectiveness of each of these modes, Ka-band has proven to be reliable, relatively high-bitrate, and currently has more established nodes throughout the solar system. These factors initially drew us to a hybrid X-band/Ka-band communications architecture, but upon further consideration, Laser communication achieves a dramatically higher-order-of-magnitude increase in link bitrate. This increase in planned link sizing, coupled with the apparent robustness of Laser communication demonstrated by JPL Psyche's DSOC and Artemis II's live video streaming capability [20], allowed our team to feel confident in trusting Laser communication to conduct long-range, high-bitrate communications between Earth, Mars, and future crewed missions to the red planet.

I. Subsystem Development

I.1 Communications

I.1.1 Communications Requirements

To begin with the communications design layout, a diagram showing the mission data flow was developed (Figure 8) to display how information travels during both uplink and downlink through the communications architecture. The diagram highlights the major sources, transition points, and mission data end-users, defining the operational procedures and paths that the communications subsystem must support.

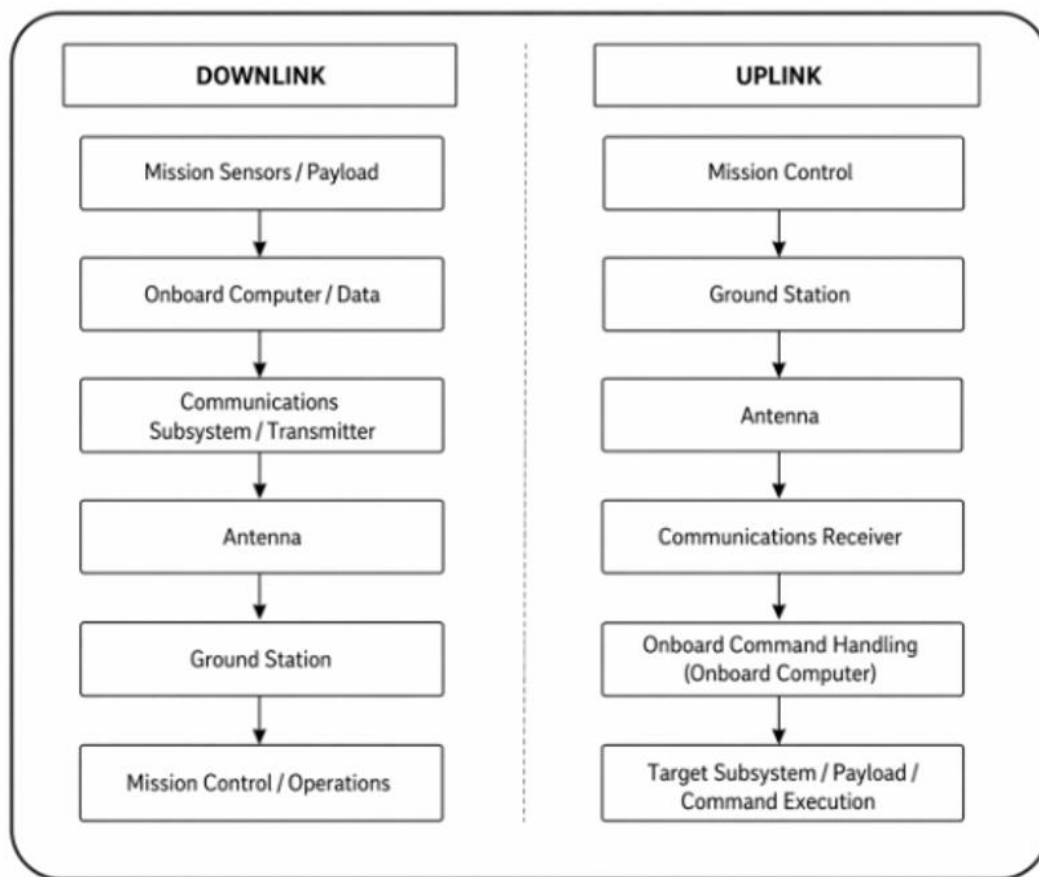


Figure 8: The communications data flow during both the uplink and downlink operations for the HEIMDALL mission. The diagram displays how the payload data travels from the mission sensors through the onboard computer system, communications subsystem, spacecraft and spacecraft antenna, and ground station to mission control, while data follows the reverse path for command execution.

Following the data flow diagram in Figure 8, a table displaying the communications requirements was created (Table 12) to convert the mission uplink and downlink needs into specific requirements. The table formulates the communications functions by statement, origin, and destination, data rate, access, delay, availability, and reliability so that further budget and architecture decisions can be based directly on clear mission requirements.

Table 12: This table shows the communications requirements for the HEIMDALL mission, covering the major downlink and uplink functions and the performance requirements that define the communications architecture.

Req No.	Requirement Statement	Direction, Source - End User	Data Rate	Access/Delay	Availability /Reliability	Criticality
R15	Science payload data shall be downlinked for mission analysis.	Downlink, Payload -Mission Control	2 Mbps / 0.9 GB per contact hr	Per pass/< 24 hr	High/High	CRITICAL
R16	Status telemetry shall be available $\geq 99\%$ of contact time with a maximum latency of 10 minutes per pass.	Downlink, OBC/subsystems - Mission Control	5 kbps	Per pass/< 10 min	High/High	Medium
R17	Fault telemetry shall be prioritized for response in the case of an anomaly.	Downlink, OBC/FDIR - Mission Control	Low, event-specific	Near real-time	Very high/Very high	CRITICAL
R18	Critical commands shall be uplinked for emergency actions	Uplink, Mission Control - OBC	Very low	Minimal delay	Very high/Very high	CRITICAL
R19	Payload/configuration commands shall be uplinked for instrument tasks.	Uplink, Mission Control - OBC	Low	During contact/< 1 hr	High/ High	Medium
R20	Routine commands shall be uplinked for nominal mission operations.	Uplink, Mission Control - Payload/subsystem	Low	Scheduled contact/< 1 hr	High/High	Low
R21	Relay communication via Mars crew assets shall be supported.	Uplink/Downlink, Mission Control ↔ Spacecraft (via Mars relay)	Low – Medium 1 kbps – 1 Mbps	Relay delay ≤ 5 s	High/High	High

Each of the requirements shown in Table 12 supports a different aspect and needs of the HEIMDALL mission and helps provide clarity in the communications architecture on both the spacecraft operations and science levels. The science payload downlink requirement R15 is mission critical because it ensures that the essential mission data can return to the ground successfully for further analysis, whereas the status/health telemetry requirement R16 allows continuous spacecraft health and subsystem performance monitoring. Fault telemetry R17 is also important because it supports the detection and response of possible anomalies, allowing spacecraft issues to be prioritized, identified, and resolved. During uplink, the critical command requirement R18 allows for emergency actions to be transmitted with minimal delay, while payload and configuration commands R19 support instrument task application and normal operation mission reconfiguration. Next, the routine command requirement R20 establishes that day-to-day spacecraft operations can be carried out accurately and reliably. Lastly, the relay communication requirement R21 is important because it ensures satellite communication through crewed Mars mission infrastructure or relay spacecraft when direct connection with Mission Control is delayed, limited, or unavailable. All together, these communications requirements authorize the performance needs that the communications subsystem must comply with to ensure spacecraft safety and mission success. The constraints for this subsystem can be found in Appendix 3. The communications subsystem requirements are directly tied to the communication's functional requirements (Table 3) since this communication architecture has the potential capacity to enable a line-of-sight between Mars, Earth, and future missions being launched to the red planet.

I.1.2 Alternate Communications Architecture

In selecting our communications architecture, primary consideration was given to ground station availability, along with prioritizing communication spectra providing high bitrate at deep space range (1AU). NASA's Deep Space Network (DSN) currently allows for S-, X-, and Ka-band RF communication for uplink and downlink, with varying antenna sizes at its three main locations, Goldstone (USA), Madrid (Spain), and Canberra (Australia). These ground sites are positioned approximately 120 degrees apart along Earth's rotational axis, allowing for near-constant transmission with assets in non-polar inclinations. Utilizing this framework and given HEIMDALL's planned positioning at Earth-Sun L4, both burst (short window) and continuous RF communication is possible with no known eclipse time or blackout.

Focusing on RF transmission, HEIMDALL's placement circumvents the need for relay satellites, but with dedicated gimbals, our mission could orient RF antennae to any point in the equatorial plane, with some possible variation in inclination. This provides an uneclipsed line of sight for low-interference transmission to anywhere in the inner solar system and beyond. In our current structural plan, the HGA could also be tasked with outer solar system transmission, in a burst-mode configuration. Continuous transmission tasking with ground stations would prevent this flexibility, but by utilizing the full 100W power of our planned system we found that burst transmission would be viable as a means of daily or otherwise cyclical downlink of scientific and CD&H data, along with tasking uplink in an identical aiming set (so long as uplink and downlink with ground stations are coordinated in a previous link or on a set schedule).

For the purposes of processing scientific data and given the incorporation of DSOC technology in our communications architecture, it is recommended that NASA JPL be entrusted with the downlink data, to be compared with that of ESA VIGIL for a more robust understanding of solar phenomena.

I.1.3 Individual Link Data Rates

For our satellite permanent station at Earth-Sun Lagrange Point 4, and acting as a communications relay in deep space, four potential link budgets must be specified. These include ground station uplink to HEIMDALL, deep space mission uplink to HEIMDALL, and downlinks to each node, respectively.

Data rates are dependent on the mode of communication selected (X-band or Laser), range of link, interference, and power availability. Our location at L4 allows for minimal interference on downlink, as there are no bodies in proximity to deflect radio or light waves, and no significant sources of interference within a range of 0.25 AU (nearest source being Venus) at any point in our heliocentric orbit. On uplink, interference would only become a major consideration for deep space missions transmitting from nearby planets or when nearing L4-Mars conjunction; this specific instance is avoidable via Earth-based uplink architecture, as is Earth-Mars conjunction via HEIMDALL relay.

Data rates are determined via component limitations and published data from previously flown missions at similar distance. This method was preferred to manual calculation as link data rates are heavily dependent on power, components, and modulation scheme of transmitting nodes which would require significant investment to upgrade at

ground stations and are unknown for future missions. In this regard, our team felt that trusting published data was a reliable means of data rate calculation. For example, DSN stations have antenna gain in the range of 56-74 dBi in X-band uplink, and power supply estimates ranging from 20kW to 400kW [19]. These powerful transmitters essentially nullify our requirements for reception area, as signals can be tuned to communicate effectively with any size parabolic antenna on HEIMDALL. Similarly, calculating laser communication parameters may be beyond the scope of the initial Request for Proposal, and as NASA JPL both pioneered the technology and is the source of all relevant data, should be trustworthy for our purposes.

Importantly, we classify HEIMDALL in these estimates as the ‘top’ node, so all incoming transmissions are considered uplink, and all outgoing transmissions are considered downlink.

Table 13: NASA Deep Space Network and JPL Psyche references for nominal bitrate of X-band and DSOC transmissions.

Reference	X-band Data Rate (1AU)	DSOC Data Rate (1AU)
NASA DSN 70m Antenna/JPL Table Mountain Facility (DSOC)	12 kbps	1.8 kbps
JPL PSYCHE	0.89-3.5 Mbps	25 Mbps

Noticeably, the lack of atmospheric interference of JPL Psyche at 1AU allows for much faster bitrates than ground stations with even higher power.

To ascertain whether these values are comparable with our selected equipment, only the X-band HGA need be analyzed for signal strength at 1AU, and for comparable gain to JPL Psyche’s HGA.

Antenna Gain Function:

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \left(\frac{\pi D^2}{4} \eta \right)}{\left(\frac{c}{f} \right)^2} = \left(\frac{\pi D f}{c} \right)^2 \eta \quad (6)$$

Utilizing a heritage parabolic HGA of diameter 2.1m, transmission frequency of 10GHz, aperture efficiency of 0.55, and λ equal to 0.03m, idealized gain was calculated to be approximately 44.3 dB. This is on par with heritage deep-space systems and consistent with pointing accuracies achieved by previous missions.

Friis Transmission Formula:

$$SNR = \frac{E_b R}{N_0} = \frac{P_t L_l G_t L_s L_a L_r L_p G_r}{k T_s} \quad (7)$$

Table 14: Friis Transmission Formula Variables and reference values taken from SMAD and MRO operational parameters.

Variable	Description	Downlink (L4-Mars)	Downlink (L4-Earth)
SNR	Signal to Noise Ratio	280.22 dB-Hz	278.18 dB-Hz
E_b	Energy Per Bit	Dependent on FTF	Dependent on FTF
R	Data Rate (bits/s)	2 Mbps	2 Mbps
N_0	Noise Density	Dependent on FTF	Dependent on FTF
P_t	Transmit Power (W)	100 W	100 W
L_l	Line Losses	0.5 dB	0.5 dB
G_t	Transmitter Gain	44.3 dB	44.3 dB
L_s	Free Space Path Loss	195.6 dB	186.9 dB
L_a	Atmospheric Attenuation	(negligible for X-band)	(negligible for X-band)
L_r	Rain Loss	(negligible for Mars)	0.1-0.4 dB/km
L_p	Pointing Error Loss Budget	1 dB	1 dB
G_r	Receiver Gain	45.2 dBi	74 dBi
k	Boltzmann's Constant	1.38E-23 J/K	1.38E-23 J/K
T_s	System Noise Temperature	135 K	135 K

For Earth-side communication, we assumed no rain fade, minimal atmospheric losses, and took all other variables from the Deep Space Network 70-meter antenna.

For Mars-side communication, again we assumed no rain fade, minimal atmospheric losses, and took all other variables from the Mars Reconnaissance Orbiter's HGA X-band transmission suite [16].

Effective Isotropic Radiated Power

$$EIRP = P_t L_t G_t \quad (8)$$

Assuming transmission power of 100W, line losses of 0.5dB, and transmission gain of 44.3 dB, we found our effective isotropic radiated power on downlink to be 98.3 dBm.

These values align with reported parameters for JPL Psyche’s X-band communications system [17], thus allowing us to clarify a minimum uplink and downlink bitrate for transmission in both systems. Importantly, these values are minimized for HEIMDALL-Spacecraft links in Table 15, as a lack of atmospheric and photon interference could allow for even higher bitrates in the range of 1 Gbps [20].

Table 15: Uplink and Downlink bitrates for transmission in X-band (10 GHz) and DSOC (Near-Infrared).

Link	Ground Station	Spacecraft
X-band Uplink (to HEIMDALL)	1.8 kbps	Not Intended
X-band Downlink (from HEIMDALL)	180+ kbps	Not Intended
DSOC Uplink	1.8 kbps	25+Mbps
DSOC Downlink	25 Mbps	25+Mbps

I.1.4 Link Design and Sizing

In the process of selecting communications equipment, HEIMDALL conducted numerous investigations of spectra, components, and modulation schemes to establish a flight-proven, under-budget, and high-bandwidth communications platform at a range of 1 AU.

Table 16 lists an array of standard space communications spectra along with their frequencies, first historical usage, and long-range performance. For our purposes, S-band radio loses efficacy at a much shorter range than we require. X-band, Ka-band, and near-infrared lasers all satisfy range requirements. Ka-band, unfortunately, experiences significant rain fade, which would limit our ability to communicate reliably with ground stations on Earth. These factors led to our selection of X-band/Laser Hybrid communications architecture.

Table 16: Various RF frequency bands commonly used in long-range space communications, along with history of use and maximum range performance.

Name	Frequency Range	First Flight	Demonstrated Range	Maximum Data Rate at Range
S-Band	2-4 GHz	1967 (USB aboard Apollo 4)	0.00215 AU	1.6 kbps
X-Band	8-12 GHz	1965 (LES-1)	10+ AU	1-5 kbps
Ka-Band	26.5-40 GHz	1993 (NASA ACTS)	2.5 AU	331 kbps
Laser (Near-Infrared)	300×10^{12} Hz	2023 (JPL Psyche)	2.58 AU	6.25 Mbps

In addition to the data rates and ranges described in Table 11, X-band radio communication limits the bitrate at long ranges compared to Ka-band, but is far longer flight-proven and robust in demonstrated efficacy. S-band radio is severely range-limited, so much so that after HEIMDALL exits Earth's sphere of influence, any use case for S-band would be eliminated from contention. Laser communication in the near-infrared wavelengths is superior to all radio communications platforms in nearly every regard but requires much more finely tuned pointing accuracy. In this respect, our team surmised that Laser would be a preferable medium for large data packages and non-essential duties, such as video transmission and mission updates from any future long-range missions, e.g., crewed and uncrewed missions to Mars.

I.1.5 Reasons for Selection

Historical examples of X-band radio frequency communications subsystems can be found throughout the history of spacecraft design and development. From the first flight in the 1960s to modern-day telecom and scientific missions, X-band has proven itself a reliable workhorse for health, telemetry, scientific data, and general communications functions [19]. For these reasons, along with its absence of rain fade, HEIMDALL will utilize X-band transmission for all critical functions and daily transmissions, relying on heritage amplifiers, transponders, and antennae to ensure reliability and guaranteed success of recurring mission tasks.

Recently used for live video feed on Artemis II, near-infrared laser transmission was first demonstrated at deep space ranges aboard JPL Psyche in 2023 [20]. In this mission, primarily tasked with studying an asteroid via long-range rendezvous, Deep Space Optical Communications (DSOC) was shown to be an effective means of standalone communication for both uplink and downlink of large data packages.

DSOC was also employed for simultaneous RF-Laser hybrid communications in a pragmatic demonstration of a high-bitrate laser link for scientific data, and a low-bitrate X-band link for telemetry and health of the spacecraft. Importantly, in this demonstration, the vastly different modes of communication in RF-Laser Hybrid allowed for two distinct channels of uplink/downlink to be operated at full bitrate simultaneously, essentially doubling spacecraft tasking and operational capacity from a mission control perspective. In this series of demonstrations, RF and Laser were received simultaneously at the same antenna by fixing a photon-counting light receiver onto a high-gain ground-based antenna to simplify pointing accuracy for the spacecraft and test for potentially unseen sources of interference. No such interference was noted, and full use of both channels was demonstrated numerous times at varying distances from Earth [17].

It is the opinion of our team that X-band/Laser hybrid communications architecture is the optimal balance between reliability and performance necessary to maximize our data rates on both uplink and downlink for HEIMDALL. While future missions may find Laser communication to be reliable and effective as a replacement for RF systems, due to the newness of DSOC, at this time, we feel that reserving X-band RF for mission-critical functions is pertinent to ensure the continued efficacy of our mission well into the 2040's and beyond.

I.2 Attitude Determination & Control System

The requirements and constraints for the Attitude Determination and Control System (ADCS) were driven by several mission needs, including the pointing requirements for onboard scientific instruments, communication alignment with Earth and future Mars missions, and fine attitude control for station keeping. The Requirements R7 through R10 ensure that the ADCS subsystem works and responds to the demands that Sun-Earth L4 will impose on the satellite (orientation, rotation, etc.). R7 being tied to SR2 to ensure that HEIMDALL points in the right direction in the event of CMEs or SEPs, and R8 being linked to CR1 and helping to establish the right communication orientation needed for data transmission, as described in section I.1. The constraints for the ADCS subsystem can be found in Appendix 3.

Table 17: Preliminary values for ADCS requirements

Requirement	Preliminary Value
Determination accuracy	$\pm .01^\circ (3\sigma)$
Determination range	360°
Control accuracy	$\pm .05^\circ (3\sigma)$
Control range	360°
Control jitter	$\leq .001^\circ/\text{s}$
Control drift	$\leq .01^\circ/\text{hr}$
Control settling time	$\leq 300\text{s for } 30^\circ \text{ slew}$

I.2.2 Defining ADCS Control Modes

The ADCS for HEIMDALL must support the spacecraft payload, which includes a coronagraph, helio-spheric imager, telescope/spectrometer, and magnetometer, while also supporting communications relay operations near the Sun–Earth L4 point. Due to HEIMDALL’s wide range of needs, the ADCS system will be comprised of five main modes: science, communication relay, safety, acquisition, and slew. [21]. Because the coronagraph and helio-spheric imager require a stable, accurate line of sight pointing, science mode is the most demanding control mode and drives the overall ADCS design. In this mode, HEIMDALL will maintain high pointing accuracy, very low jitter, and low drift during extended observations. The telescope/spectrometer also requires accurate and stable pointing for measurements. In communications relay mode, the spacecraft must maintain sufficient pointing accuracy for antenna alignment. In safe mode, only coarse Sun pointing is required to preserve spacecraft power and thermal safety. In acquisition mode, the spacecraft must determine and stabilize its attitude after disturbances or mode changes. In slew mode, the spacecraft will reorient operational attitudes and unload stored wheel momentum as needed. Overall, with these five modes, HEIMDALL will be able to achieve all scientific requirements and communication-oriented objectives.

I.2.3 Quantifying Disturbance Environment

The dominant disturbance acting on HEIMDALL is expected to be solar radiation pressure. Because the spacecraft operates near the Earth–Sun L4 point and includes exposed surfaces such as solar arrays, communications hardware, and coronagraph structures, solar radiation pressure will create a continuous external disturbance torque on the vehicle. Using a preliminary estimate with a solar flux of 1372 W/m^2 , a reflectance factor of 0.6, an illuminated area of 2.38

m², and a .5m offset between the center of pressure and center of mass gives a solar radiation pressure force of approximately 1.74×10^{-5} N and a resulting disturbance torque of approximately 8.7×10^{-6} N·m. For design margin, the ADCS may be sized for solar radiation pressure disturbances approaching 10^{-4} N·m. A summary of equations and variables to calculate solar radiation pressure disturbances can be found in Table 18.

Table 18: Solar Radiation Disturbance Torque Calculation Parameters

Variable	Description	Value
F_s	Solar Flux	$1372 \frac{W}{m^2}$
c	Speed of light	$3 \times 10^8 \frac{m}{s}$
A	Illuminated area	$2.38 m^2$
q	Reflectance Factor	0.6
i	Sun incidence angle	0°
$\cos i$	Cosine of incidence angle	1
d	Offset between center of pressure and center of mass	0.5 m
$F_{srp} = \frac{F_s}{c} A(1 + q) \cos i \text{ (N)}$		
$T_{srp} = F_{srp} d \text{ (N} \times \text{m)}$		

Other common external disturbances are less significant for this mission. Aerodynamic disturbance torque is negligible because HEIMDALL operates in deep space rather than in an atmosphere. Magnetic disturbance torque is also negligible because the spacecraft is too far from Earth or another polar body for a magnetic field to influence the spacecraft. Gravity gradient torque is much smaller than it would be for a low Earth orbit spacecraft, making it not part of the design at Earth Sun L4. Internal disturbances, however, remain important, which include reaction wheel vibration, structural flexing, thruster misalignment, and thruster mismatch, all of which can contribute to pointing jitter. Because the coronagraph is sensitive to line-of-sight disturbances, these internal effects must also be minimized through careful component placement and structural design.

Overall, the HEIMDALL disturbance environment is dominated by solar radiation pressure torque, with the preliminary value being $T_{srp} \approx 9 \times 10^{-6} \text{ N} \cdot \text{m}$. This disturbance drives the need for adequate reaction wheel momentum storage and a dedicated fine adjustment thruster subsystem for momentum unloading and station keeping.

I.2.4 Preliminary ADCS Hardware Selection

A preliminary design of HEIMDALL's attitude determination system includes two star trackers, one IMU, and six coarse Sun sensors. For attitude control, HEIMDALL uses four reaction wheels as primary actuators. To support momentum unloading and fine station keeping, a separate propulsion subsystem consisting of eight monopropellant thrusters is included.

The star trackers provide high-accuracy attitude knowledge required for the coronagraph, helio-spheric imager, and telescope/spectrometer, while the IMU provides angular rate information between updates. The six Sun sensors support safe mode and acquisition, with one mounted on each face of the cubic spacecraft bus to provide near-complete Sun visibility. The reaction wheels provide precise three-axis control with one wheel redundancy and are sized to counter continuous solar radiation pressure disturbance torque, support required slew maneuvers, and store accumulated angular momentum over time. A suitable reaction wheel for HEIMDALL is the Collins Aerospace RDR-68. To support momentum unloading and fine station keeping, HEIMDALL also includes a separate fine attitude control and station keeping propulsion subsystem consisting of eight Moog MONARC-5 monopropellant thrusters arranged in balanced pairs near the spacecraft edges or corners.

Special consideration must be given to the magnetometer, since it is sensitive to spacecraft-generated magnetic interference. Unlike the optical instruments, the magnetometer does not strongly drive fine pointing accuracy, but it does affect component placement. To reduce magnetic contamination, the magnetometer should be mounted on a boom or positioned as far as practical from the reaction wheels, power electronics, and other magnetic noise sources. A preliminary selection of components for the ADCS subsystem is shown in Table 19.

Table 19: Preliminary Component Selection for the ADCS Subsystem

Component	Manufacturer	Type	Quantity	Power Consumption
RDR 68	Collins Aerospace	Reaction Wheel	4	Steady state <20W Max. Torque 90W
ASTRO CL	Jena-Optronik	Star Tracker	2	<1.2 W
LR-450 IMU	Northrop Grumman	Gyroscope / IMU	2	<15W
Coarse Sun Sensor	Bradford Space	Sun Sensor	6	Passive (0W)
MONARC-5	Moog	Attitude Control	8	18W each

I.2.5 Defining Determination and Control Algorithms

Fusion between the star trackers and the IMU will be required to get an accurate estimation of the spacecraft's attitude and angular rate during operations. The star trackers will provide precise and absolute attitude positioning, while the IMU provides continuous rate information. The Sun sensors are used primarily in safe mode and acquisition mode, where coarse Sun pointing is sufficient to preserve spacecraft safety and power generation. A closed-loop PD or PID control system can be used to command the reaction wheels during operations when needed. In science mode, this control system maintains high pointing accuracy and low jitter to satisfy the needs of the coronagraph. In communications relay mode, it maintains the pointing needed for antenna alignment. During slew mode, the controller commands the required spacecraft reorientation and settling behavior. When stored wheel momentum approaches operational limits, the MONARC-5 thrusters are pulsed to unload the wheels and perform fine station-keeping corrections. This approach combines precise wheel-based pointing with propulsion-assisted long-term control.

I.2.6 Iterate and Document

The final HEIMDALL ADCS design will be developed iteratively, ensuring that all needs of the payload are satisfied. The coronagraph, heliospheric imager, and telescope/spectrometer drive the requirements for pointing accuracy, stability, and low jitter, while the magnetometer drives the requirement for low magnetic interference, which will be obtained through careful placement and integration into the spacecraft. The reaction wheels and station-keeping thrusters must balance the external environmental forces at L4 and satisfy the payload performance needs. As a result of this iterative design process, the final ADCS system will be capable of meeting both the scientific requirements of the mission and the spacecraft control requirements.

I.3 Command & Data Handling

The Command and Data Handling (C&DH) subsystem will be in charge to receive and sending information to the ground station (Either being on Earth or Mars) as well as having a communication link to future missions being sent to space. Requirements R12 and R13 (found in Appendix 1) will work to ensure that SR1 and SR2 science requirements are fulfilled, since the computer will be in charge of processing the collected information from the payload instruments and send it to either ground station or missions traveling through space. C&DH requirements R11, R13, and R14 will work to maintain a near constant communication link for the whole 15-year mission duration. The C&DH subsystem constraints are found in Appendix 3. The on-board computer (OBC) chosen for HEIMDALL was the Voyager Technologies IPC-7000 with a Proton400k dual-core processor. Since HEIMDALL will be located far away from Earth, the radiation-hardened assurance is one of the most critical values for the OBC since it must survive significant amounts of radiation during its operational life [2]. The IPC-7000 has a TID value of 100 krad, making it one of the best options for our mission [15]. The architecture and functionality of the OBC will be described as follows in sections I.3.2 to I.3.6:

I.3.2 Clock, bit sync, and timing

The IPC-7000 will maintain the master spacecraft time using its internal hardware oscillator [15]. Since HEIMDALL will be stationary at L4 for a duration of 12 years, oscillator drift can be a problem. To account for that, the C&DH system will use time-synchronization commands from the ground station during scheduled TT&C uplink windows. To ensure accurate timestamping of science telemetry, the C&DH subsystem will distribute a 1 pulse per second (1PPS) hardware synchronization signal across the data bus. The purpose of this 1PPS is to ensure that the payload instruments and the TT&C subsystem are operating on the exact same microsecond. Regarding bit synchronization, data transfer between the solid-state buffer and the transmitter will utilize a common interface to avoid synchronization issues. Also, the C&DH subsystem will provide the accompanying clock signal alongside the telemetry data stream to ensure bit sync and to prevent data loss at the radio interface.

I.3.3 Storage

To satisfy the onboard storage constraints, the system will use a dedicated solid-state buffer (SSB) with a maximum capacity of 1 TB [15]. Another feature that the HEIMDALL team liked about this OBC is that it comes with dual user-programmable Virtex 7 FPGAs that perform real-time 12-bit JPEG compression on the raw solar images before they are routed to the SSB [15].

I.3.4 C&DH Hardware-Level Recovery

To ensure that HEIMDALL keeps its functionalities in a dangerous environment, the C&DH subsystem implements a multi-tiered fault detection, isolation, and recovery architecture, making use of both hardware interventions and ROM-stored flight software sequences. HEIMDALL needs a reliable way to keep functioning during a 15-year mission to L4. A redundant system of on-board computers is determined to be the best way to accomplish the mission's success. HEIMDALL will have a primary and secondary OBC. The primary IPC-7000 OBC will operate alongside an external Solid-State Power Controller (SSCP) acting as a hardware watchdog timer. The active OBC must send a continuous 5-volt heartbeat signal to the SSPC. In cases where a dangerous CME damages the active OBC, stopping the heartbeat, the SSPC will automatically cut power to the primary board and route the power to the spare OBC (which is another IPC-7000), initiating a clean reboot and seamless continuation of normal operations.

I.3.5 Fault Detection and Recovery

The Proton400k guarantees high reliability with an error rate of 1×10^{-4} error/day [15]. The flight software includes a ROM-stored safe mode sequence; if a primary radio fault is detected, the active IPC-7000 will automatically command the external RF switch to route telemetry through the backup receiver. When any major fault is triggered, the ROM-stored sequence will automatically execute load-shedding. With this, the OBC will immediately cut power to the high-draw payload instruments to preserve power while the OBC and the ADCS subsystem work together to re-establish a stable sun-pointing attitude and Earth communication link.

I.3.6 Daily Telemetry Volume

The continuous monitoring of HEIMDALL and all the information gathered from the solar payload equipment will generate approximately 7 GB of science telemetry per 24-hour period. This number equated to a daily data volume of 57,344 Megabits (Mb). With this architecture, the HEIMDALL team attempted to facilitate a safer and doable way to monitor space weather while having an active line of communication between L4, Earth, Moon, and Mars. A C&DH diagram shown in Figure 9 illustrates the various pathways taken by data streams throughout the system.

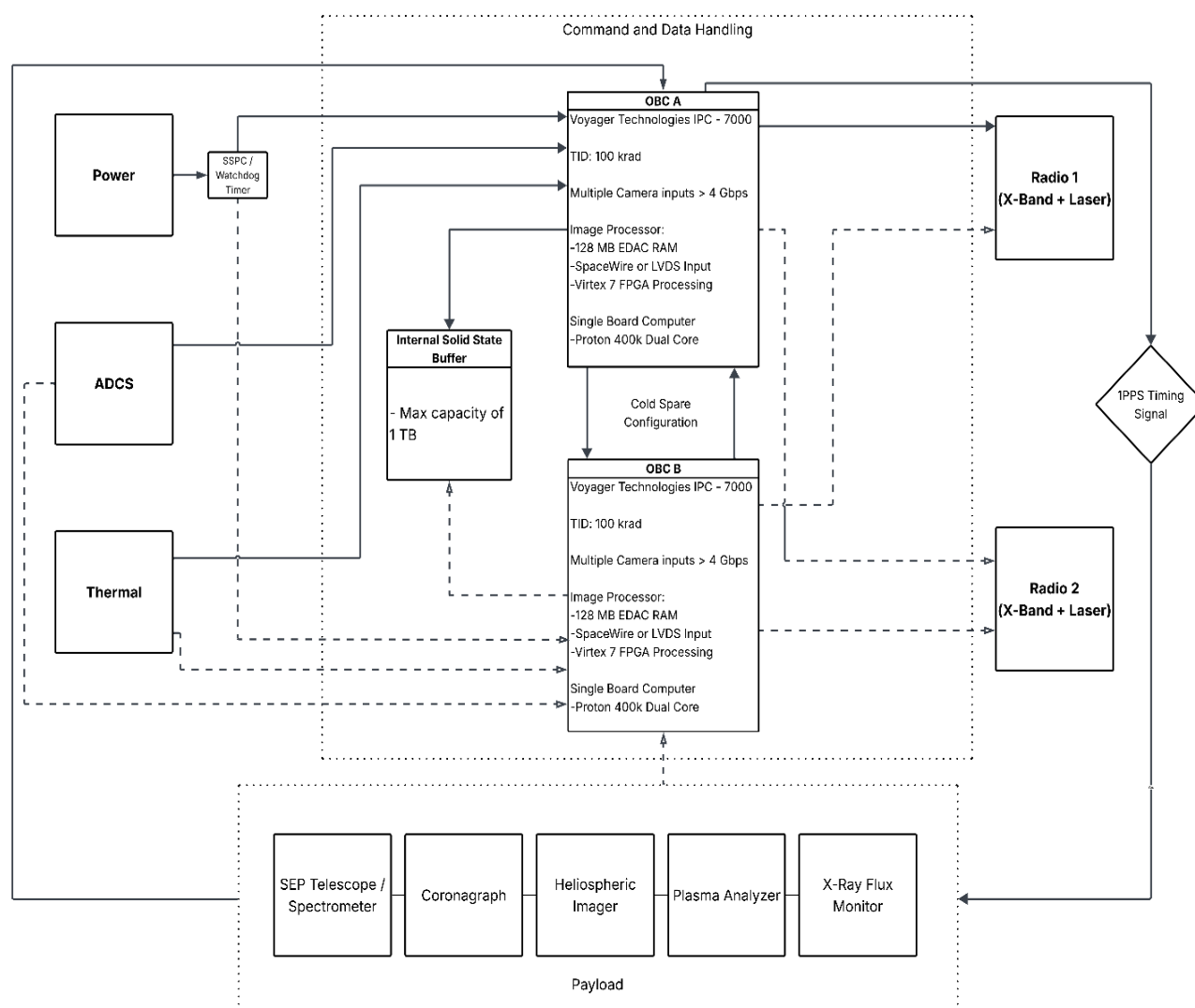


Figure 9: Processes that the OBC has to manage in all phases of the HEIMDALL mission. This diagram also shows how the redundancy between the two computers works, with the dashed lines indicating the cold OBC spare connections.

I.4 Propulsion

The propulsion subsystem is one of the most important areas of the HEIMDALL satellite; requirements R23 and R25 will work to maintain HEIMDALL at L4 for the whole mission duration, fulfilling the science and communication requirements seen in Tables 2 and 3 (requirements SR1, CR3, and CR4). The constraints for the propulsion subsystem can also be found in Appendix 3. For the phasing maneuver prior to Sun-Earth L4 insertion, the second stage of Falcon 9 can be used for all significant changes in ΔV . The Falcon 9 second stage can reignite its engine, allowing for multiple burns to occur, providing precise orbital insertion within its ΔV budget. Using the second stage as a boost stage for the phasing maneuver will allow HEIMDALL to conserve fuel and mass, which can be allocated to other scientific, communications, or bus systems. Additionally, the conservation of mass can be used as a cost-saving measure, as the HEIMDALL project is primarily cost-constrained.

To choose a suitable method of propulsion for HEIMDALL, the first thing that needs to be determined is the required ΔV that the propulsion subsystem must be able to provide. To simplify the calculation, it is assumed that the propulsion subsystem mass budget is separate from the attitude and control subsystem, even if the final method of propulsion could be able to be used for both ΔV changes and attitude adjustments. Once the required ΔV is determined, different propulsion methods can be explored for their feasibility and relevance to HEIMDALL's mission. From there, they can be compared to determine which method would be most suitable for the 15-year mission duration that is anticipated.

I.4.1 Station Keeping ΔV Determination

Sun-Earth L4 gives HEIMDALL the advantage of being placed into a region of space that is gravitationally stable. This inherent stability allows for the required ΔV per year to be minimal, on the order of a few meters per second per year. Because HEIMDALL is seeking to perform a mission that has no heritage background and there are no active missions to L5 yet, which would share similar ΔV requirements, determining an appropriate ΔV on which to base station keeping maneuvers requires significant assumptions absent a full n-body gravitational analysis.

Research began with looking at current Sun-Earth L1 satellites. L1 is located along the axis connecting the Sun and Earth and is located between them 1.5 million kilometers (0.01 AU) from Earth. Unlike L4, L1 is a point of unstable equilibrium, meaning that without routine station keeping, any object in a Lissajous orbit will eventually drift away

and fall out of L1. With this in mind, if ΔV data can be found for satellites in L1, it is reasonable to assume that the required ΔV for L4 would be at most equal, and most likely less than that of L1 [8]. The Advanced Composition Explorer (ACE), Solar Helio-spheric Observatory (SOHO), and the Global Geoscience Wind (WIND) missions are all currently orbiting L1, and have been since the late 90's and early 2000's [5]. The typical annual station keeping costs have been around $1.0 \frac{m}{s}$ for ACE and WIND and have been less for SOHO [5].

While using $1.0 \frac{m}{s}$ per annum would be sufficient for station-keeping estimations, there is going to be some built-in ΔV to account for the frequency of maneuvers that would be required to maintain an orbit precise enough for laser communications and coronagraph observations. Both systems rely on stable positioning in space to be able to operate effectively, and to be able to maintain that stability, more frequent corrections will be needed. ACE, SOHO, and WIND perform corrections about every three months [5]. For increased orbit stability, HEIMDALL will conduct corrective burns every 1-2 months to limit the accumulation of ΔV drop. With this increase in frequency, a total annual ΔV budget of $2.0 \frac{m}{s}$ will be used. In a worst-case scenario, this will account for the necessary maneuvers to maintain 15 years of mission operation, and in a best-case scenario, will allow HEIMDALL to operate beyond its design life, given that other systems are still operational.

Another source of ΔV that can be calculated is due to solar radiation pressure. Solar radiation pressure is the force caused by the exchange in momenta between the photons emitted by the Sun and [a] satellite's surface [9]. After gravity, solar radiation pressure is usually the most significant source of orbit disturbance for deep space missions, hence it must be accounted for. Solar radiation pressure has been calculated using the values and equations found in Table 18. The force calculated can then be used to find acceleration using Newton's second law:

$$a_{sc} = \frac{F_{SR}}{m_{total}} \quad (12)$$

Where:

a_{sc} = Acceleration of the Spacecraft

m_{total} = Total Mass of the Spacecraft

This acceleration found in Equation 13 can be multiplied by the number of seconds in 15 years to compute the total ΔV change induced by solar radiation pressure over the course of the entire mission. This ΔV is combined with the

$2 \frac{m}{s}$ per year estimation and iteratively solved to get a wet mass for the spacecraft. Once convergence happens, the result is the mission ΔV necessary for station keeping, as well as the wet mass of the spacecraft. This final ΔV is requirements R24 and R25 and can then be used to determine the most suitable type of propulsion for station keeping.

I.4.2 Propulsion Methods

There are two classes of propulsion that have historically been used on spacecraft: Chemical and Electric. Chemicals can be broken down further into two types: monopropellant and bipropellant, which each have their own advantages depending on the application. All three types of propulsion have been explored for HEIMDALL to weigh the merits of each and cross-compare their effectiveness. It is from this comparison that a recommendation will be made.

I.4.3 Electric

Electric propulsion, specifically Hall effect thrusters, is unique in that its propellant mass requirement is very low, while its long-term ΔV capabilities and specific impulse are extremely high. This makes it potentially suitable for two types of missions: High ΔV missions where burn time is not a constraint, and missions with low ΔV needs that have a minimal mass constraint. HEIMDALL would fall into the second mission type, since the annual ΔV requirement is $\sim 2 \frac{m}{s}$ and minimizing wet mass would reduce launch costs. Despite the mass reduction hall effect thrusters would provide, the major drawback comes from the class name: Electricity is always necessary to enable the ionization of the propellant and its acceleration through an electric and magnetic field. For HEIMDALL, based on the calculated ΔV and subsequent propellant mass calculation, an early estimate of the power required to provide sufficient performance to HEIMDALL was 3000 W. Adding this into the existing power budget of ~ 425 W gives ~ 3500 W of power necessary during station-keeping operations. Calculating the required solar panel area and mass was done using:

$$P = JA\eta \cos \theta f_d \quad (14)$$

Where:

$$P = 3500 = \text{Power, } W$$

$$\eta = 0.25 = \text{Effective Solar Array BOL Efficiency}$$

$$\theta = 0 = \text{Sun Incidence Angle, deg}$$

$$f_d = 0.08 = \text{Array Degradation Factor at EOL}$$

Solving for the area and adding 10% margin results in a solar panel area of 14.14 m^2 . Using an array density of $5.1 \frac{\text{kg}}{\text{m}^2}$ results in the total array mass of 72.13 kg . If the area is instead solved by using the non-electric power budget of $\sim 500 \text{ W}$, a total array mass of $\sim 10 \text{ kg}$ is calculated. This means that for electric propulsion to be a viable option for HEIMDALL, the amount of mass saved must be greater than 62 kg for the extra solar panel area and mass to not outweigh the propellant mass saving.

Table 20: Hall effect thruster calculation results.

Electric Hall Effect								
Thruster Name	I_{sp} (s)	Thrust Force per Thruster (N)	Power per Thruster (W)	Number of Thrusters	Total Spacecraft Mass (kg)	Total Propellant Mass (kg)	Total Mission ΔV ($\frac{\text{m}}{\text{s}}$)	Total Impulse Time (Hours)
ST-25	1500	0.014	250	4	645.184	2.944	54.964	175.426
BHT-600	1400	0.039	600	4	645.394	3.154	54.956	62.973
SHT1500	1800	0.08	1350	2	644.693	2.453	54.983	61.401
ST-100	1700	0.107	1500	2	644.838	2.598	54.977	45.907
BHT-6000 HT	2029	0.325	5000	1	644.417	2.177	54.993	30.229
BHT-6000 HI	2708	0.298	6000	1	643.871	1.631	55.015	32.969

As shown in Table 20, Hall effect thrusters give a clear propellant mass advantage for HEIMDALL, requiring between 1.5 and 3.2 kg of propellant for all candidates to meet the mission ΔV of $\sim 55 \frac{\text{m}}{\text{s}}$. This mass saving benefit, however, is reduced by the large, sustained power demand and long impulse times, both of which place strain on the electric system. This results in the feasibility of electric propulsion for HEIMDALL being driven by the power requirement rather than propellant mass. To get estimates of solar panel sizing and mass from Equation 14, the ST-100 was used for its balance between power, propellant mass, and impulse time.

I.4.4 Chemical

Two forms of chemical propulsion were explored for HEIMDALL: monopropellant and bipropellant. Monopropellant uses a liquid, typically hydrazine, as the propellant, which is passed over a solid catalyst that causes the propellant to

rapidly decompose, heat up, and expand, resulting in thrust. Bipropellant uses two separate liquids, a fuel, such as monomethyl hydrazine (MMH), and an oxidizer, such as dinitrogen tetroxide (N₂O₄), which are stored in separate tanks and combined in a thruster's combustion chamber. As the two liquids are mixed, they react with each other and ignite, resulting in a large increase in temperature and pressure, which generates thrust. Between the two forms, bipropellant is more efficient and capable of much higher performance, but has greater system complexity, having to store and precisely balance two liquids rather than one for monopropellant. For HEIMDALL to benefit from the increased efficiency that bipropellant offers, there needs to be a substantial difference in the propellant mass compared to monopropellant.

Table 21: Monopropellant thruster calculation results.

Monopropellant								
Thruster Name	I_{sp}	Thrust Force per Thruster (N)	Number of Thrusters	Total Spacecraft Mass (kg)	Total Propellant Mass (kg)	Total Pressurant Mass (kg)	Total Mission ΔV ($\frac{m}{s}$)	Total Impulse Time (Hours)
Rafael 25N	220	25	2	662.442	20.049	0.128	54.313	0.196
MT-8A	212	4	2	663.204	20.805	0.133	54.285	1.227
MT-2	218	20	2	662.628	20.233	0.129	54.307	0.245
MRE-15	228	66	1	661.734	19.346	0.123	54.339	0.149
MRE-5.0	232	22	2	661.399	19.013	0.121	54.352	0.223

As shown in Table 21, while the monopropellant thrusters offer a simple system for integration, it comes at the cost of propellant mass, with the required mass for a ΔV of $\sim 54.3 \frac{m}{s}$ being in the 19 to 21 kg range. The total impulse time for all thrusters is low compared to electric propulsion, which would allow for large, quick corrections if necessary. The simplicity of integration, only having to factor in one propellant liquid, makes monopropellant an attractive option for HEIMDALL, even with the increase in mass. Monopropellant also has the benefit of being one of the primary chemical propulsion types for reaction control systems, which would allow the propellant budget to be shared between the two if necessary. From the thrusters found, the MRE-5.0 minimizes the propellant mass due to having the largest I_{sp} while also maintaining a total impulse time under one hour. Table 21 also budgets in the mass for a pressurant to keep the system at an optimal pressure. The gas used here is helium, which has a heritage background, is stable, and can be heavily pressurized reliably, limiting the required tank size. Using the ideal gas law and a propellant tank pressure of 22 Bar, the calculated amount of pressurant needed is about 0.125 kg for the mission duration, a nearly negligible amount of mass compared to the rest of HEIMDALL.

Table 22: Bipropellant thruster results.

Bipropellant								
Thruster Name	I_{sp} (s)	Thrust Force per Thruster (N)	Total Spacecraft Mass (kg)	Total Propellant Mass (kg)	Fuel Mass (kg)	Oxidizer Mass (kg)	Total Mission ΔV ($\frac{m}{s}$)	Total Impulse Time (Hours)
S10-13	292	10	657.351	15.111	5.681	9.430	54.502	0.982
BT-6	295	21.5	657.198	14.958	5.623	9.334	54.507	0.457
Halcyon	300	22	656.949	14.709	5.530	9.179	54.517	0.446
DST-11H	307	22.24	656.613	14.373	5.404	8.970	54.529	0.441
5 lbf Columbian	284	22.24	657.776	15.536	5.841	9.696	54.486	0.441

As shown in Table 22, bipropellant offers a mass advantage over monopropellant, with the amount of propellant, including both fuel and oxidizer, required for a ΔV of $\sim 54.5 \frac{m}{s}$ being in the 14.3 to 15.6 kg range. This lower propellant mass results in a smaller tank volume than monopropellant, but with the additional complexity of having a tank for both oxidizer and fuel. Additionally, the impulse time for the selected thrusters is longer than that of monopropellant in the range of 0.4 to 1.0 hours. This is a result of only using one thruster and can be easily reduced by adding additional thrusters at the cost of implementation complexity and increased risk. From the thrusters found, the DST-11H minimizes total propellant mass at 14.3 kg while also having the tied lowest impulse time of 0.44 hours.

I.4.5 HEIMDALL Propulsion Recommendation

After evaluating Hall effect, monopropellant, and bipropellant propulsion options, the recommended station-keeping propulsion system for HEIMDALL is the MRE-5.0 monopropellant thruster from Northrop Grumman. Even though electric propulsion would result in the least propellant mass, its power demands would increase the burden placed on the electrical system and require a substantial increase in the solar array size, offsetting the propellant savings. Coupled with the long impulse times, which would limit HEIMDALL's ability to react to unforeseen events, electric propulsion was ultimately decided against. Bipropellant provides an increase in efficiency compared to monopropellant, but this comes with an increase in the complexity of the propellant system due to the necessity of separate tanks for fuel and oxidizer, additional feed hardware, and more constrained integration requirements. For HEIMDALL, a propulsion system that has a balance between acceptable propellant mass and simple integration rules out bipropellant.

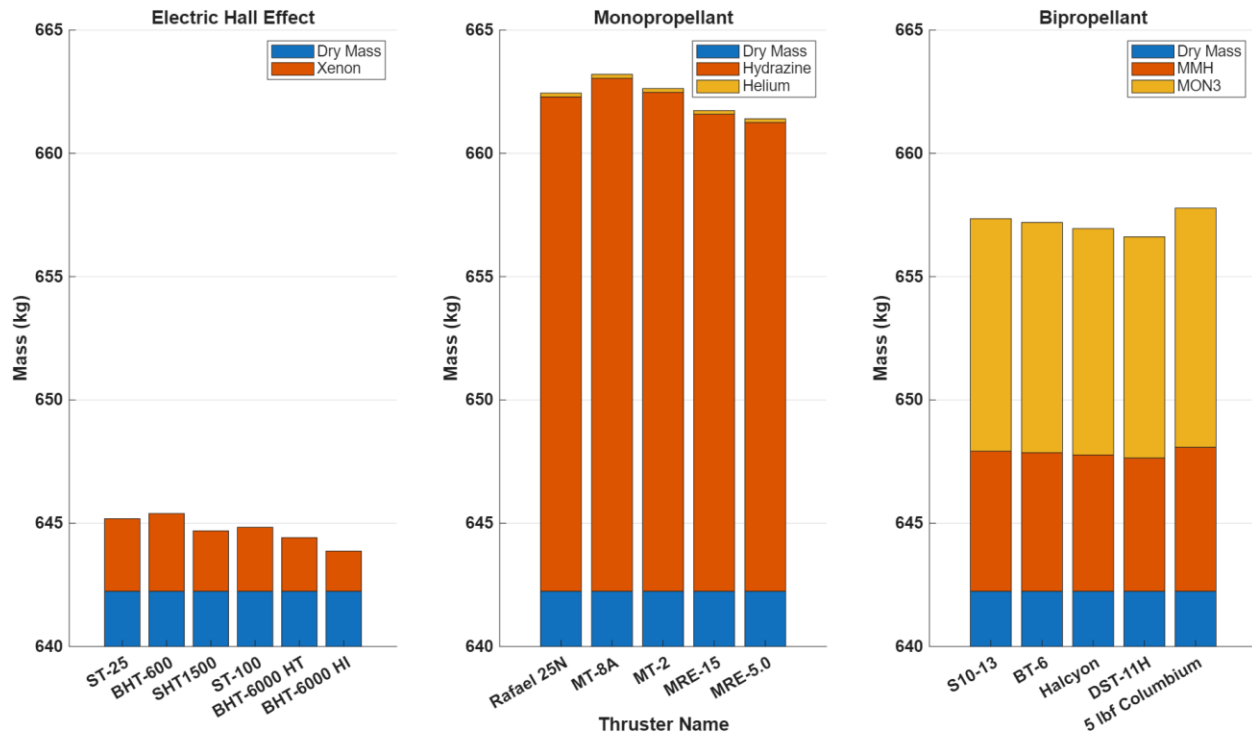


Figure 10: Propellant mass distribution by propulsion type.

Using the trades found in Table 21 and from Figure 10, the MRE-5.0 was selected because it is the most efficient of the compared thrusters, with an I_{sp} of 232 seconds. This results in the lowest propellant mass of 19.01 kg while achieving the mission ΔV of $54.352 \frac{m}{s}$. The MRE-5.0 also provides a middle ground on the required total mission impulse length of 0.223 hours, or 13 minutes and 22 seconds, when compared to the other monopropellant options. This gives the MRE-5.0 a good balance between the total propellant mass, maneuver responsiveness, and integration simplicity, making it the ideal choice for HEIMDALL station-keeping maneuvers.

I.5 Power

I.5.1 Spacecraft Power Requirements

A component-based electrical power budget was developed for HEIMDALL by reviewing each subsystem's hardware selections and creating different operating modes that the satellite will go through. The recommended approach is to prepare an operating power budget from the payload and bus-subsystem loads with varying operating modes, paying close attention to the average and peak power draw [1]. The HEIMDALL team pieced together an average daily operating power subtotal from all the components, coming out to:

$$P_{subtotal} = 136.01 + 56.2 + 40 + 10 + 0 + 0 + 2 + 149 = 393.2 \text{ W}$$

A 20% design margin was applied to account for any power losses and inefficiencies: Bringing the total average daily requires spacecraft power to:

$$P_{req} = 393.21 + 78.642 = 471.9 \text{ W}$$

Two additional profiles were made for the spacecraft:

$$\text{Station} - \text{keeping power: } 425 \text{ W} ; \text{ Deployment power: } 320.4 \text{ W}$$

Both the additional modes are lower than the average daily required power and thus can be covered by the baseline power procedure. Table 23 below breaks down the percentage of total daily power drawn for each subsystem.

Table 23: Subsystem percent of the total power. This table shows the percentage that each subsystem used in HEIMDALL represents with respect to the total power needed.

Total Power (W)	471.852
Subsystem	Subsystem Percent of Total Power (%)
Payload	28.8
ADCS	11.9
C&DH	8.5
Power	2.1
Propulsion	0
Structure	0
Thermal	0.4
Communications	31.6
Power margin	16.7

Table 24: Power consumption by each subsystem. The amount of power that each subsystem of the HEIMDALL satellite is being shown here, as well as the daily power output, theoretical max power output, station keeping power, and deployment power.

Subsystem	Average Daily Power Output (W)	Component (Type)	Component Average Daily Power Output (W)	Theoretical Max Power Output (W)	Station Keeping Power (W)	Deployment Power (W)
Payload	136.01	WAVES Heritage (Radio Burst Monitor)	8	8	0	0
		PLA on VIGIL (Solar Wind Ion/Plasma Analyzer)	14	14	0	0
		VIGIL Heritage X-ray Monitor (X-ray Flux Monitor)	30	30	0	0
		MAG/FGM Pair (Fluxgate magnetometer)	1.71	1.71	0	0
		PMI VIGIL (EUV imager/magnetograph)	40	40	0	0
		CCOR-1 (White-light Coronagraph)	25	25	0	0
		SOLOHI (Heliospheric Imager)	13.5	13.5	0	0
		STIS (Energy Particle Telescope/Spectrometer)	3.8	3.8	0	0
ADCS	56.2	ASTRO CL (Star Tracker)	1.2	2.4	1.2	0
		LR-450 IMU (Gyroscope / IMU)	15	30	15	15
		RDR 68 (Reaction Wheel)	40	360	40	40
		MONARC-5 (Attitude Control)	0	144	144	6
		Coarse Sun Sensor (Sun Sensor)	0	6	6	6
C&DH	40	IPC-7000 (CPU)	40	80	40	40
Power	10	STARRAY T-STAR (Deployable Solar Array)	0	0	0	0
		STARBUCK-MINI (SmallSat PCDU / EPS)	10	10	10	10
Propulsion	0	MRE-5.0 (Monopropellant)	0	76	76	0
Structure	0	DCUBED HDRM (Solar Panel Deployment)	0	128	0	128
Thermal	2	Minco Polyimide Thermofoil Kapton Film Heater	2	10	2	2
Communications	149	TESAT IDT (Transponder)	20	20	20	20
		MRO TWTA (Amplifier)	50	100	0	0
		Laser transmitter	4	4	0	0
		Laser system	75	100	0	0
Power Margin	78.642	20% Power Margin		181.74	70.84	53.4
Total Daily Power (W)	471.852		Power (W)	1388.15	425.04	320.4

The HEIMDALL team pieced together an average daily operating power subtotal from all the components, coming out to:

$$P_{subtotal} = 136.01 + 56.2 + 40 + 10 + 0 + 0 + 2 + 149 = 393.2 \text{ W}$$

A 20% design margin was applied to account for any power losses and inefficiencies: Bringing the total average daily requires spacecraft power to:

$$P_{req} = 393.21 + 78.642 = 471.9 \text{ W}$$

Two additional profiles were made for the spacecraft:

$$\text{Station – keeping power: } 425 \text{ W}$$

$$\text{Deployment power: } 320.4 \text{ W}$$

Both the additional modes are lower than the average daily required power and thus can be covered by the baseline power procedure.

Table 25 breaks down the percentage of total daily power drawn for each subsystem.

Table 25: Subsystem percent of the total power. This table shows the percentage that each subsystem used in HEIMDALL represents with respect to the total power needed.

Total Power (W)	471.852
Subsystem	Subsystem Percent of Total Power (%)
Payload	28.8
ADCS	11.9
C&DH	8.5
Power	2.1
Propulsion	0
Structure	0
Thermal	0.4
Communications	31.6
Power margin	16.7

I.5.2 STK Time Estimations

Using STK to simulate our scenario, we get the total solar time of 99840 hours for the duration of 12 years, and the total eclipse time of zero hours. The difference in solar time and eclipse time is expected as HEIMDALL orbits the

Sun at L4. HEIMDALL will be exposed to the Sun for most of its operational lifetime. At L4, no objects will block sunlight from reaching HEIMDALL. For this time estimation, we are only looking at the satellite starting when it is already at L4 until the end of its operational lifetime. Before it reaches L4, the satellite is protected and placed inside the upper stage of the launch vehicle. The vehicle is responsible for delivering the satellite to its destination.

I.5.3 Power Source

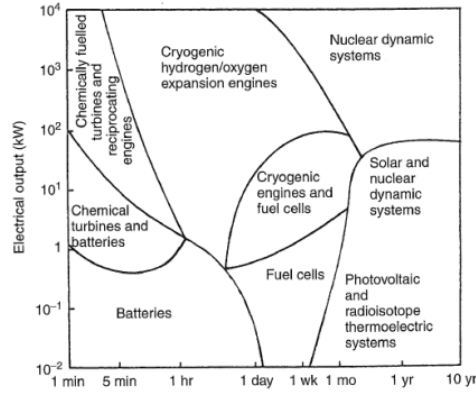


Figure 11: Power output options for mission duration [22].

The required electric power for our mission is ~470 watts. The two power sources suitable for our mission are solar arrays and radioisotope thermal generators. The selected primary power source is a photovoltaic solar-array system with secondary backup battery storage, as HEIMDALL is expected to have near-continuous Sun exposure at Earth–Sun L4. Radioisotope thermoelectric systems are possible for our power requirements and time frame, but having to deal with radioactive material adds major manufacturing cost and complexity, which goes beyond the scope of our mission, especially when sunlight is continuously available, and the power levels are modest to complete the objective of our mission.

I.5.4 Solar Array Estimations

The solar array must be sized to provide sufficient electrical power to meet the spacecraft's demands during daily operations. The general solar equation is given below [26].

$$P_{sa}T_d = \frac{P_dT_d}{\eta_d} + \frac{P_eT_e}{\eta_e} \quad (1)$$

where P_{SA} is the required solar-array output during daylight, P_d and P_e are the spacecraft's power requirements during the daylight and eclipse, respectively, T_d and T_e are the durations of daylight and eclipse, and η_d and η_e are the daytime and nighttime path efficiencies.

During our mission, the eclipse duration is negligible, $T_e \approx 0$. As our mission has an Earth sun-synchronous orbit. This eliminates the second term of the general solar equation. The daytime efficiency of getting power directly η_d is assumed to be 0.85. This assumption accounts for losses in power conditioning, distribution, and conversion with electrical power subsystems. Inputting the average required spacecraft power as the daylight load from Table 1, the required solar-array output becomes:

$$P_{SA} = \frac{471.852}{0.85} = 555.12 \text{ W}$$

Thus, the solar array must generate approximately 555 W of electrical power under nominal conditions to meet the satellite's demand.

Table 26: T-Star solar array from Rocket Lab's standardized array (STARRAY). It provides a beginning-of-life power output as a function of voltage configuration, panel size, and the number of wings per panel [23].

T-Star BOL Vmp and size options with wing power

Voltage Option	Substrate Size (mm)	Wing Power (Pmp, beginning of life, AMO 28°C and 0° angle of incidence)			
		1 Panel per Wing	2 Panels per Wing	3 Panels per Wing	4 Panels per Wing
Vmp of 45V BOL	541 x 1198	183 W	366 W	547 W	727 W
	667 x 1198	229 W	457 W	683 W	909 W
	800 x 1198	275 W	548 W	819 W	1090 W
	800 x 1479	320 W	638 W	955 W	1270 W
	800 x 1619	365 W	729 W	1091 W	1451 W
	800 x 1760	411 W	821 W	1228 W	1633 W
	800 x 1900	457 W	911 W	1364 W	1814 W
Vmp of 62V BOL	800 x 1198	263 W	525 W	786 W	1045 W
	800 x 1479	329 W	656 W	981 W	1305 W
	800 x 1619	394 W	787 W	1177 W	1566 W
	800 x 1900	460 W	917 W	1373 W	1825 W

Looking at the Rocket Lab T-Star data sheet, the solar panel data sheet gives the power provided per wing. Using the power required for the solar array of 555 W and then applying the end-of-life factor to find the power required to ensure the 555 W will be available at the end of its life.

$$P_{BOL, \text{vendor}} \geq \frac{555.12}{(0.985)^{12}} = 665.5 \text{ W}$$

The spacecraft layout gives two symmetric wings; each wing only needs half of the power requirement at the beginning of life, or 332.75 W per wing. Looking at Table 28, it gives two primary choices: the 800×1619 mm, 1 panel per wing, 365 W each, and the 541×1198 mm, 2 panels per wing, 366 W each. Totaling the area for these solar panels gives 1.3m^2 per wing or 2.6m^2 total, and $.65\text{m}^2$ or 2.6m^2 in total, respectively. The selected solar panel will produce more power than what is required for the satellite. Looking at the two options, we determined that the single panel per wing makes more sense as our design doesn't have a width constraint, so the longer panels can fit into the design and, in the process, eliminate the complexity of deploying multiple panels per wing.

I.5.5 Solar Array Deployment

Solar array deployment is a critical function of a satellite operation, enabling the transition from a compact launch configuration to a fully operational power-producing satellite. The surface area of solar panels is too large, so the decision is made to have solar panels extend off the sides of the satellite. Due to launch vehicle size constraints, the solar arrays, which take up large surface areas, need to be stowed away against the spacecraft body and are later deployed once in the final orbit.

The deployment is commonly achieved by a hold-down and release mechanism (HDRM), which restrains the array during launch and actuates once activated. Once released, stored strain energy drives the panel extension. The Rocket Lab STARRAY system supports one or more panels per wing. Deployment geometry directly impacts available power, so it is integral to get a correct deployment.

I.5.6 Battery Capacity Determination

Even in the absence of a regular eclipse, a minimum battery capacity is required to support irregular operations and improve system robustness. A conservative assumption is made that the battery will support an hour of full average daily satellite operation without solar input. Using the spacecraft's power requirement: $P_d = 472$ W, the required energy storage is:

$$C_r = 471.852 \times 1 = 471.852 \text{ W} \cdot \text{hr}$$

Now accounting for the selected depth of discharge of 50%:

$$C_{battery} = \frac{C_r}{DOD} = \frac{471.852}{.5} = 943.7 \text{ W} \cdot \text{hr}$$

Totaling the battery capacity to be approximately .94 kW · hr.

Table 27: Saft Total Energies and their lithium-ion batteries [25]

	ENERGY				EXTENDED LIFE & TEMPERATURE	
	VL 34570 xlr	MP 144350 xlr	MP 174865 xlr	MP 176065 xlr	MP 174565 xtd	MP 176065 xtd
Form factor	Cylindrical D	Prismatic	Prismatic	Prismatic	Prismatic	Prismatic
Nominal voltage	3.65 V	3.65 V	3.65 V	3.65 V	3.65 V	3.65 V
Nominal capacity	5.4 Ah	2.6 Ah	5.3 Ah	6.8 Ah	4.0 Ah	5.6 Ah
Max. continuous discharge current	11.0 A	5.0 A	10.0 A	14.0 A	8.0 A	11.0 A
Max. pulse discharge rate	21.0 A	10.0 A	21.0 A	27.0 A	16.0 A	22.0 A
Max. charge current	5.4 A	2.6 A	5.0 A	6.8 A	4.0 A	5.6 A
Cycle life	>3600	1100	950	1800	2700	2700
(Cycled to 70 % of the cells original capacity)	(100 % DoD, C/2-C/2, +20°C)	(100 % DoD, C-C/2, +20°C)	(100 % DoD, C-C/2, +20°C)	(100 % DoD, C-C/2, +20°C)	(100 % DoD, C-C/2, +25°C)	(100 % DoD, C-C/2, +25°C)
Typical weight	130 g	66 g	121 g	150 g	97 g	135 g
Discharge temperature range	-35 / +60°C	-35 / +60°C	-35 / +60°C	-35 / +60°C	-40 / +85°C	-40 / +85°C
Charge temperature range	-30 / +60°C	-30 / +60°C	-30 / +60°C	-30 / +60°C	-30 / +85°C	-30 / +85°C

Typical values relative to cells stored for one year or less at +30°C max. Performances vary according to discharge characteristics (current, duration, frequency), temperature conditions, storage conditions prior to usage and applications acceptable minimum voltage.



Looking at Table 27, a battery manufacturer for spacecraft, the battery that best fits our application is the MP 174565 xtd. The battery gives 2,700 cycles at 100% DOD, which still surpasses what is needed for our mission.

The voltage is 3.65 V and capacity is 4.0 Ah [1], so calculating:

$$E = V \times Ah = 3.65 \times 4.0 = 14.6 \text{ Wh}$$

Meaning for our satellite, we need a multi-cell battery pack, which can be calculated by dividing the required power by the single battery, getting you 64.6 batteries needed. Rounding up to 65 cells and multiplying by the mass of 97 grams gives you around 6.3 kg for the mass of the battery pack, which will later be used in the final mass estimation.

I.6 Structures & Mechanisms

The structural subsystem is key to ensuring the safety of all the subsystems integrated into the HEIMDALL satellite. Requirements R1 and R2 are directly tied to communication requirement CR3, which fulfills one of the most important communication mission objectives (see Figure 1). The structural subsystem constraints are found in Appendix 3.

I.6.2 Material Discussion

There are many materials that can be selected; however, finding the perfect equilibrium between mechanical properties, price, manufacturing, weight, and radiation shielding can present complications. The 6000 and 7000 series aluminum alloys are the most common option for satellite structures, since these materials are highly accessible in the global market, making it one of the most cost-effective options, however, compared to modern composites and alloys, the density of these 6000 and 7000 aluminum alloys is relatively high, which will result in a mass penalty for our launch and, as a consequence, more budget will have to be destined to the HEIMDALL bus and less to the payload equipment [10]. The good news is that there is one aluminum alloy that can be potentially used as the primary bus material. Al-Li 2195 is a copper-lithium-silver-magnesium-zirconium alloy that yields a massive 20% weight reduction and shows good mechanical properties that can help HEIMDALL to survive its 15-year mission duration. Al-Li 2195 demonstrates a superior fatigue crack growth resistance compared to 6000 and 7000 series aluminum alloys, which can help HEIMDALL not only to survive direct impacts against small debris located at L4, but also against the violent acoustic vibration and high-frequency shock of launch vehicle separation [11].

Carbon fiber reinforcement polymers (CFRP) can help the HEIMDALL team to reduce the bus structure weight and at the same time, provide an overall great protection against space debris found at L4. CFRPs are highly anisotropic, meaning that these materials exhibit direction-dependent physical properties. This property can be used as an advantage since orienting the continuous carbon fibers along the primary load paths experienced during launch. CFRP's count with nearly zero coefficient of thermal expansion, which can ensure that sensitive optical and communication payload instruments remain perfectly aligned despite the massive temperature ranges found at L4 [12].

For more critical, highly stressed local areas, Ti-6Al-4V Titanium alloy is a great material for extreme space environments. Ti-6Al-4V has a relatively low thermal conductivity compared to aluminum, which makes it ideal for thermal isolation standoffs, preventing detrimental heat transfer from high-heat areas to more critical components of the HEIMDALL satellite [13]. The properties of some potential materials are organized in Table 28:

Table 28: Material options for the HEIMDALL satellite and its performance characteristics.

Material	Aluminum 6061-T6	Aluminum-Lithium 2195	Ti-6Al-4V	CFRP
Density (g/m ³)	2.7	2.56	4.43	1.6 – 1.9
Tensile Strength (MPa)	210	570	950 – 1380	1500
Thermal Expansion	High	High	Excellent	Excellent
Cryogenic Performance	Good	Excellent	Excellent	Good
Radiation Resistance	Excellent	Excellent	Excellent	Poor
Pyrotechnic Shock Tolerance	Good	Excellent	Excellent	Vulnerable
Manufacturability	Excellent	Good	Complex	Complex
Raw Material Cost (USD/kg)	\$4 – 7\$	\$15 – 25\$	\$40 - \$150	\$90 - \$200

I.6.3 Structural Architecture

The proposed spacecraft bus for HEIMDALL employs a hybrid structural architecture that balances mass efficiency, mechanical robustness, and long-term radiation as well as temperature resistance. The primary load-bearing thrust tube and base plate must be constructed from Aluminum-Lithium 2195. This section is the central spine of the spacecraft, and Al-Li 2195 aligned perfectly because of its great thermal expansion, tensile strength, and weight. This advanced alloy provides the necessary strength to handle the massive low-frequency axial and lateral loads experienced during liftoff. Not only that, but having an excellent cryogenic performance, Al-Li 2195 alloy will not become brittle in deep space shadows, as well as the parts facing the Sun. This alloy also provides baseline shielding and structural mass that contributes to radiation dose reduction, all while being significantly cheaper and faster to machine than titanium or other composite materials. The secondary structure (shear panels, instrument decks, exterior walls, payload instrument mounting) should be constructed using aluminum honeycomb cores with CFRP face sheets. Since HEIMDALL will be exposed to harsh conditions during its 15-year mission duration, advanced high-performance resin systems are needed for the CFRPs. Matrices such as cyanate were selected over epoxy due to superior radiation and moisture resistance. This composite sandwich configuration reduces the overall mass of the spacecraft while maintaining a rigid structure. The high bending stiffness shifts natural frequencies and reduces amplitude response under the massive acoustic noise during launch. The CFRP panels, with excellent thermal expansion, ensure that the optical payload as well as the communication systems remain perfectly aligned despite the extreme temperature ranges that HEIMDALL will experience. To maintain structural integrity for a 15-year mission duration, these exterior panels must be integrated with z-graded metallic foils and charge dissipation coatings to shield

against ionizing radiation. Lastly, critical joint interfaces, kinematic mounts, separation rings, and thermal standoffs should be machined out of titanium Ti-6Al-4V. One of the downsides of titanium is its cost but since these components only represent a small portion compared to other sections of the spacecraft and that is why the HEIMDALL team decided to go for this material. Crucially, the acoustic impedance mismatch between the dense Ti-6Al-4V mounts and the lightweight CFRP panels will promote a partial reflection and transmission loss of high-frequency separation shock waves at material interfaces. This result will offer more protection during liftoff and launch vehicle separation to the inner payload components [14].

The combination of these materials into a unified, hybrid structure will provide a spacecraft bus with high probabilities of survival in the hazardous conditions found in L4. This architecture minimizes as much as possible dry mass, launch cost, and radiation exposure while maintaining a dynamic rigidity to survive launch acoustics, separation shocks, and the impact of space debris.

I.6.4 HEIMDALL's Structural Concept

A cubic spacecraft bus configuration will be used for the HEIMDALL satellite with an approximate length of 1.15 meters by 1.15 meters, falling into the medium-to-large category for satellites. The ram side of HEIMDALL will always be pointing directly toward the Sun; therefore, most of the helio-physics equipment will be located on this face of the satellite. The equipment that can be found on this face will be the Helio-spheric Imager, X-Ray Flux Monitor, Coronagraph, Sun Sensor, and an EUV Imager. Devices such as the Helio-spheric Imager will have a rail system that moves the sensor out of the spacecraft in order to have a better view of the Sun. The plasma analyzer, along with the magnetometer, will be located on the right-hand face of the satellite. Similar to the heliospheric imager, the magnetometer will be deployed out of the satellite with the help of a rail system since this device is highly sensitive to the possible magnetic field generated by HEIMDALL. Opposite the Sun-facing side of the satellite will be a radiator system to dissipate heat into deep space. Next, the bottom face of the satellite will have the laser communication equipment with an integrated damping system to increase the stabilization of the signal being sent to Earth, crewed (or uncrewed) missions, or Mars. The communications antenna will be located at the top face of the satellite. The two solar arrays will be located at the left and right sides of the satellite. During the phases when the satellite is inside the launch vehicle, the solar panels will be folded along the left and right faces. When HEIMDALL executes phase 5 of the mission, these solar panels will be deployed to face directly towards the Sun and start generating power for

subsequent tasks and operations. Inside the satellite are all the main computers, battery, ADCS system, thermal dissipation controls, and reaction wheels. At each corner of the satellite, small RCS thrusters will be placed to help the control system induce larger rotations and allow for better adjustability when exact positioning is needed, when it comes to communicating, receiving information, or analyzing the solar weather. Propulsion implementation on HEIMDALL is as follows: two thrusters on the sun-facing side and two thrusters on the opposite side, with the 71 L propellant and 15.2 L pressurant tank placed as close to the center of mass as possible. The location of the thrusters is to allow them to add or remove energy from HEIMDALL's orbit without increasing its orbital velocity. If corrections need to be made to HEIMDALL's orbital velocity, the RCS thrusters will be able to provide this correction. Placing the propellant and pressurant tanks as close to the center of mass of HEIMDALL as possible will help to counteract the change in the center of mass experienced as the tanks drain, adding to the ease of stability of the satellite. Figure 12 shows a potential 3D model for the final structural concept for the HEIMDALL satellite based on volumetric placement:

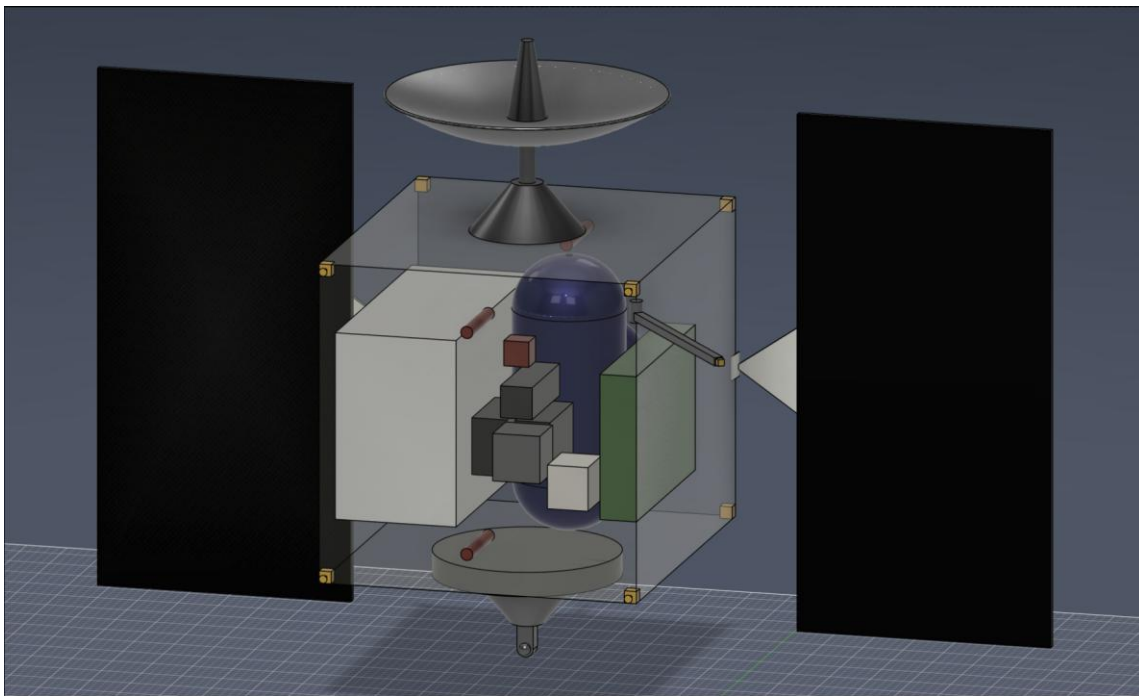


Figure 12: 3-D model for the HEIMDALL satellite.

I.7 Thermal Subsystem

I.7.1 Requirements and Constraints – Defining Temperature Limits

Our thermal control system (TCS) must operate to maintain all satellite components within their respective temperature ranges throughout the entire mission. Scientific instruments, batteries, and communication equipment all have narrow temperature limits, meaning that these critical components must be strictly thermally controlled and cared for to ensure mission success. Batteries typically operate within a strict range from about -5°C to 25°C , where general remote sensing payload instruments, and telecommunications can operate from about -10°C to 50°C [1]. Other components of the satellite, like the structure or solar arrays, have less restrictive temperature limits, but still need to be accounted for when designing the thermal system. These temperature limit requirements help us determine an operable thermal envelope for HEIMDALL while allowing us to define the needs for the thermal control system. Data for the typical operating ranges of spacecraft components is gathered from [1] and [3], shown below in Table 29:

Table 29: Typical Operating Temperatures

Component	Operating Range (degrees Celsius)
Batteries	-5 to +25
Solar Arrays	-105 to +110
Propulsion Systems	+7 to +65
Communications	-10 to +50
Thermal Control:	
Multilayer Insulation (MLI)	-160 to +250
Heaters and Heat Pipes	-35 to +60
Radiators	-95 to +60
Scientific Instruments:	
Coronagraph	-35 to +60
Heliospheric Imager	-100 to -50
Telescope/Spectrometer	-35 to +5
Magnetometer	-80 to +80

Thermal requirements R5 and R6 will aim to maintain the right temperatures so that both the communication and science payload can work to fulfill science/communication requirements SR2 and CR4, respectively. These drive the purpose of the TCS to make sure all satellite components are well kept within their respective temperature limits.

Satellites in LEO tend to experience a temperature range of -100 degrees Celsius to 260 degrees Celsius (ESA, 2004). This range helps give a reliable estimate to work off, even if HEIMDALL will be operating at L4. This specific space environment removes the convective heat transfer part of thermal mechanisms, which helps simplify thermal equilibria. Due to HEIMDALL's extreme distance away from Earth, approximately 1 Astronomical Unit (AU), both the Earth-reflected solar radiation (albedo) and Earth-emitted infrared (IR) radiation (planet shine) are considered negligible. The only *outside* heating that the satellite will be absorbing is the direct solar flux from the Sun. This means that heat dissipation and absorption will only happen due to thermal radiation within the satellite and the outside environment of space. To put it simply, our satellite is a lone body in a vacuum where the Sun is the only factor that can and will contribute to HEIMDALL's heat absorption.

The additional heat generation of the satellite will be due to internal components such as electronics, power systems, and other instruments generating heat. Some of this heat generation must be dissipated outside the satellite to avoid overheating, or else it could lead to instrument failures. A worst-case hot environment develops as the side of the satellite is continuously exposed to the Sun's radiation, along with the additive internal heat generation. On the other hand, a worst-case cold environment develops on the side with no exposure to the Sun, by getting reined in by the cold depths of deep space. Being able to create a thermal system that adheres to these thermal constraints is a high priority for HEIMDALL.

To meet the thermal requirements and constraints of HEIMDALL, a passive thermal control system will be primarily implemented. NASA's 2024 State-of-the-art of Small Spacecraft Technology [2] explains how a passive TCS maintains the temperature of components without using any power. Applications of TCS are considered state-of-the-art for larger satellites, so the Technology Readiness Level (TRL) can be considered less than 9 for smaller applications. A combination of using multi-layer insulation (MLI) blankets, heat pipes, and deployable radiators can be used to passively control and move unwanted heat. This allows excessive heat to be moved to radiative surfaces on the satellite to dissipate it out to the environment through radiation. For HEIMDALL, this may be able to help alleviate the worst-case cold problems such that the heat pipes can transfer the unwanted heat from components to the satellite surface facing cold deep space, where the radiators are located; this could help momentarily heat up the cold side, if needed, before the heat is removed to space by the radiators. However, these radiators usually dissipate heat instantaneously, so there would be no heat retained to be able to heat up the cold side of the satellite. Furthermore, this doesn't

necessarily serve as a proper heating system for components that can get too cold or if the satellite itself is struggling with the temperatures. Therefore, active heating methods, such as a heater, could be used to help maintain components like batteries within their temperature ranges. Additionally, specific surface thermal control coatings will be selected to regulate the satellite's passive heat absorption and dissipation.

I.7.2 Thermal Control Mechanisms

The primary thermal control system for HEIMDALL will be with MLI blankets. These are specifically aluminized Kapton MLI blankets. These blankets consist of 10-20 layers of Aluminized Kapton and Dacron mesh spacers. The emissivity reduction for this blanket is approximately 0.01-0.03, and the heat flux reduction is 70%-90%. These MLI blankets would be applied to all external spacecraft surfaces where thermal isolation is required. These blankets would likely contribute to a mass of about 0.1-0.3 kg/m². Where MLI blankets are not used or effective, a thermal control coating will be applied, such as AZ-93 Zinc Oxide-based white coating. This coating has a solar absorptivity of 0.17-0.20 and an infrared emissivity of 0.90-0.92. This coating has a negligible mass. As heat is absorbed or internally generated, HEIMDALL will use constant conductance heat pipes to transfer sun-side heat and heat from electrical components to the radiator's surface. These have a mass of about 0.33 kg/m, which consists of aluminum with a 1.27 cm diameter. Additionally, radiators will be used as the primary heat rejection mechanism. HEIMDALL will use AZ-93 high-emissivity coated aluminum radiator panels oriented toward deep space. Emissivity will be approximately 0.9. At 300K, the heat rejection capability will be approximately 300-400 W/m². The added mass of these radiators is 0.2-0.6 kg.

These systems are passive and require no power; however, they are not sufficient for worst-case cold scenarios, especially given the small temperature range of some components, such as batteries. For this reason, HEIMDALL will also need to use some active thermal control, such as a Kapton film heater. The Minco Polyimide Thermofoil Heater will be sufficient to be able to maintain thermal control in cold conditions. The power density of this heater is defined by the NASA SOA-listed equipment as: 0.8-5.1 W/cm². The temperature capability for this heater is -200 degrees Celsius to 200 degrees Celsius. At peak power, the heater would require 5-10 W and would incorporate controlled thermostats or temperature sensors. The added mass for the heater would be <50 g.

The total mass of the HEIMDALL thermal control system, including both passive and active components, is estimated to be between 0.5 and 1.5 kg. This does not include the mass for the heat pipes or MLI blankets, as the actual size needed is unknown for this project. The total mass of these components will not be significant towards the full mass of HEIMDALL. This estimate is consistent with typical spacecraft design guidelines, where the thermal subsystem accounts for approximately 2–5% of the total spacecraft mass.

The thermal control system is primarily passive and therefore requires no continuous electrical power. Active thermal control is provided by low-power Kapton heaters used to regulate temperature-sensitive components, particularly satellite batteries. The full thermal system is designed to operate at a peak power of approximately 5–10 W, with an average power consumption of 1–5 W depending on operating conditions and duty cycle, and a mass of about 1.5 kg.

Overall, the thermal system maintains low mass and power requirements while ensuring that all spacecraft components remain within their allowable temperature limits throughout the mission. Table 30 below shows a summary of the proposed thermal system:

Table 30: Thermal Mechanism Descriptions

Thermal Mechanism	Passive or Active	Description
Multi-Layer Insulation (MLI) Blankets	Passive	10-20 layers of Aluminized Kapton and Dacron mesh spacers - Emissivity: 0.01-0.03 - Heat Flux Reduction: 70-90% - Mass: 0.1-0.3 kg/m²
AZ-93 Zinc Oxide-based White Coating	Passive	- Absorptivity: 0.17-0.20 - Infrared Emissivity: 0.90-0.92 - Mass: negligible
Aluminum Heat Pipes	Passive	Mass: 0.33 kg/m (1.27 cm diameter)
AZ-93 High-Emissivity Coated Aluminum Radiator Panels	Passive	- Emissivity: ~ 0.9 - Heat Rejection Capability: 300-400 W/m² - Mass: 0.2-0.6 kg
Minco Polyimide Thermofoil Kapton Film Heaters	Active	- Power Density: 0.8-5.1 W/cm² - Temperature Capability: -200 to +200 degrees Celsius - Peak Power: 5-10 W (1-5 W average) - Mass: < 50 g

I.7.3 Power Dissipation Scenarios

The thermal control of the satellite can be determined by balancing the energy shown in Equation (9):

$$q_{solar} + q_{albedo} + q_{planetshine} + Q_{gen} = Q_{stored} + Q_{out,rad} \quad (9)$$

Where:

q_{solar} = The amount of solar heating that is absorbed by the spacecraft due to distance to the sun, surface area facing the sun, and solar absorptivity of the spacecraft surface.

q_{albedo} = The amount of solar heating reflected by the planet (Earth) and absorbed by the Earth's surface area and solar absorptivity.

$q_{planet\ shine}$ = The infrared (IR) heating absorbed from the planet due to the planet's surface area, and IR emissivity.

Q_{gen} = The heat generated by the spacecraft due to the power dissipation of spacecraft components.

Q_{stored} = The heat stored by the spacecraft due to thermal capacitance of the spacecraft.

$Q_{out,rad}$ = The heat emitted due to radiation due to surface area made as the radiator space, IR emissivity of surface, and difference in temperature between spacecraft radiator and heat sink where it's dissipating to (deep space); also includes heat lost through insulation or surface not made to function as radiators.

With HEIMDALL operating at L4, the albedo and planet shine heat absorption can be neglected since the spacecraft is not close enough to Earth to be affected by these conditions. Additionally, the system will be assumed to be in a steady state, so the temperature of the satellite is not changing with time, making Q_{stored} zero. So, the equation reduces to Equation (10):

$$q_{solar} + Q_{gen} = Q_{out,rad} \quad (10)$$

The full suite of the payload system has an estimated total power of 284 W, which is the dominant factor of HEIMDALL's internal heat generation, Q_{gen} . Other subsystem power generation will add to the total heat generation, but for the initial estimation, this will be the total equilibrium internal heat generated. Most of the electrical power

generated will ultimately dissipate as heat through radiation in this estimated system, so the internal heat generated was assumed to be equal to the total power of the satellite. The satellite's geometry was assumed to use a gray body, which also approximated the shape as an equivalent sphere with the same surface area as its cylinder. The absorptivity and emissivity coefficients were assumed to be constant for the satellite surface coatings, 0.3 and 0.9, respectively. For the hot case, the upper limit of heat generation was approximated to be 650 W, and for the cold case, the lower limit of heat values was 284 W. The estimated gray body satellite has an approximate cylindrical area of $A = 8\pi m^2$ (1 m radius, 3 m height), and the equivalent sphere has an area of $A_c = 2\pi m^2$. The Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} W/m^2K^4$. The solar flux, G_s , has three cases; hot, cold, and equilibrium. The hot case is $G_{s,hot} = 1418 \frac{W}{m^2}$, the cold case is $G_{s,cold} = 1326 \frac{W}{m^2}$, and equilibrium is $G_{s,equilibrium} = 1362 \frac{W}{m^2}$. Preliminary temperature calculations were performed by manipulating the simplified thermal equilibrium Equation 10 to get Equation (11):

$$T_{hot} = \left[\frac{A_c G_s \alpha + Q_{total}}{A \sigma \epsilon} \right]^{\frac{1}{4}} \text{ and } T_{cold} = \left[\frac{Q_{total}}{A \sigma \epsilon} \right]^{\frac{1}{4}} \quad (11)$$

$$\text{Equilibrium Temperature} = -56.01^\circ C$$

$$\text{Worst Case Hot Temperature} = -47.54^\circ C$$

$$\text{Worst Case Cold Temperature} = -151.16^\circ C$$

These are the estimated temperature scenarios of the satellite, not including solar arrays, that were calculated in MATLAB. Meaning that HEIMDALL must remain within this temperature range near equilibrium and away from worst-case scenarios. The equilibrium temperature can be explained as relatively low because of the negligible effects of Earth's albedo and planet's shine. The cold case estimated low temperatures, resulting in the need for some active thermal control systems on top of the passive control. These values align with our mission such that HEIMDALL is at L4, while only receiving solar radiation from the Sun, where the Earth does not contribute to any heat absorption. The satellite is positioned in a way where one side is always facing the hot effects of the Sun, while the other side is receiving the effects of the cold vacuum of space.

I.7.4 Special Considerations

With HEIMDALL located at L4, the extreme cold temperatures come into effect. The anti-sunward side will be experiencing very cold temperatures, estimated at the worst case to about -151.2 degrees Celsius. The Minco Polyimide Thermofoil Heater, an active thermal control system, will account for this worst-case cold scenario since it is made to handle a temperature range of -200 to 200 degrees Celsius. This heater will most likely be used to keep batteries within their limits, along with other communication/scientific equipment that has narrow temperature ranges.

HEIMDALL considers the limited range of temperature constraints and requirements with systems with high power dissipation or low thermal conductivity. The payload is likely responsible for the largest amount of heat generation and dissipation, so it requires its own heat transport control system through heat pipes to radiator surfaces. Batteries, communications, and other ADCS components have more strict thermal ranges; These must be abided by for successful operations using insulation and active heaters. Conductive mounting of components can be used as a mechanism to keep parts connected to the structure and allow them to receive help from the thermal control system.

Most importantly, thermal mechanisms were chosen to protect the coronagraph, heliospheric imager, and energetic particle telescope/spectrometer to ensure our mission accomplishes the minimum scientific observables. These three instruments are the most vital to HEIMDALL's mission, and their respective temperature ranges had to be taken care of.

At L4, HEIMDALL is in a thermal environment that is dominated by the Sun's solar radiation. In this near-perfect vacuum, thermal convection and Earth radiation effects are negligible. Knowing this allows for heat collection, transportation, and dissipation to be carefully monitored and conducted to be able to balance HEIMDALL at a temperate equilibrium. This ensures all HEIMDALL's components stay within their respective temperature ranges, preventing any components from overheating or undercooling.

J. Size, Weight, and Power Budget

HEIMDALL is seeking to use heritage equipment wherever possible to reduce satellite development time and costs. This aligns with the AIAA RPF requirements and constraints, which note that “it is highly desirable to use technologies that are already demonstrated on previous programs or [are] currently in the NASA technology development portfolio” [30]. The size, weight, and power table found in Appendix 4 summarizes HEIMDALL’s equipment in two sections: science & communication, and the subsystems. The left side of the table shows the equipment type, proposed instrument, and where it fits into either the mission goal or which subsystem it is a part of, while the right side includes the instrument properties. This format allows the size, weight, and power table to show how each component supports either the mission science and communication objectives or the subsystem architecture, while including how each component contributes to the overall mass, volume, and power requirements. Despite HEIMDALL aiming to use entirely heritage equipment, there was information that could not be found for certain pieces of equipment. Despite best efforts, the following equipment have official information missing and estimates were used based on available data for similar equipment; dimensions for STIS, an energetic particle telescope; dimensions for PLA from VIGIL, a solar wind ion/plasma analyzer; dimensions for PMI from VIGIL, an EUV imager/magnetograph; dimensions for the Saft Total Energies MP 176065 xtd, HEIMDALL’s battery; and dimensions and weights for much of the thermal subsystem which will be discussed.

For subsystems such as thermal, propulsion, and structures, where the overall mass is highly dependent on the final satellite configuration, historical data were used to get an overall system estimate using mass percentages from large communication satellites in geostationary orbit [26]. While this does not get masses for specific components, it does allow for a good idea of the expected mass of the subsystem, which can be incorporated into the overall satellite mass. See Appendix D Table 1 of [26] for exact mass percentages used in thermal, propulsion, and structures calculations.

K. Mission Timeline

For HEIMDALL to complete all the science and communication objectives established by the AIAA report, the 2024 National Academies of Sciences' Decadal Survey, and NASA's Moon to Mars objectives, the development, testing, launching, and operational time must follow the mission schedule listed in Appendix 5. A tentative Gantt chart was created to illustrate HEIMDALL's phase duration (Figure 13):

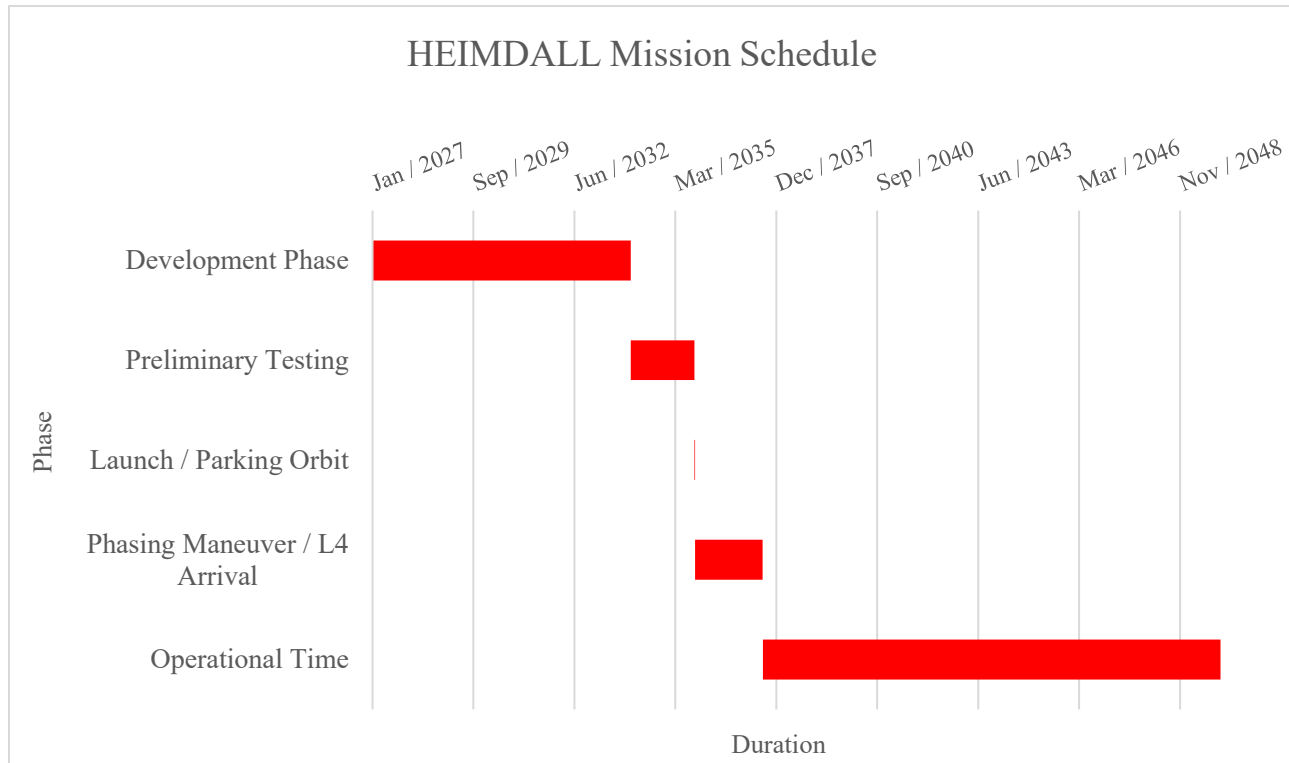


Figure 13: HEIMDALL's phase duration Gantt chart.

L. Budget

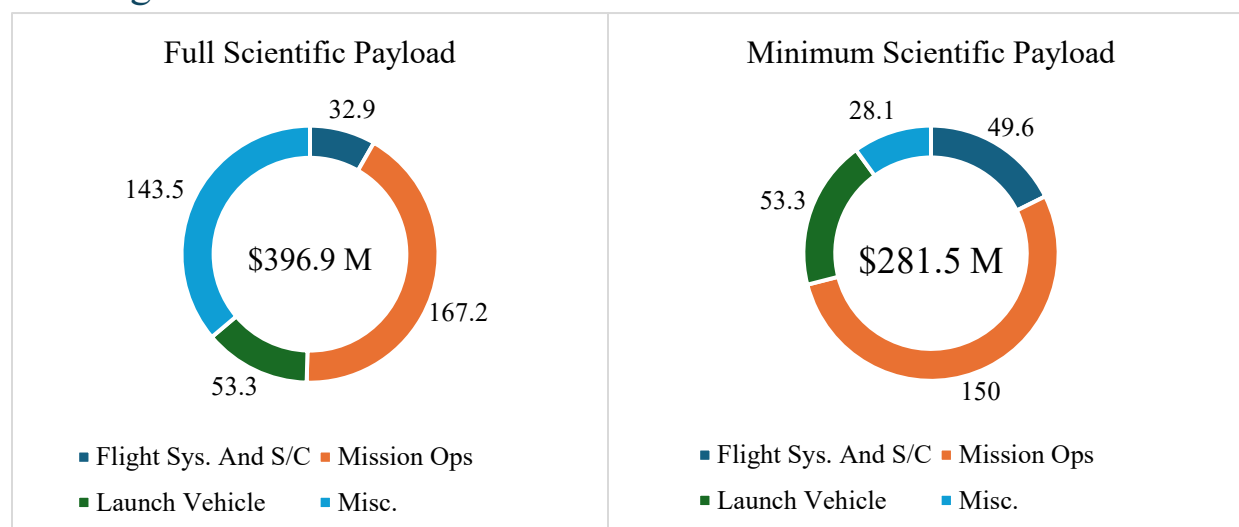


Figure 14: HEIMDALL's budget Analysis.

The biggest limiting factor in terms of the budget for HEIMDALL will be operational costs once the satellite is in position and transmitting scientific data. Using NASA's Project Cost Estimating Capability (PCEC) software, two budgets were produced. While both meet the proposal's requirements of staying under a \$400 Million dollar maximum, one supports the full scientific payload defined prior, while the other estimates a price using the minimum payload with its minimum data transmission rate. As shown in the figure above, the majority of the financial burden comes from the Mission Operation Systems (MOS), specifically, phase E. HEIMDALL currently meets the expected operation time frame, providing communication between Earth and Mars between 2036 and 2049, but HEIMDALL is fully capable of continued operation past the required timeline with continued funding. The estimated budget for the full and minimum scientific payload is given in Table 35 for further breakdown of each mission phase.

Table 31: Full Payload Cost Estimate

ROBOTIC SPACECRAFT WBS

FY2025 \$M				Units Conversion Factor: 1.000						
				Inflation Factor: 1.333						
WBS #	Level	Line Item Name/Description		Non-Recurring	Recurring Production	Non-Allocated	Operations	TOTAL	Fee + Burden	TOTAL w/Fee + Burden
0	1	System Name		\$ 48.4	\$ 92.6	\$ 55.9	\$ 167.2	\$ 364.1	\$ -	\$ 364.1
1.0	2	Project Management		\$ 3.0	\$ 4.9	\$ -	\$ -	\$ 8.0	\$ -	\$ 8.0
2.0	2	Systems Engineering		\$ 8.1	\$ 17.8	\$ -	\$ -	\$ 25.9	\$ -	\$ 25.9
3.0	2	Safety and Mission Assurance		\$ 4.2	\$ 9.3	\$ -	\$ -	\$ 13.6	\$ -	\$ 13.6
4.0	2	Science/Technology		\$ -	\$ -	\$ 2.5	\$ -	\$ 2.5	\$ -	\$ 2.5
5.0	2	Payload(s)		\$ 2.2	\$ 4.8	\$ -	\$ -	\$ 7.0	\$ -	\$ 7.0
5.01	3	Payload Management		\$ 0.4	\$ 0.6	\$ -	\$ -	\$ 1.0	\$ -	\$ 1.0
5.02	3	Payload System Engineering		\$ 1.7	\$ 3.7	\$ -	\$ -	\$ 5.4	\$ -	\$ 5.4
5.03	3	Payload Product Assurance		\$ 0.1	\$ 0.2	\$ -	\$ -	\$ 0.3	\$ -	\$ 0.3
5.10	3	Instruments - EMPTY ROLLUP		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5.x	3	Payload I&T		\$ 0.1	\$ 0.2	\$ -	\$ -	\$ 0.3	\$ -	\$ 0.3
6.0	2	Flight System \ Spacecraft		\$ 24.3	\$ 32.9	\$ -	\$ -	\$ 57.2	\$ -	\$ 57.2
6.01	3	Flight System Project Management		\$ 0.2	\$ 0.3	\$ -	\$ -	\$ 0.4	\$ -	\$ 0.4
6.02	3	Flight System Systems Engineering		\$ 5.8	\$ 12.6	\$ -	\$ -	\$ 18.4	\$ -	\$ 18.4
6.03	3	Flight System Product Assurance		\$ 1.2	\$ 2.8	\$ -	\$ -	\$ 4.0	\$ -	\$ 4.0
6.10	3	Spacecraft		\$ 16.9	\$ 16.4	\$ -	\$ -	\$ 33.3	\$ -	\$ 33.3
--	4	Structures & Mechanisms		\$ 0.4	\$ 1.1	\$ -	\$ -	\$ 1.5	\$ -	\$ 1.5
--	4	Thermal Control		\$ 1.3	\$ 0.8	\$ -	\$ -	\$ 2.1	\$ -	\$ 2.1
--	4	Electrical Power & Distribution		\$ 0.4	\$ 2.3	\$ -	\$ -	\$ 2.7	\$ -	\$ 2.7
--	4	GN&C		\$ 4.0	\$ 3.8	\$ -	\$ -	\$ 7.8	\$ -	\$ 7.8
--	4	Propulsion		\$ 0.9	\$ 1.2	\$ -	\$ -	\$ 2.1	\$ -	\$ 2.1
--	4	Communications		\$ 4.1	\$ 2.2	\$ -	\$ -	\$ 6.3	\$ -	\$ 6.3
--	4	C&DH		\$ 4.7	\$ 4.8	\$ -	\$ -	\$ 9.5	\$ -	\$ 9.5
--	4	Entry, Descent, and Landing		\$ 1.1	\$ 0.3	\$ -	\$ -	\$ 1.3	\$ -	\$ 1.3
--	5	Parachutes		\$ 0.0	\$ 0.0	\$ -	\$ -	\$ 0.0	\$ -	\$ 0.0
--	5	Thermal Protection System		\$ 1.1	\$ 0.2	\$ -	\$ -	\$ 1.3	\$ -	\$ 1.3
6.x	3	Flight System I&T		\$ 0.2	\$ 0.9	\$ -	\$ -	\$ 1.1	\$ -	\$ 1.1
7.0	2	Mission Operations System (MOS)		\$ 6.0	\$ 21.1	\$ -	\$ 167.2	\$ 194.4	\$ -	\$ 194.4
--	3	MOS/GDS Development (Phase B-D)		\$ 6.0	\$ 21.1	\$ -	\$ -	\$ 27.2	\$ -	\$ 27.2
--	3	Mission Ops & Data Analysis (Phase E)		\$ -	\$ -	\$ -	\$ 167.2	\$ 167.2	\$ -	\$ 167.2
8.0	2	Launch Vehicle/Services		\$ -	\$ -	\$ 53.3	\$ -	\$ 53.3	\$ -	\$ 53.3
9.0	2	Ground Data System (GDS)		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10.0	2	System Integration, Assembly, Test & Check Out		\$ 0.4	\$ 1.7	\$ -	\$ -	\$ 2.1	\$ -	\$ 2.1
11.0	2	Education & Public Outreach		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
							Reserves %	TOTAL w/Reserves		
							9%	\$ 396.9	TOTAL w/Fees, Burdens, &	
									\$ 396.9	

Table 32: Minimum Payload Cost Estimate

ROBOTIC SPACECRAFT WBS

Units Conversion Factor: 1.000 Inflation Factor: 1.333													
FY2025 \$M		Level	Line Item Name/Description	Non-Recurring		Recurring Production		Non-Allocated		Operations	TOTAL	Fee + Burden	TOTAL w/Fee + Burden
0		1	System Name	\$	28.0	\$	81.2	\$	55.2	\$	117.2	\$	281.5
1.0		2	Project Management	\$	2.6	\$	6.1	\$	-	\$	-	\$	8.8
2.0		2	Systems Engineering	\$	2.8	\$	4.6	\$	-	\$	-	\$	7.4
3.0		2	Safety and Mission Assurance	\$	1.4	\$	3.0	\$	-	\$	-	\$	4.3
4.0		2	Science/Technology	\$	-	\$	-	\$	1.9	\$	-	\$	1.9
5.0		2	Payload(s)	\$	1.4	\$	2.9	\$	-	\$	-	\$	4.4
5.01		3	Payload Management	\$	0.6	\$	1.4	\$	-	\$	-	\$	2.1
5.02		3	Payload System Engineering	\$	0.6	\$	1.0	\$	-	\$	-	\$	1.5
5.03		3	Payload Product Assurance	\$	0.1	\$	0.3	\$	-	\$	-	\$	0.5
5.10		3	Instruments - EMPTY ROLLUP	\$	-	\$	-	\$	-	\$	-	\$	-
5.x		3	Payload I&T	\$	0.1	\$	0.2	\$	-	\$	-	\$	0.3
6.0		2	Flight System \ Spacecraft	\$	11.9	\$	37.7	\$	-	\$	-	\$	49.6
6.01		3	Flight System Project Management	\$	2.2	\$	5.0	\$	-	\$	-	\$	7.2
6.02		3	Flight System Systems Engineering	\$	2.0	\$	3.2	\$	-	\$	-	\$	5.2
6.03		3	Flight System Product Assurance	\$	1.1	\$	2.4	\$	-	\$	-	\$	3.5
6.10		3	Spacecraft	\$	6.3	\$	26.1	\$	-	\$	-	\$	32.4
--		4	Structures & Mechanisms	\$	0.3	\$	0.3	\$	-	\$	-	\$	0.6
--		4	Thermal Control	\$	0.8	\$	0.4	\$	-	\$	-	\$	1.2
--		4	Electrical Power & Distribution	\$	0.4	\$	2.6	\$	-	\$	-	\$	3.0
--		4	GN&C	\$	1.3	\$	1.6	\$	-	\$	-	\$	2.9
--		4	Propulsion	\$	0.7	\$	0.4	\$	-	\$	-	\$	1.1
--		4	Communications	\$	-	\$	19.1	\$	-	\$	-	\$	19.1
--		4	C&DH	\$	1.8	\$	1.5	\$	-	\$	-	\$	3.2
--		4	Entry, Descent, and Landing	\$	1.1	\$	0.2	\$	-	\$	-	\$	1.3
--		5	Thermal Protection System	\$	1.1	\$	0.2	\$	-	\$	-	\$	1.3
6.x		3	Flight System I&T	\$	0.4	\$	0.9	\$	-	\$	-	\$	1.3
7.0		2	Mission Operations System (MOS)	\$	7.4	\$	25.9	\$	-	\$	117.2	\$	150.5
--		3	MOS/GDS Development (Phase B-D)	\$	7.4	\$	25.9	\$	-	\$	-	\$	33.3
--		3	Mission Ops & Data Analysis (Phase E)	\$	-	\$	-	\$	-	\$	117.2	\$	117.2
8.0		2	Launch Vehicle/Services	\$	-	\$	-	\$	53.3	\$	-	\$	53.3
9.0		2	Ground Data System (GDS)	\$	-	\$	-	\$	-	\$	-	\$	-
10.0		2	System Integration, Assembly, Test & Check Out	\$	0.4	\$	1.0	\$	-	\$	-	\$	1.3
11.0		2	Education & Public Outreach	\$	-	\$	-	\$	-	\$	-	\$	-
										TOTAL		TOTAL w/Fees, Burdens, &	
										Reserves		Reserves %	
										0%		\$ 281.5	
												\$ 281.5	

M. Risk Analysis

Due to our mission lacking a crewed component, we evaluated associated risk through the lens of mission criticality, rather than a standard dual approach of functional failure points alongside health and safety considerations. A system of categorizing hazards to identify areas of further research is defined. Categories of risk are assigned by first identifying classifications of hazard by subsystem impact, hazard origin, and mission phase, outlined in Table 33.

Table 33: Hazard Categorization Key

ID	PR	ST	CO	SO	TH	EL	MG	RD	SW	EV
Category	Propulsion	Structure	Comms	Science Observation	Thermal	Electrical	Management	Radiation	Software	Environment

Following this categorization, our team worked together to identify fault points in previous deep-space, lunar, Earth-orbital, and high-altitude rocket-launched missions. Using historical examples as a starting point, we then analyzed novel fault points and created a combined table. Table 34 shows a selection of hazards pertinent to HEIMDALL, their categorization, and a brief description of their impact on our mission.

Table 34: Selected Hazards and their Descriptions

Hazard Type	Hazard	Description
ST	Launch Vibration Damage	High acoustic and vibration loads during launch could damage sensitive instruments if not sufficiently isolated.
RD	Solar Radiation Damage	Solar energetic particle events and long-term radiation exposure could damage avionics or degrade instrument performance.
TH	Thermal Control Failure	Failure of heaters, radiators, or insulation could cause spacecraft components to overheat or freeze, leading to degraded performance or hardware damage.
SW	Onboard Software/System Malfunction	Software errors or system faults could interrupt spacecraft operations, causing temporary loss of communications or science data.
CO	Communications Failure	Failure of the communication subsystems (transmitter, receiver, or antenna hardware) could result in loss or degradation of the communication link with Earth.

A risk matrix with an intensity scale key is built to estimate the likelihood and severity of various hazards and visualize their overall threat to mission integrity, as shown in Figure 15. Each of the 15 hazards we measured is placed in its respective location on the risk plot. The bracketed numbers are the hazards, in no particular order, and the two-letter labels are the categories from Table 34. Probability and Severity Weighted Criteria and Examples in Appendix 2 Table 38 demonstrate a rating system for indexing likelihood and severity on a ten-point scale.

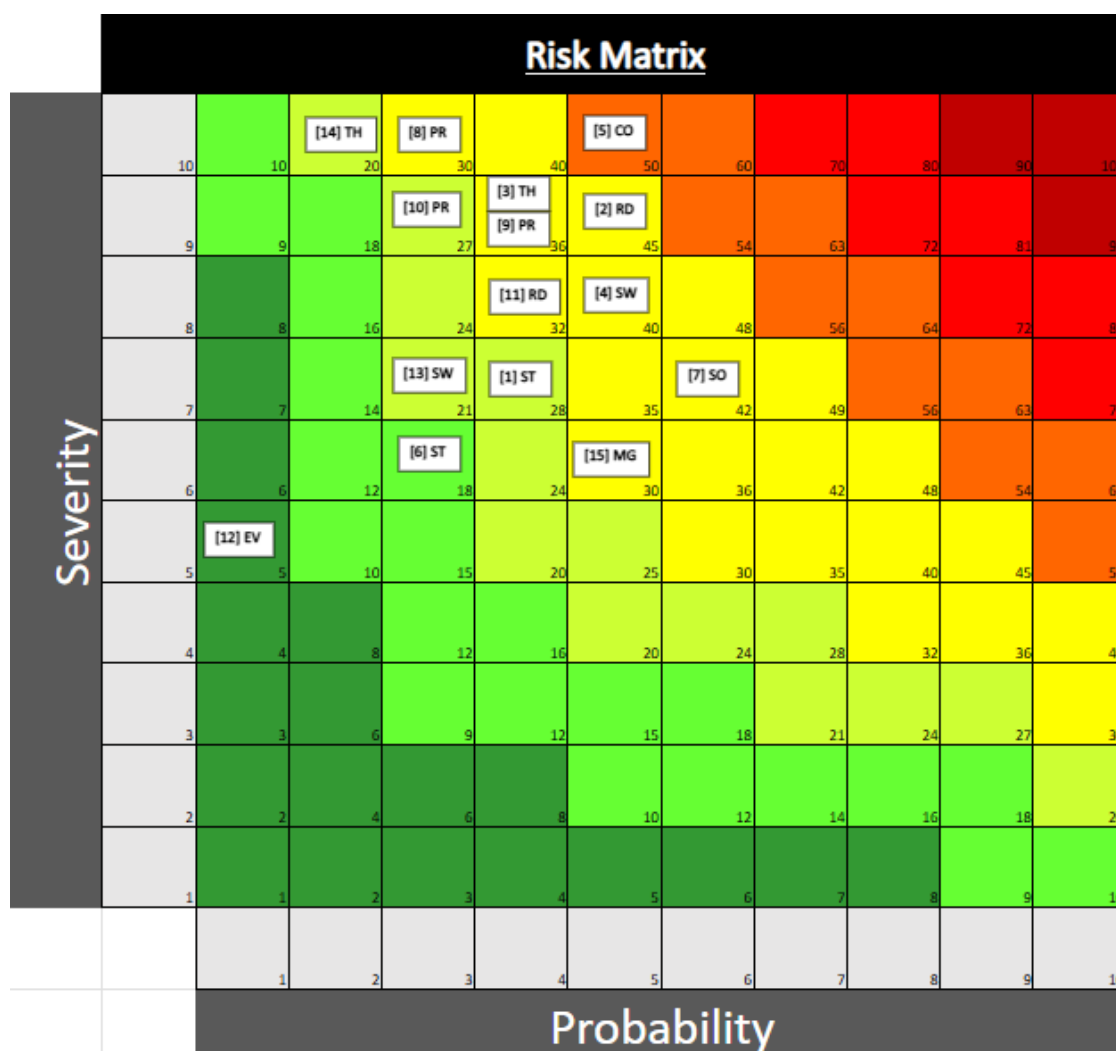


Figure 15: Risk Matrix Hazards on Plot

Total risk is assessed by multiplying the assessed severity score by the assessed likelihood score, resulting in a score between 1 and 100. This score is then color-coded for emphasis and tabulated alongside risk type, ID, and mitigation strategy. Score color ranges span green (1-29), yellow (30-49), orange (50-69), and red (70-100), with warmer tones

representing less acceptable risk. Mitigation strategies ranged from design considerations and manufacturing processes to autonomous system deployment and component redundancy. Table 35 represents the three highest rated hazards, out of 15 researched hazards, their impact on the mission, risk level, and mitigation strategies. The rest of the hazards can be observed in Table 39 in Appendix 2.

Table 35: Top 3 Hazards, Risk Level, and Mitigation Strategy

Impact	Risk Level	Mitigation Strategy
EM interference could disrupt signal clarity for incoming transmissions, jeopardizing critical comms and s/c instructions which may be time-sensitive or unrepeatable.	42	EM shielding of critical systems, booms distancing antennae from EM-emitting components, insulating material shielding high-voltage systems
Degradation or failure of spacecraft electronics and sensors, reducing mission lifetime and scientific data return.	45	Radiation shielding, automatic safe-mode planning, utilizing solar data for predictive modeling of radiation spikes.
Loss or interruption of communication with Earth could prevent transmission of scientific data and telemetry and limit mission operations.	50	MGA/LGA backup comms, redundancy in comms hardware, ability to substitute Mars-side/Earth-side systems in case of emergency.

N. Software Use and AI Disclosure

Throughout the duration of this design project, a combination of engineering software tools was used for computation, analysis, modeling, and documenting. Microsoft Excel was used to organize data and conduct calculations. Microsoft PowerPoint and Word were also used for organizational purposes and to present technical findings. [MATLAB](#) was used to develop models for calculations as well as to generate relevant figures. MATLAB scripts created for calculations can be found at the following [GitHub](#). Ansys [Systems Tool Kit](#) (STK) was used to determine, model, and simulate our satellite's orbits and trajectory. Additionally, a NASA budgeting tool, [Project Cost Estimating Capability](#) (PCEC), was provided that supported mass and cost estimations for the satellite. [Fusion 360](#) was used in the creation of the volumetric representation of HEIMDALL. [Lucid Chart](#) was used to create most of the diagrams and figures found throughout the proposal.

Artificial Intelligence (AI) tools of [ChatGPT](#), [Gemini](#), and [Claude](#) were used to provide limited assistance with clarifications, explanations, and source findings. They were *not* used for defining or conclusive results for this project. All results and conclusions came from the verification of project members and published findings. AI was also used in the generation of the team patch based on the 3D model created in Fusion 360.

O. Conclusion

Over the course of our mission, HEIMDALL will launch from Earth, complete a heliocentric phasing maneuver, and reposition to Earth-Sun L4, placing a semi-permanent link in future communications pathways from Earth to Mars and allowing for further understanding of space weather and heliophysics. HEIMDALL, in conjunction with ESA VIGIL, will supply humanity with binocular observation posts from which to better observe and predict space weather. This advancement will open the door to much further manned exploration of our solar system and more in-depth analysis of the most impactful object in it.

Through the enumeration of requirements and constraints, concept of operations, and historical analyses, we have determined a mission structure that, if put into action, will address numerous long-term goals held by NASA and the international space community. Upon reaching L4, HEIMDALL shall utilize a suite of flight-proven components to observe our Sun, transmit collected data, and serve as a communications node for Earth, Mars, and future missions to the red planet. By utilizing newly developed Deep Space Optical Communications, HEIMDALL ensures that supported data rates will not be made obsolete as the need for communication in space grows in both nodes and data size.

After conducting numerous trade studies, science, technology, and methodology comparisons, our team has developed an integrated satellite that meets multiple requirements set out by the National Academies of Science, Engineering, and Medicine's *The Next Decade of Discovery in Solar and Space Physics*, in addition to NASA's *Moon to Mars Strategy and Objectives*.

Importantly, by utilizing NASA's own PCEC software, HEIMDALL was designed from the beginning to fall within AIAA's RFP cost budget of \$400 million USD. This constraint was met by meticulous discussion on the necessity of each piece of equipment included in our subsystem design, the inclusion of cost-friendly alternatives, and a reliance on cost-locked, heritage technology, which could be manufactured and integrated with minimal research and development expenditure. The mission architecture specified in this document is a comprehensive and realistic approach to address AIAA's request for proposal. Project HEIMDALL provides a plausible approach to facilitating communications in deep space during Earth-Mars conjunction, as well as an effective means of further characterizing solar phenomena from a novel vantage point.

Appendix 1: Requirements Legend

Table 36: All subsystem requirements.

Structural Subsystem Requirements	
Being located at L4 will expose HEIMDALL to extreme radiation and cosmic rays that the structural material will have to survive for a mission duration of 15 years.	R1
The structural subsystem will require materials with high fracture toughness for the outside layers since these will be exposed to all sorts of space particles, small meteoroids, etc.	R2
The HEIMDALL bus must survive the impact and vibrations during launch and launch vehicle separation; the structural materials have to be able to absorb as much energy so that the internal components receive as little damage as possible.	R3
Thermal Subsystem Requirements	
The thermal subsystem shall radiate collected internal bus heat into the space environment via radiators mounted on the anti-Sun face of the satellite.	R4
The thermal subsystem shall maintain all internal components within their specified operational temperature during all phases of the mission. The temperature should be between -30 °C and 130 °C.	R5
The thermal mechanisms shall protect the coronagraph, heliospheric imager, and energetic particle telescope/spectrometer to meet the minimum scientific observables, requirements, and constraints.	R6
ADCS Requirements	
The ADCS subsystem shall maintain a control jitter of no more than and a control drift of no more than 0.01 deg/hr to support stable observations by HEIMDALL's scientific and communication instruments	R7
The ADCS subsystem shall provide three-axis attitude determination accuracy between $\pm 0.01^\circ$ (3σ) and control accuracy of $\pm 0.05^\circ$ (3σ) to support communications relay operations and science pointing needs throughout HEIMDALL's operational lifetime.	R8
The ADCS subsystem shall support all five operational modes of HEIMDALL: science mode, communication relay mode, momentum unloading mode, safe mode, and acquisition mode.	R9
The ADCS subsystem shall support autonomous onboard attitude determination and control without continuous real time ground station intervention.	R10
Command & Data Handling Requirements	
The OBC should have a reliable redundancy system in the event of extreme solar radiation that might damage scientific components.	R11
The OBC shall be able to process all the data collected and perform real-time JPEG compression on raw solar images prior to routing data to storage.	R12
The C&DH subsystem shall distribute a continuous 1 pulse per second hardware synchronization signal across the data bus to ensure timing accuracy between the payload and the TT&C subsystems	R13
The system shall be capable of applying time-synchronization commands from the ground station during scheduled TT&C uplink windows.	R14

Communication Requirements	
Science payload data shall be downlinked for mission analysis.	R15
Status telemetry shall be available $\geq 99\%$ of contact time with a maximum latency of 10 minutes per pass.	R16
Fault telemetry shall be prioritized for response in the case of an anomaly.	R17
Critical commands shall be uplinked for emergency actions.	R18
Payload/configuration commands shall be uplinked for instrument tasks.	R19
Routine commands shall be uplinked for nominal mission operations.	R20
Relay communication via Mars crew assets shall be supported.	R21
Power Requirements	
The power subsystem shall provide a total power output of 472 W with a 20% margin	R22
Propulsion Requirements	
The propulsion subsystem shall provide all station keeping impulses required to maintain the satellite near Sun-Earth L4 for the full mission lifetime.	R23
The propulsion subsystem shall provide a minimum station keeping ΔV capability of 30 m/s over 15 years to account for gravity losses.	R24
The propulsion subsystem shall provide additional delta-v capability for solar radiation pressure correction and trajectory dispersions.	R25
The propulsion subsystem shall account for a minimum 2% unusable propellant residual.	R24
The propulsion subsystem shall maintain partial maneuver capability after the failure of a single thruster.	R25
The propulsion subsystem shall be compatible with launch vehicle hazardous-material and payload safety requirements.	R26

Appendix 2: Scientific Payload and Risk Analysis

Table 37: Min/Max Values for Scoring and TRL Lookup Table for Scientific Payload Trade Study

Min/Max helpers (from Candidate Data)			TRL Lookup (editable)	
			TRL	Score (0-1)
			9	1
			8	0.9
			7	0.8
			6	0.6
			5	0.4
			4	0.2
			3	0.1
			2	0.05
			1	0
Metric	Min	Max		
Mass	0.023	303		
Power	0.48	340		
Size	0.000005292	337.1963		

Table 38: Probability and Severity Weighted Criteria and Examples and Key

Key							
Category	Range	Definition	Example*	Category	Range	Definition	Example*
Probability	1	Extremely Unlikely	Asteroid Impact on Ground	Severity	1	No Effect	Software bug before assembly
	2	Unlikely	Collision in LEO		2	Self-Correcting	Light Breeze on Takeoff
	3	Improbable	Failure on Pad		3	Temporary Effect	Software Update in Transit
	4	Possible	First Stage Failure		4	Mild Effect	EM interference in LGA
	5	Plausible	Second Stage Failure		5	Component Threat	Cold Weld of Dampener after Takeoff
	6	Real Possibility	Acoustic Damage on Takeoff		6	Subsystem Threat	Science Instrument Soft Failure
	7	Slightly Probable	Cosmic Dust Encounter at L4		7	System Threat	Single Antenna Failure
	8	Probable	Launch Delay		8	Spacecraft Threat	Solar Panel Debris Impact
	9	Highly Probable	Radiation Exposure		9	Mission Threat	Total Comms Failure
	10	Certain	Intense G-Forces		10	Mission Critical	Uncontrolled Craft Spin

Table 39: Full Hazards List

ID	Hazard Type	Hazard	Description	Probability	Severity	Impact	Risk Level	Mitigation Strategy
1	ST	Launch Vibration Damage	High acoustic and vibration loads during launch could damage sensitive instruments if not sufficiently isolated.	4	7	Payload damage during launch could cause partial or total loss of mission instruments and prevent the spacecraft from performing its science objectives	23	Natural Frequency Analysis in development process, lightweight acoustic insulation, anti-oscillation s/c construction
2	RD	Solar Radiation Damage	Solar energetic particle events and long-term radiation exposure could damage avionics or degrade instrument performance.	5	9	Degradation or failure of spacecraft electronics and sensors, reducing mission lifetime and scientific data return.	45	Radiation shielding, automatic safe-mode planning, utilizing solar data for predictive modeling of radiation spikes
3	TH	Thermal Control Failure	Failure of heaters, radiators, or insulation could cause spacecraft components to overheat or freeze, leading to degraded performance or hardware damage.	4	9	Failure of the thermal control system could cause overheating or freezing of spacecraft electronics or instruments, potentially leading to subsystem shutdown or permanent hardware damage.	36	Thermal regulation redundancy in construction, permanent hot-cold polarity in s/c position allowing for unpowered, automatic thermal control
4	SW	Onboard Software/System Malfunction	Software errors or system faults could interrupt spacecraft operations, causing temporary loss of communications or science data.	5	8	Software faults in the onboard computer could interrupt spacecraft operations, require system reboot, or temporarily suspend science observations and communications.	40	Pre-launch troubleshooting/debugging, onboard computer redundancy, stop-loss programming strategies
5	CO	Communications Failure	Failure of the communication subsystems (transmitter, receiver, or antenna hardware) could result in loss or degradation of the communication link with Earth.	5	10	Loss or interruption of communication with Earth would prevent transmission of scientific data and telemetry and limit mission operations.	50	MGMA/LGA backup comms, redundancy in comms hardware, ability to substitute Mars-side/Earth-side systems in case of emergency
6	ST	Cold Welding of Component Parts	Metallic Parts with compatible crystallography may fuse together in space inseparably, potentially resulting in changed thermal/acoustic profile of s/c.	3	6	Cold welding of parts may drastically affect the thermal performance of our craft and result in poor temperature regulation. Acoustic hazard presented is minimal so long as cold welding occurs after all propulsion burns.	18	Materials analysis during construction to prevent allowable materials adjacency, polymer washers between metallic parts, widespread use of composites
7	SO	Electromagnetic Interference between components	EM interference could present minor issues in onboard computing, or major hazards in uplink comms reception.	6	7	EM interference could disrupt signal clarity for incoming transmissions, jeopardizing critical comms and s/c instructions which may be time-sensitive or unrepeatable.	42	EM shielding of critical systems, booms distancing antennas from EM-emitting components, insulating material shielding high-voltage systems
8	PR	Propulsion Leak or Rupture	Propellant leakage or pressurization fault can lead to a fire, loss of maneuverability, or catastrophic failure.	3	10	Complete vehicle loss or inability to finish mission as the loss of propellant means there won't be enough margin to get into the correct orbit.	30	Leak checks, pressure tests, check valves which testing. Doing individual engine test fires and tank testing. Doing a full wet dress rehearsal. Finally first and second stage test burns.
9	PR	Fairing Deployment Failure	Payload fairing fails to separate or separates incorrectly putting the satellite into an incorrect orbit.	4	9	Mission failure due to payload not within the correct orbit or not separating from the first stage.	36	Deployment testing: test firing pyro or other mechanical separation devices. Analyzing deployment margins.
10	PR	Stage Separation Anomaly	Failure to stage, improper separation timing, physical part failures damaging the payload	3	9	Loss of function from damage to payload or complete loss of payload	27	Separation dynamics analysis, joint qualification
11	RD	Power System Degradation	Long duration exposure to radiation or small meteoroids	4	8	Power insufficiency for instruments or satellite systems can cause failure of mission to observe, measure, and communicate	32	Large solar arrays/panels, power management systems, redundant power systems
12	EV	Space Debris Impact	High speed particles/debris impacting spacecraft surfaces and instruments	1	5	Structural damage and loss of critical subsystems	5	Redundant systems, protective spacecraft orientation/shielding
13	SW	Data Storage Problems	Measured data cannot be stored to be transferred/communicated, hardware fails or the storage is corrupted	3	7	Losing valuable mission data, unable to complete its proposed mission	21	Redundant storage systems, corruption troubleshooting
14	TH	Battery Thermal Runaway	Battery cell defects, is overcharged, internal shortage, leading to overheating and failure	2	10	Catastrophic failure of electrical system	20	Well designed thermal control system, redundant batteries
15	MG	Budgeting	The project's mission needs overrun the limited budget requirements	5	6	Running out of funds to be able to accomplish the proposed mission	30	Cost planning, heritage equipment, careful selection of parts

Appendix 3: Subsystem Constraints

Table 40: All subsystem constraints.

ADCS Constraints
The ADCS subsystem is constrained by the Sun-Earth L4 environment, where solar radiation pressure is the dominant external attitude disturbance.
Command & Data Handling Constraints
To survive a deep-space environment at Lagrangian point L4, all critical C&DH processing components must have a Total Ionizing Dose (TID) of at least 100 krad.
Propulsion Subsystem Constraints
The propulsion subsystem must operate within the satellite volume allocation for tanks, thrusters, lines, and pressurization hardware.
The propulsion subsystem must not exceed the satellite power budget during thruster valve operation or thermal-control operation.
The propulsion subsystem must be compatible with long duration storage over a 15-year mission lifetime.
The propulsion subsystem must use commercially available or heritage components where possible to reduce development risk and cost.
The propulsion subsystem must minimize total wet mass while still satisfying delta-v, impulse, redundancy, implementation, and lifetime requirements.
The propulsion subsystem must be designed so that thruster plumes minimize contamination of optical payloads, including coronagraph, heliospheric imager, or other sun observing instruments.
Structural Subsystem Constraints
The structure design must fit inside the launch vehicle and have a destined attachment support.
The materials must be on the lighter scope as we all maintain a relatively accessible price and great manufacturing.
In order to maintain good communication between Earth and Mars as well as having a direct line of sight to the Sun, the materials chosen must have a good cryogenic performance to avoid changes in dimension when experiencing different temperature ranges.
Temperature can range from -100 °C to 150 °C and the materials chosen must maintain good mechanical properties at these temperature ranges.
Thermal Subsystem Constraints
To keep power budget low, the thermal subsystem architecture prioritizes passive thermal control methods before implementing active control methods.
All exterior thermal coatings and layers must be rated to survive 14 years of deep space ultraviolet and proton radiation.
Batteries, some science instruments, and communications equipment have narrow temperature ranges, -5 to 25, -35 to 5, and -10 to 50 degrees Celsius, respectively.

Appendix 4: Size, Weight, and Power Tables

Science & Communication Goals	Equipment Type	Proposed Instrument	LxWxH (m)	Volume (m^3)	Mass (g)	Peak Power (W)
Enable Early Space Weather Detection	Coronagraph	Compact Coronagraph (CCOR-1)	0.889	0.1978738	25000	22.5
			0.5842			
			0.381			
	Energetic Particle Telescope	STIS	0.165	0.00289344	2300	3.8
			0.128			
			0.137			
	Radio Burst Monitor	WAVES Heritage	50	0.125	8000	8
			0.05			
			0.05			
	Solar Wind Ion/Plasma Analyzer	PLA on VIGIL	-	0.07875	16500	14
			-			
			-			
	Heliospheric Imager	SOLOHI Heritage	0.684	1.07875	16600	13.5
			0.455			
			0.115			
	X-ray Flux Monitor	VIGIL X-ray Monitor	0.1	2.07875	14000	30
			0.1			
			0.32			
	EUV Imager/Magnetograph	PMI VIGIL	-	0.1487	28200	40
			-			
			-			
	Fluxgate Magnetometer	MAG/FGM Pair	0.02	1.1487	650	1.71
			0.02			
			0.02			
Establish and Maintain Consistent Comms between earth and Mars.	TWTA (Traveling Wave Tube Amp)	Heritage TWTA (using MRO TWTA for reference)	0.4826	0.09262788	9500	100
			0.22225			
			0.8636			
	Transponder	TESAT IDT or SDST alternative	0.22	0.0033	2000	20
			0.1			
			0.15			
	HGA(High Gain Antenna)	MRO HGA reference	2	4	19100	0
			2			
			1			
	Data Processing	Sirius Quadcore LEON4FT	0.1	0.00015	100	4
			0.1			
			0.015			
	Data Processing	IADCS200	0.95	0.2736	470	4.5
			0.9			
			0.32			
	Transmitter	TESAT IDT	0.22	0.0033	2000	20
			0.1			
			0.15			
	Transmitter/receiver	TESAT SCOT135	0.203	0.0123	50000	75
			0.225			
			0.27			

	Equipment Type	Proposed Instrument				
			LxWxH (m)	Volume (m^3)	Mass (g)	Peak Power (W)
Subsystems	ADCS	RDR-68 Reaction Wheel	0.345	0.014045	30400	20
			0.345			
			0.118			
		RDR-68 Wheel Drive Electronics	0.052	0.001862	5000	-
			0.247			
			0.145			
		LR-450 IMU	0.218	0.005581	4540	15
			0.16			
			0.16			
		ASTRO CL Star Tracker	0.06	0.000464	640	1.20
			0.06			
			0.129			
		Coarse Sun Sensor	0.11	0.000363	1290	0
			0.11			
			0.03			
		MONARC-5 Thruster	0.418	0.000261	3920	144
			0.025			
			0.025			
	Propulsion	MRE-5.0	0.25	0.000361	3200	76
			0.038			
			0.038			
	Power	T-Star Solar Array	0.8	2.5904	13520	0
			1.619			
			1			
		Saft Total Energies MP 176065 xtd	0.27	0.017496	6300	0
			0.36			
	0.09					
	Thermal	Multi-Layer Insulation (MLI) Blankets	-	0	-	0
			-			
			-			
		AZ-93 Zinc Oxide-based White Coating	-	0	0	0
			-			
			-			
		Aluminum Heat Pipes	-	0	-	0
			-			
			-			
		AZ-93 High-Emissivity Coated Aluminum Radiator Panels	-	0	600	0
			-			
			-			
	Minco Polyimide Thermofoil Kapton Film Heaters	-	0	50	5	
		-				
		-				
	CD&H	Voyager Technologies IPC-7000	0.268	0.001054	5800	40
			0.178			
			0.01105			
	Structures	Satellite Main Body	1.15	3.04175	158000	0
			1.15			
			1.15			
	Communications	Heritage TWTA (using MRO TWTA for reference)	0.4826	0.185256	9500	100
			0.22225			
			0.8636			
		TESAT IDT or SDST alternative	0.22	0.0066	2000	20
			0.1			
			0.15			
		MRO HGA reference	2	8	19100	0
			2			
			1			
		Sirius Quadcore LEON4FT	0.1	0.0003	100	4
			0.1			
			0.015			
		IADCS200	0.95	0.5472	470	5
			0.9			
			0.32			
		TESAT IDT	0.22	0.0066	2000	20
			0.1			
			0.15			
	TESAT SCOT135 (range limited, good SWaP comparison)	0.203	0.024665	50000	75	
		0.225				
		0.27				
			Totals	23.688	535200	882.21

Appendix 5: Compliance Matrix

Table 42: Compliance Matrix.

RFP Ref.	Requirement	Compliant?	Section	Rationale
4.1	Design an integrated comprehensive mission to send one or more exploration assets to space to meet the National Academies of Science's Heliophysics priority investigations as outlined in the 2024 decadal survey for solar and space physics as well as NASA's Moon to Mars Objectives ³ and to provide communication capability for potential crewed missions to Mars.	YES	C, D, I	This requirement was a constraint imposed by the Request for Proposal (RFP)
4.2.1	Scientific instruments to support our understanding of the local cosmos, including ability to monitor and detect solar and space weather phenomena, and enable the capability to provide early warning for crewed missions.	YES	C, I	This requirement is satisfied by the science and communication objectives stated in Figure 1 and achieved by the different subsystem requirements.
4.2.2	Ability to provide line of sight between the Earth and Mars to enable near constant communication capability during the RFP specified mission period.	YES	I.1	This requirement is satisfied by the communication objectives found in Figure 1 and achieved by the different subsystem requirements.
4.4	Perform trade studies on various mission designs at the architecture and system levels to demonstrate the fitness of the chosen mission and system design. It is highly desirable to use technologies that are already demonstrated on previous programs or currently in the NASA technology development portfolio.	YES	G, H	This requirement is satisfied in sections G and H. These sections show that the equipment used for HEIMDALL has been used in previous NASA's missions.
4.5	Perform mission analysis to evaluate the flight profile for the asset(s) and describe the mission profile for observation measurement and to show how the mission provides adequate coverage as specified by the RFP and provide discussion on the communication capabilities for the design.	YES	D, I	This requirement is satisfied in section D.
4.7	Discuss selection of subsystem components and the values of each of the selections and how the design requirements or scientific objectives drove the selection of the subsystem.	YES	I.1 through I.7	This requirement is satisfied and explained in section I.
4.8	Discuss the instruments selected to address the primary science objectives, and the data collection, analysis, data processing, and transmission process to address the objectives.	YES	F, G, H, I	This requirement is satisfied and explained in section F, G, H. and I.
4.9	The cost for the mission and capability development in support of its activities shall not exceed \$400 Million US Dollar (in FY25), including development, hardware, launch, and operation cost of the mission through the primary mission phase.	YES	L	This requirement was a constraint imposed by the Request for Proposal (RFP)
4.10	The mission should launch no later than December 31, 2035, with the system designed to operate into late 2040s.	YES	D	This requirement was a constraint imposed by the Request for Proposal (RFP)

References

- [1] Wertz, J. R., & Larson, W. J. (1999). *Space Mission Analysis and Design (SMAD)* (3rd ed.). Springer.
- [2] NASA. (2025, February 14). *2024 State-of-the-art of Small Spacecraft Technology*.
<https://www.nasa.gov/smallsat-institute/sst-soa/>
- [3] Naval Postgraduate School. (1992, December 15). *Near earth asteroid rendezvous: Design project for AE 4871* (NASA-CR-195518)
- [4] Pogge, R. W. (n.d.). *Nominal solar & planetary quantities*. Department of Astronomy, The Ohio State University. Retrieved April 21, 2026, from <https://www.astronomy.ohio-state.edu/pogge.1/Ast2292/Docs/NominalConstants.pdf>
- [5] Roberts, C. (n.d.). *Long-term missions at the Sun–Earth L1 libration point: ACE, SOHO, and WIND*. NASA. Retrieved April 21, 2026, from
<https://ntrs.nasa.gov/api/citations/20110008638/downloads/20110008638.pdf>
- [6] Dankanich, J., Szabo, J., Pote, B., Oleson, S., & Kamhawi, H. (n.d.). *Mission and system advantages of iodine Hall thrusters*. NASA. Retrieved April 21, 2026, from
<https://ntrs.nasa.gov/api/citations/20140012585/downloads/20140012585.pdf>
- [7] Kuhl, C., Wright, H., Hunter, C., Guernsey, C., & Colozza, A. (n.d.). *Liquid rocket propulsion for atmospheric flight in the proposed ARES Mars Scout mission*. NASA. Retrieved April 21, 2026, from
<https://ntrs.nasa.gov/api/citations/20040086730/downloads/20040086730.pdf>
- [8] Cornish, N. (n.d.). *The Lagrange points*. Retrieved April 21, 2026, from
<https://web.archive.org/web/20150907090116/http://www.physics.montana.edu/faculty/cornish/lagrange.pdf>
- [9] Branz, F., Cappelletti, C., Ricco, A. J., & Hines, J. (2023). *Next generation CubeSats and SmallSats*. Elsevier.
- [10] SMI Composites. (n.d.). *Aluminum vs. carbon fiber: What are the differences?*
<https://www.smicomposites.com/aluminum-vs-carbon-fiber-what-are-the-differences/>
- [11] Hales, S. J., & Taminger, K. M. (2010, June 20–24). *Aerospace applications of aluminum-lithium alloy thick plate* [Conference presentation]. AeroMat 2010, Bellevue, WA, United States.
- [12] SAMPE. (n.d.). *Composites applications for space*. <https://sampe.org/composites-applications-for-space/>

- [13] Leishman, J. G. (n.d.). *Aerospace materials*. In *Introduction to aerospace flight vehicles*. Embry-Riddle Aeronautical University.
<https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/aerospace-materials/>
- [14] Kitamura, T., Yoshida, K., Seko, H., & Shi, Q. (2012). Analysis of shock response for satellite separation. In *Proceedings of the International Conference on Noise and Vibration Engineering (ISMA2012)*.
- [15] Voyager Space. (2024, February 23). *IPC-7000™ image processing computer* [Data sheet].
- [16] Taylor, J., Lee, D. K., & Shambayati, S. (2006). *Mars Reconnaissance Orbiter telecommunications*.
https://descanso.jpl.nasa.gov/DPSummary/MRO_092106.pdf
- [17] Biswas, A., et al. (2025). Overview of the deep space optical communications (DSOC) technology demonstration. *Proceedings of SPIE*. <https://doi.org/10.1117/12.3043966>
- [18] Ashby, M. F., Shercliff, H., & Cebon, D. (2019). *Materials engineering: Science, processing and design* (4th ed.). Butterworth-Heinemann.
- [19] Kegege, O., et al. (2012). *Three-dimensional analysis of Deep Space Network antenna coverage*. NASA.
<https://ntrs.nasa.gov/api/citations/20120004035/downloads/20120004035.pdf>
- [20] Glavich, T. (2015). *Deep space optical communications*. NASA.
https://www.nasa.gov/sites/default/files/atoms/files/tglavich_dsoc.pdf
- [21] Messmann, D., et al. (2019). Hardware-in-the-loop and software-in-the-loop testing of the MOVE-II CubeSat. *Aerospace*, 6(12), 130. <https://doi.org/10.3390/aerospace6120130>
- [22] Fortescue, P., Stark, J., & Swinerd, G. (2003). *Spacecraft systems engineering* (3rd ed.). John Wiley & Sons.
- [23] Rocket Lab. (n.d.). *STARRAY (StarRay) customizable solar array solutions datasheet*. Rocket Lab Space Systems.
- [24] Mallon, K. R., Assadian, F., & Fu, B. (2017). Analysis of on-board photovoltaics for a battery electric bus and their impact on battery lifespan. *Energies*, 10(7), 943. <https://doi.org/10.3390/en10070943>
- [25] Saft (TotalEnergies). (2023). *Saft lithium product selector guide*.
- [26] Zandbergen, B. T. C. (2020). *Spacecraft bus design and sizing*. Delft University of Technology.
- [27] Santana-Ros, T., Micheli, M., Faggioli, L., Cennamo, R., Devogèle, M., Alvarez-Candal, A., Oszkiewicz, D., Ramírez, O., Liu, P.-Y., Benavidez, P. G., Campo Bagatin, A., Christensen, E. J., Wainscoat, R. J., Weryk, R., Fraga, L., Briceño, C., & Conversi, L. (2022). Orbital stability analysis and photometric

- characterization of the second Earth Trojan asteroid 2020 XL5. *Nature Communications*, 13, Article 447.
<https://doi.org/10.1038/s41467-022-27988-4>
- [28] European Space Agency (ESA). (n.d.). *VIGIL mission overview*.
https://www.esa.int/Space_Safety/VIGIL/VIGIL_mission_overview
- [29] Posner, A., Arge, C. N., Staub, J., St. Cyr, O. C., Folta, D., Solanki, S. K., Strauss, R. D. T., Effenberger, F., Gandorfer, A., Heber, B., Henney, C. J., Hirzberger, J., Jones, S. I., Köhl, P., Malandraki, O., & Sterken, V. J. (2021). A multi-purpose heliophysics L4 mission. *Space Weather*, 19, e2021SW002777.
<https://doi.org/10.1029/2021SW002777>
- [30] *AIAA Design Competitions*. (2026, April). AIAA - Shaping the Future of Aerospace. <https://www.aiaa.org/get-involved/university-students/design-competitions/#design-compe>
- [31] National Environmental Satellite, Data, and Information Service. (n.d.). *Space weather next*. National Oceanic and Atmospheric Administration. Retrieved May 13, 2026, from <https://www.nesdis.noaa.gov/our-satellites/future-programs/space-weather-next>
- [32] National Aeronautics and Space Administration. (2026, March 19). *Parker Solar Probe*. NASA Science. Retrieved May 13, 2026, from <https://science.nasa.gov/mission/parker-solar-probe/>
- [33] National Aeronautics and Space Administration. (2025, December 12). *STEREO A & B*. NASA Science. Retrieved May 2, 2026, from <https://science.nasa.gov/mission/stereo/>
- [34] National Aeronautics and Space Administration. (2026, May 11). *Mars Reconnaissance Orbiter*. NASA Science. Retrieved May 11, 2026, from <https://science.nasa.gov/mission/mars-reconnaissance-orbiter/>
- [35] National Aeronautics and Space Administration. (2026, May 11). *MAVEN*. NASA Science. Retrieved May 11, 2026, from <https://science.nasa.gov/mission/maven/>
- [36] National Aeronautics and Space Administration. (2026, April 17). *ESCAPADE*. NASA Science. Retrieved May 11, 2026, from <https://science.nasa.gov/mission/escapade/>